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(54) **FILM SECURING APPARATUS AND METHOD**

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See application file for complete search history.

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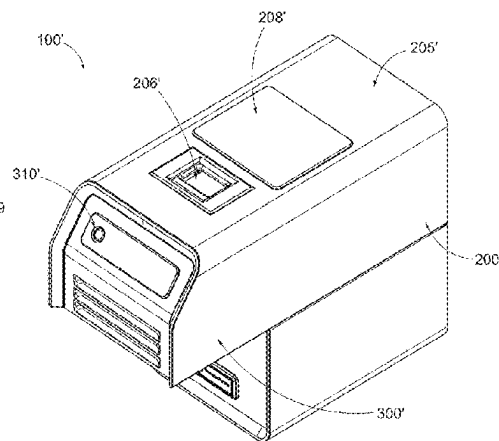
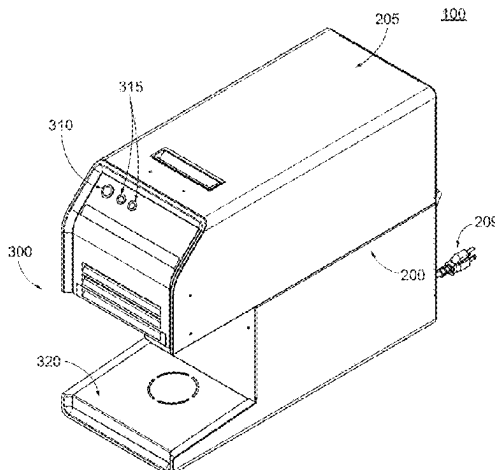
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(57) **ABSTRACT**

An apparatus may include a film support roller supporting a film roll and a housing that may include a sealing portion having an aperture sized to receive a top portion of a container. The housing may define a film path leading from the film roll to the sealing portion. A guide support assembly may have a ramp and at least one guide truss positioned along the film path. The ramp and the guide truss may guide the film across the aperture within the sealing portion, covering the aperture. The motor may advance the film along the film path. The heating element may emanate energy. The controller may cause the motor to advance the

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film onto the ramp and cause the heating element to emanate energy to cause the film to seal the top portion of the container.

11 Claims, 29 Drawing Sheets

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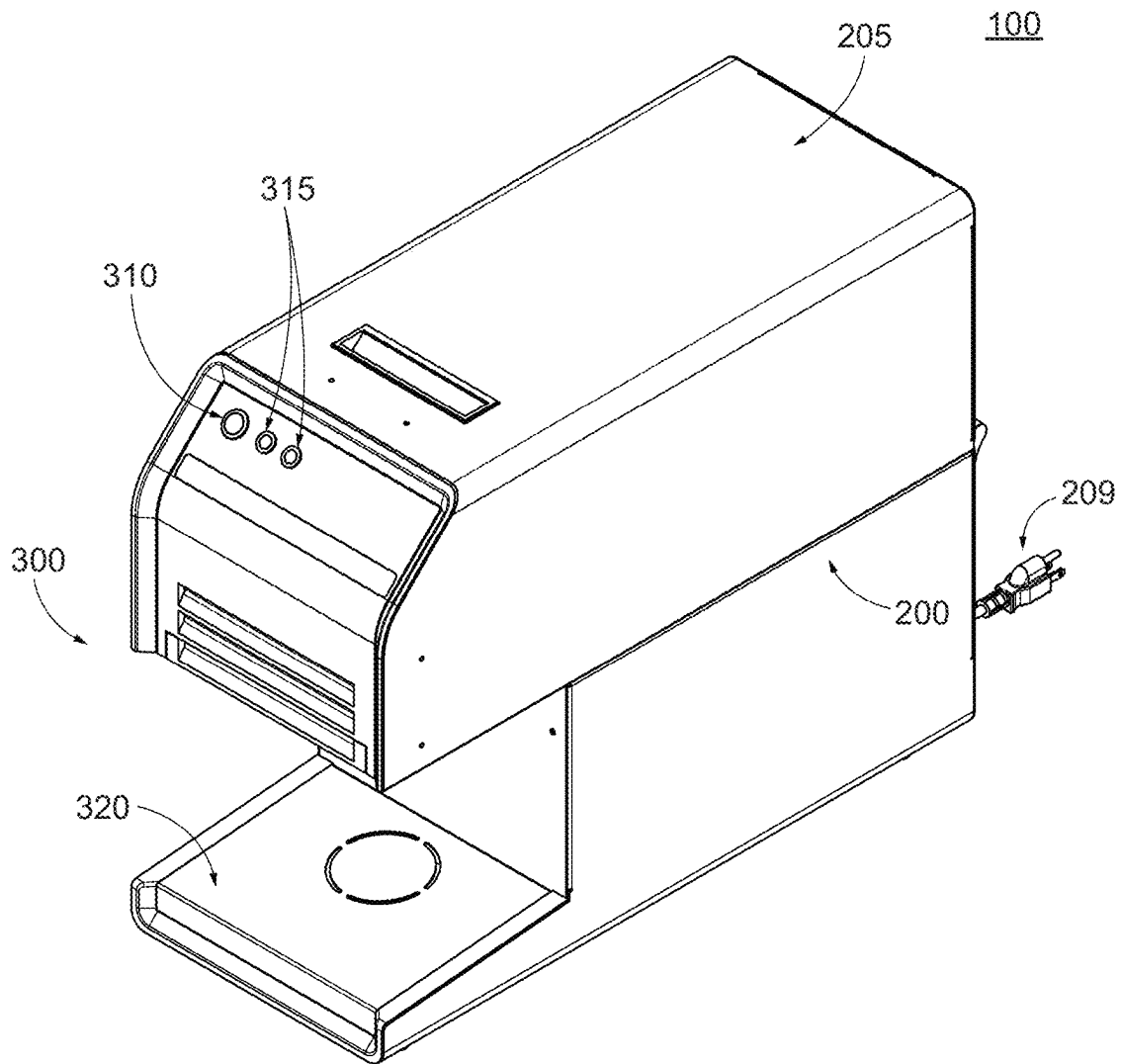


FIG. 1A

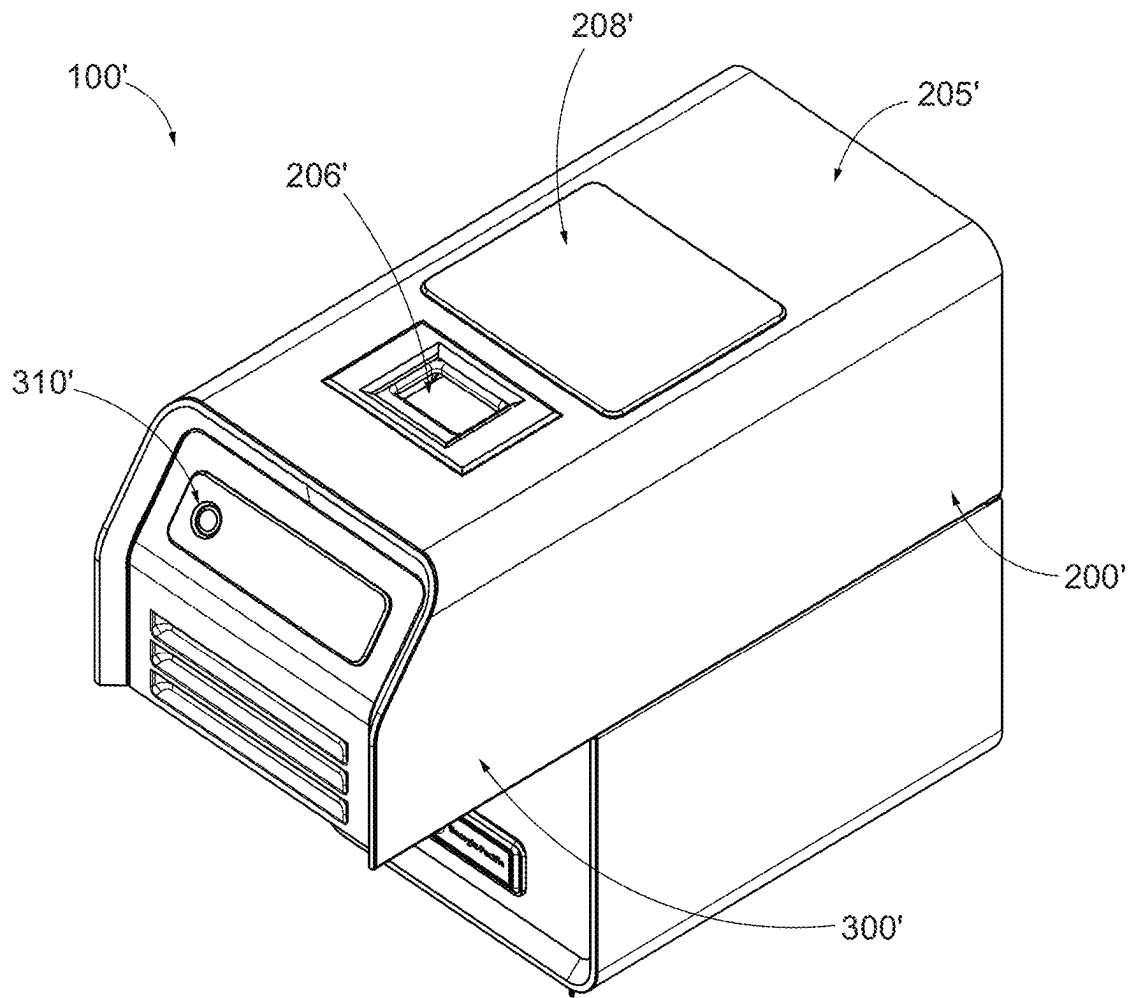


FIG. 1B

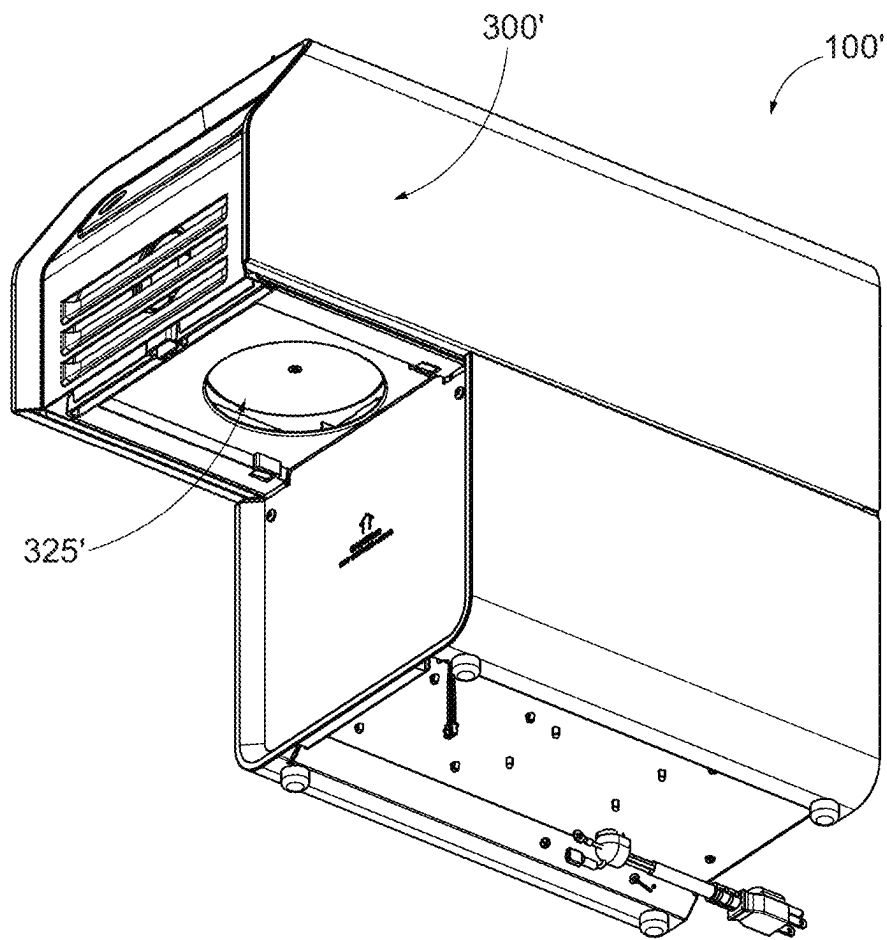


FIG. 1C

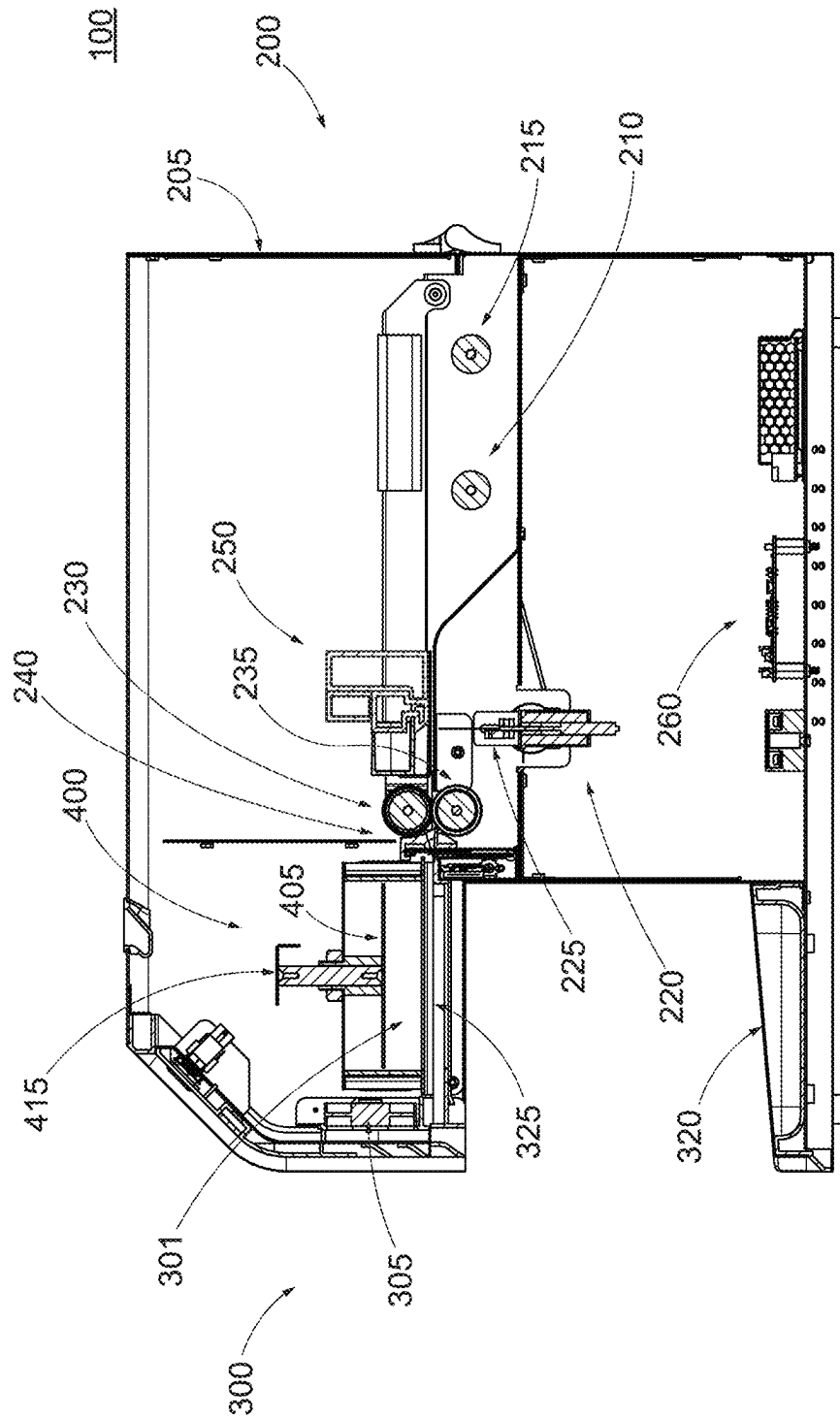


FIG. 2A

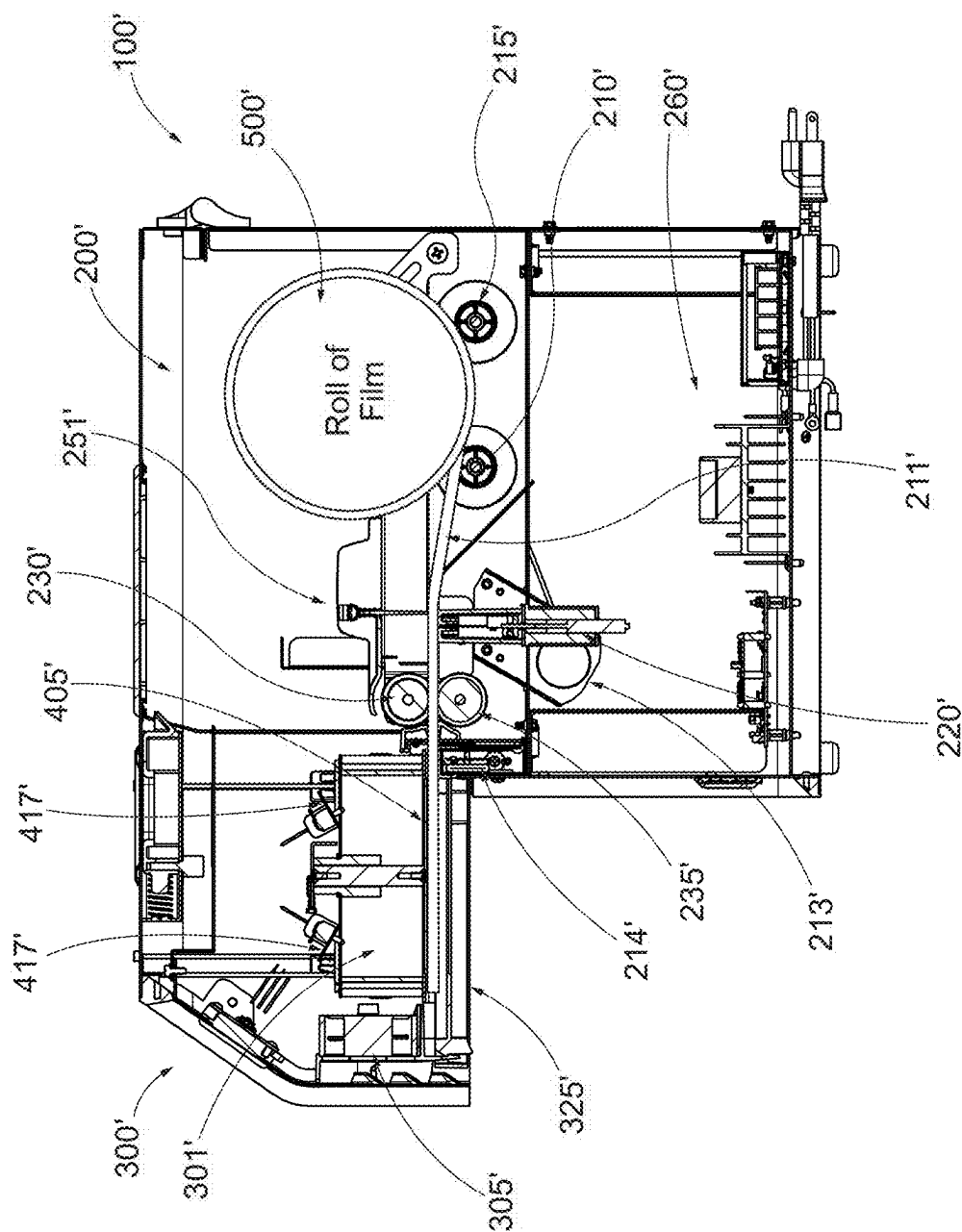


FIG. 2B

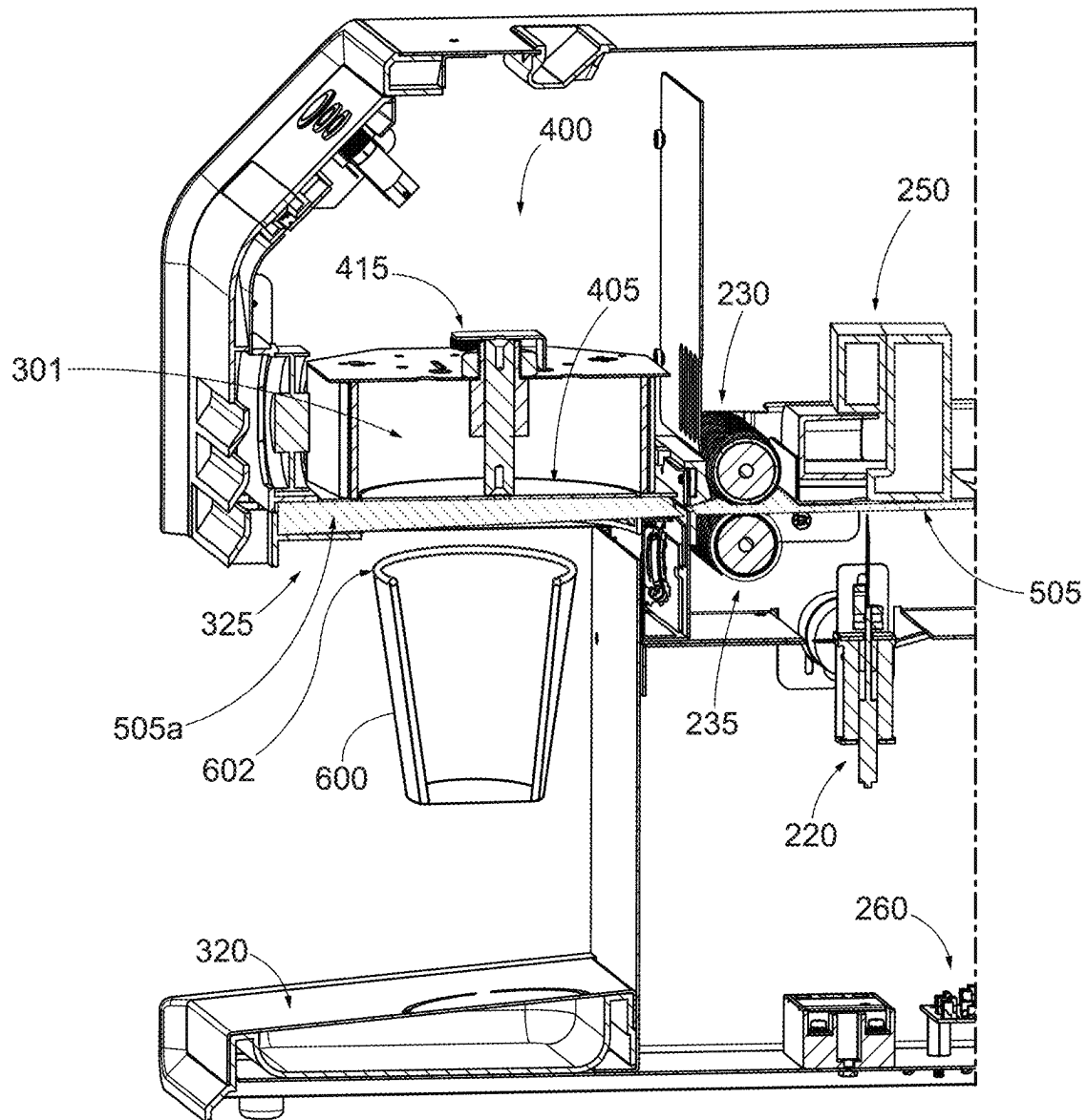


FIG. 3A

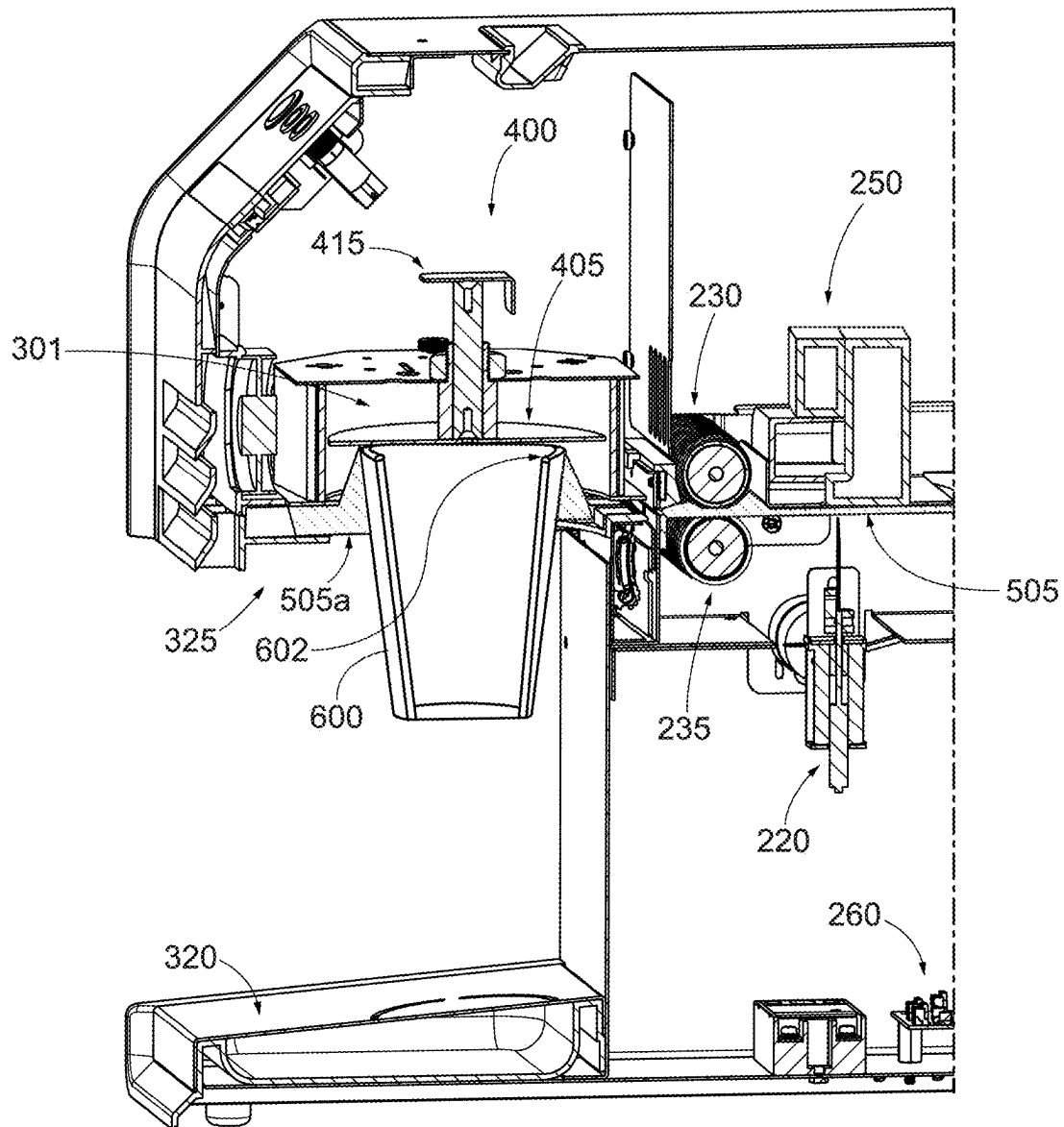


FIG. 3B

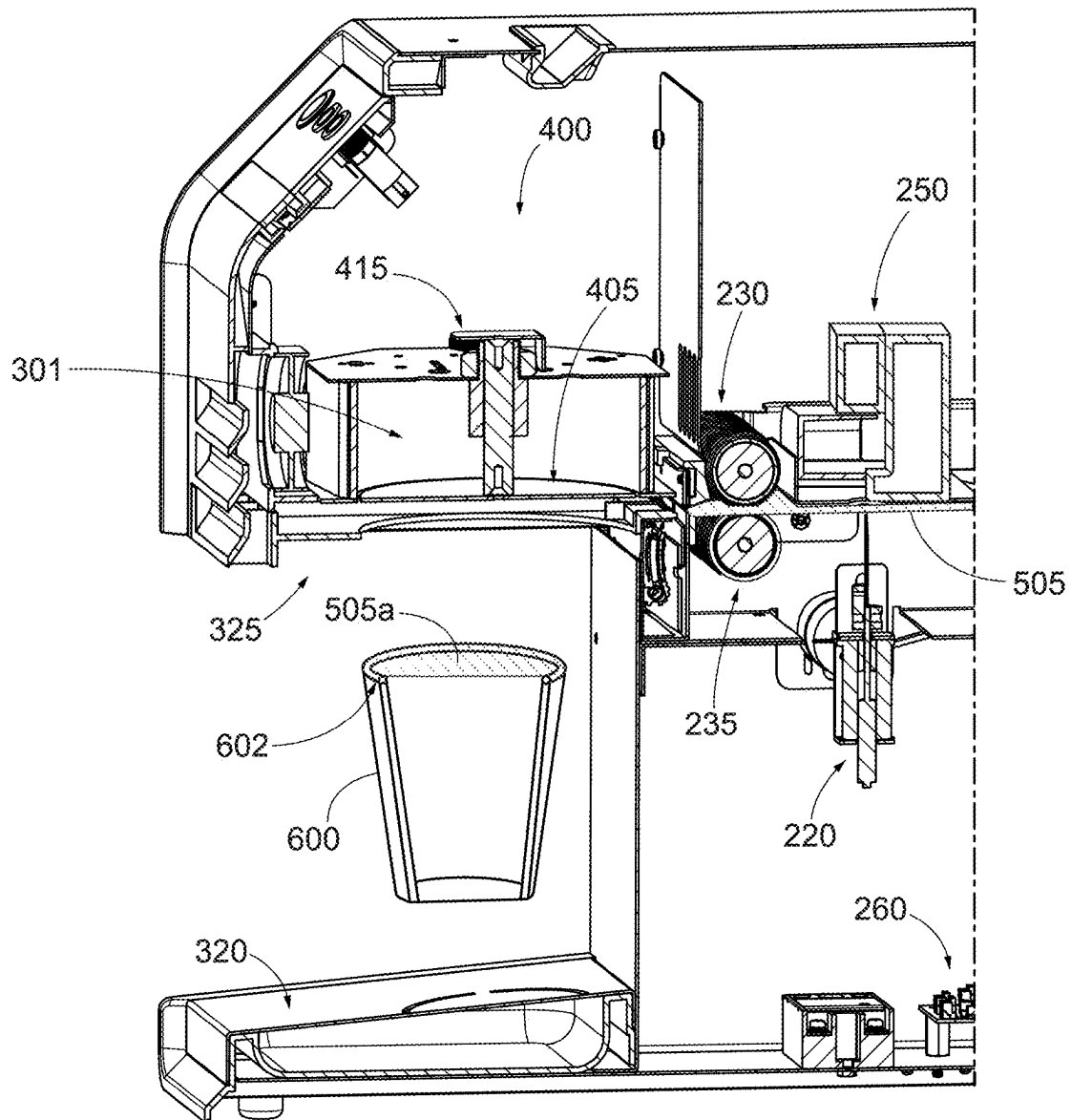


FIG. 3C

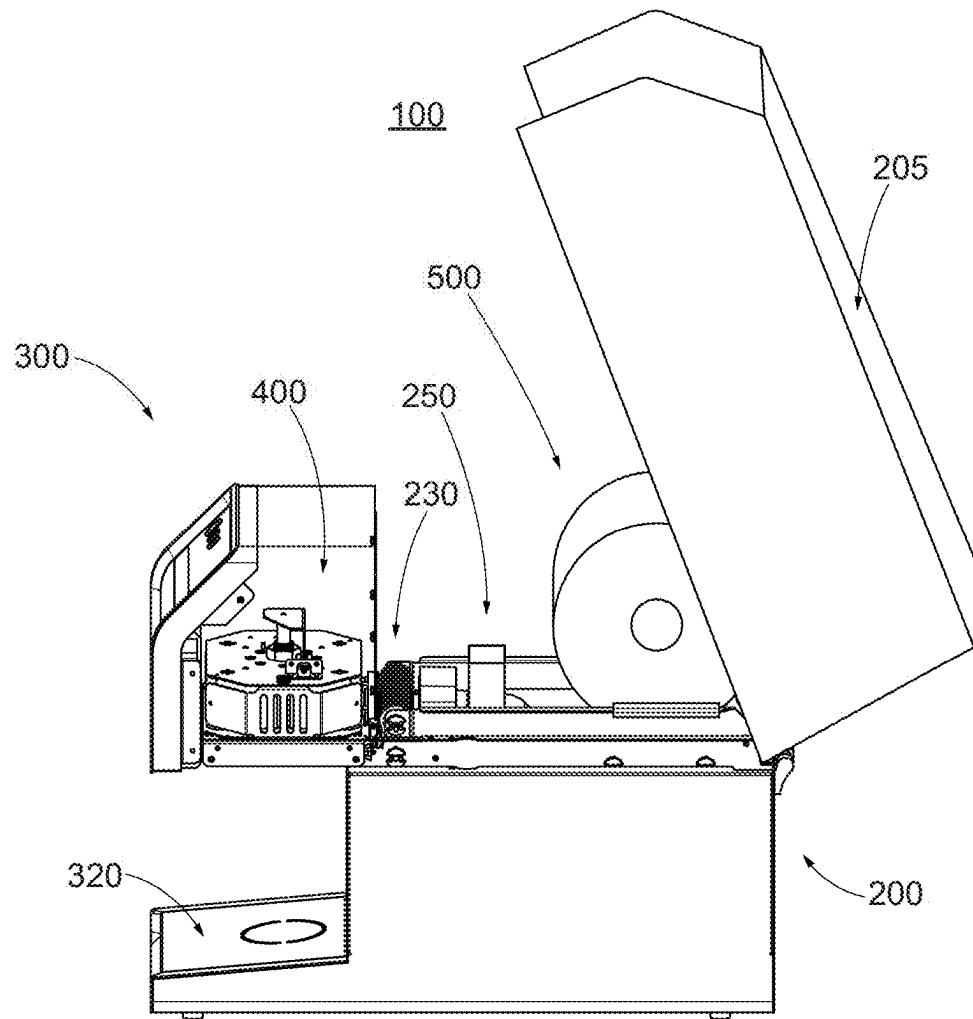


FIG. 4A

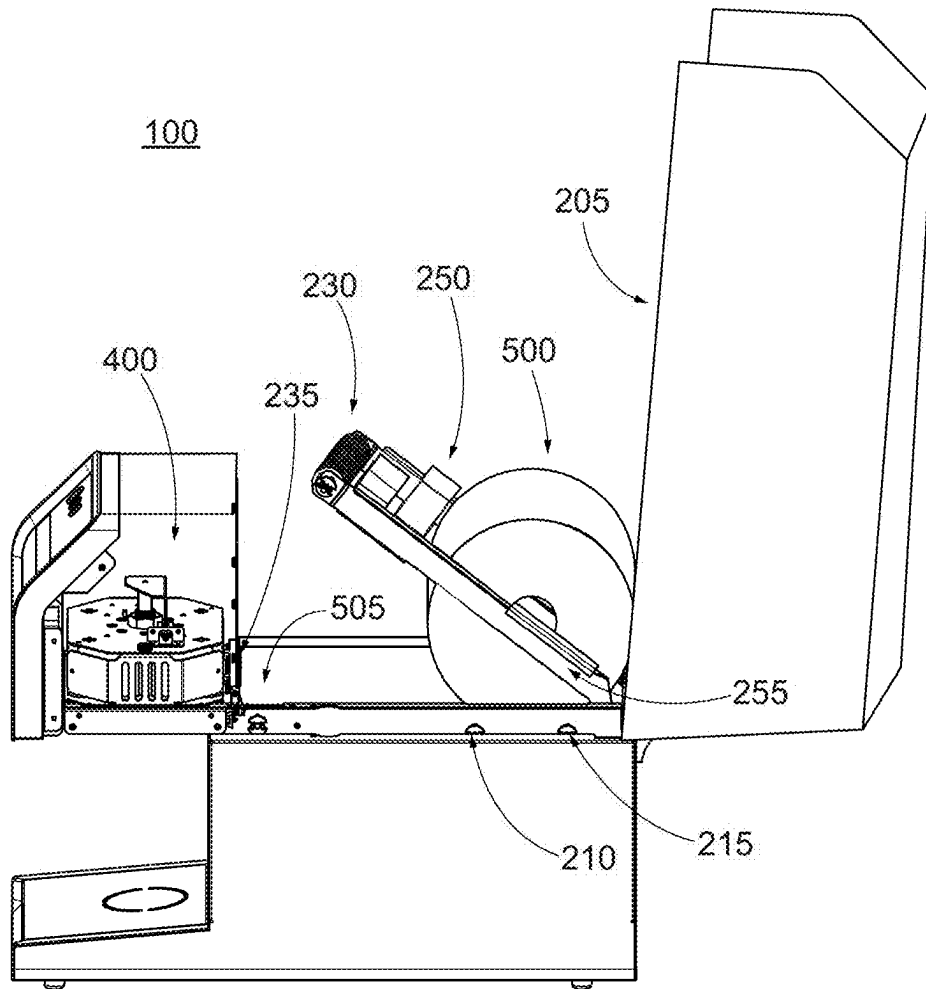


FIG. 4B

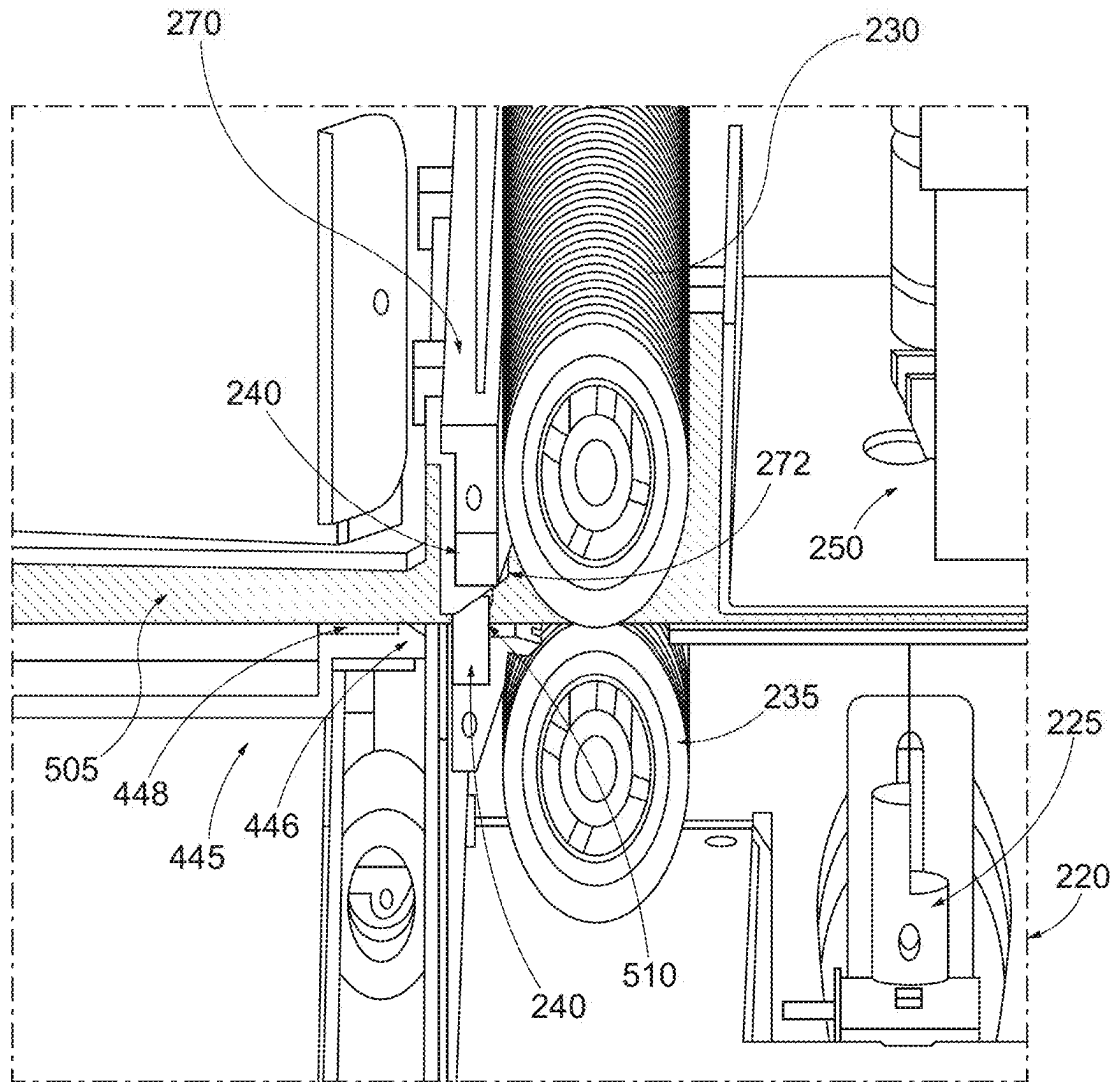


FIG. 5A

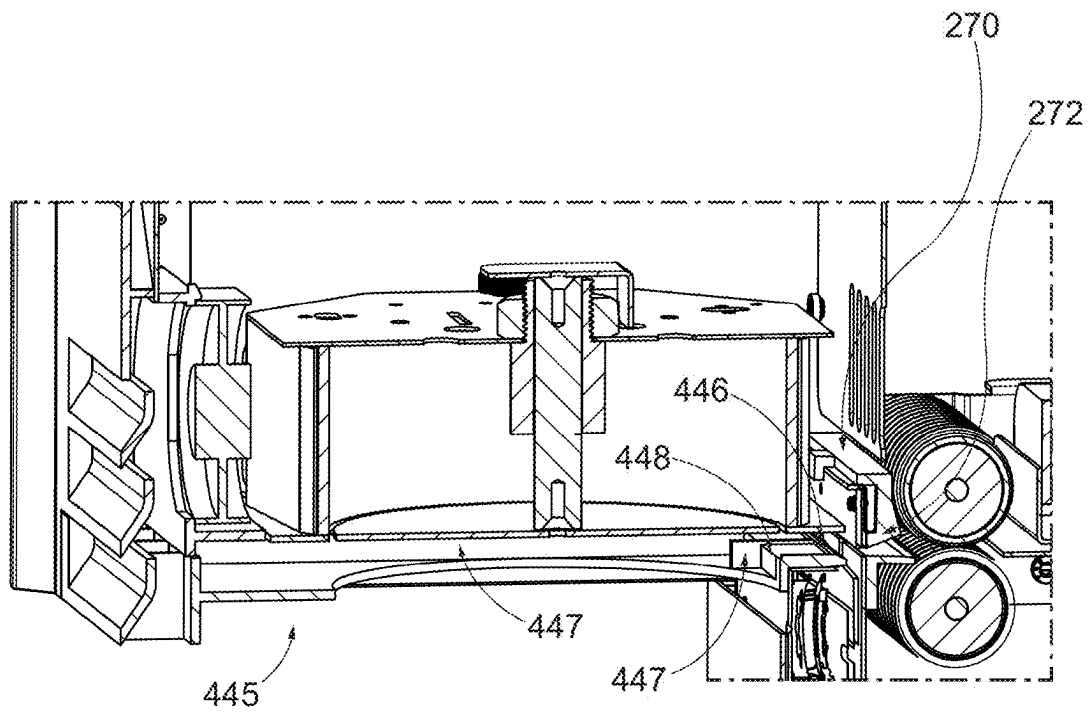


FIG. 5B

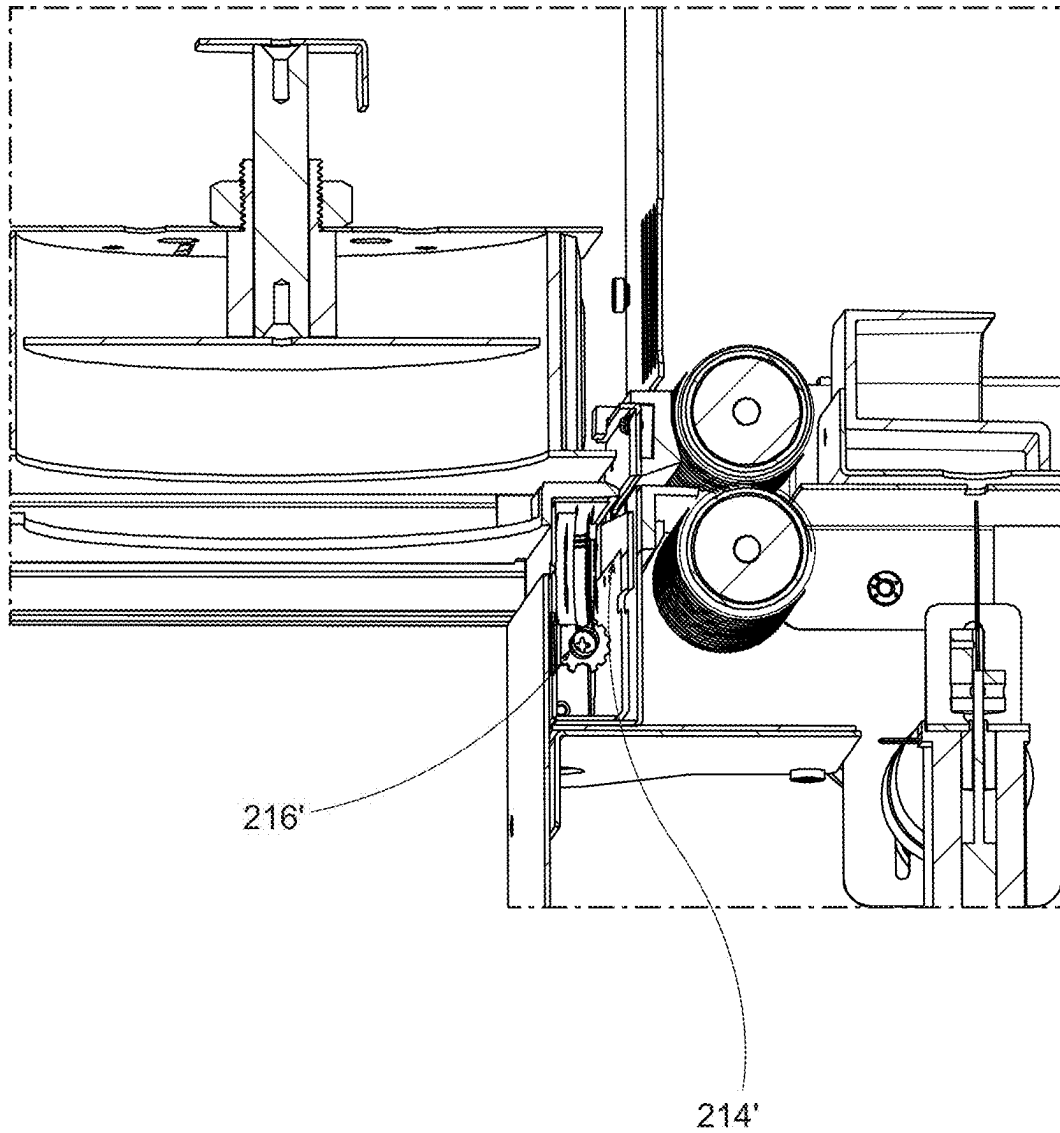


FIG. 5C

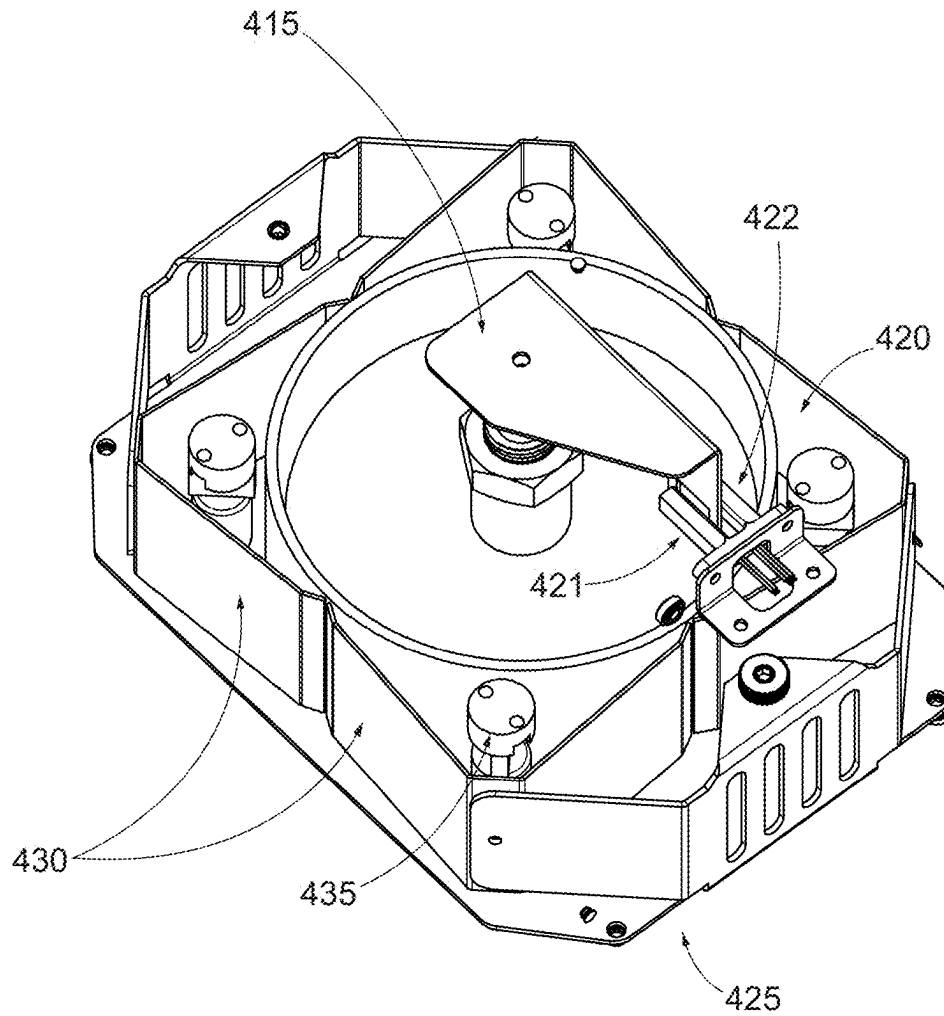


FIG. 6A

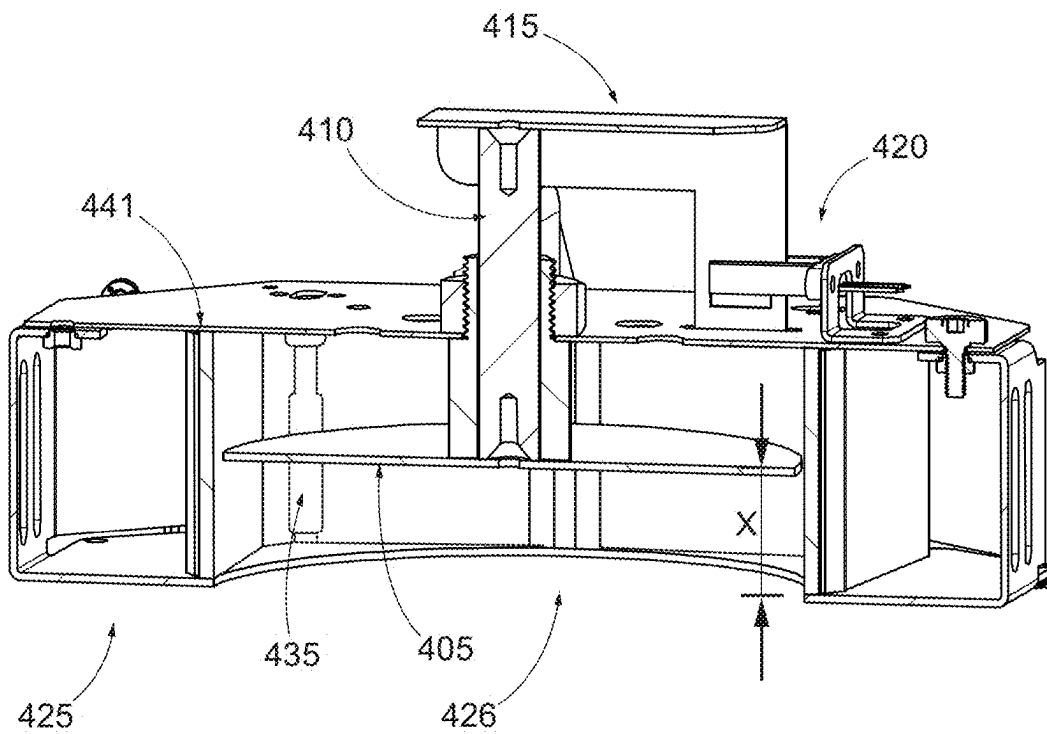


FIG. 6B

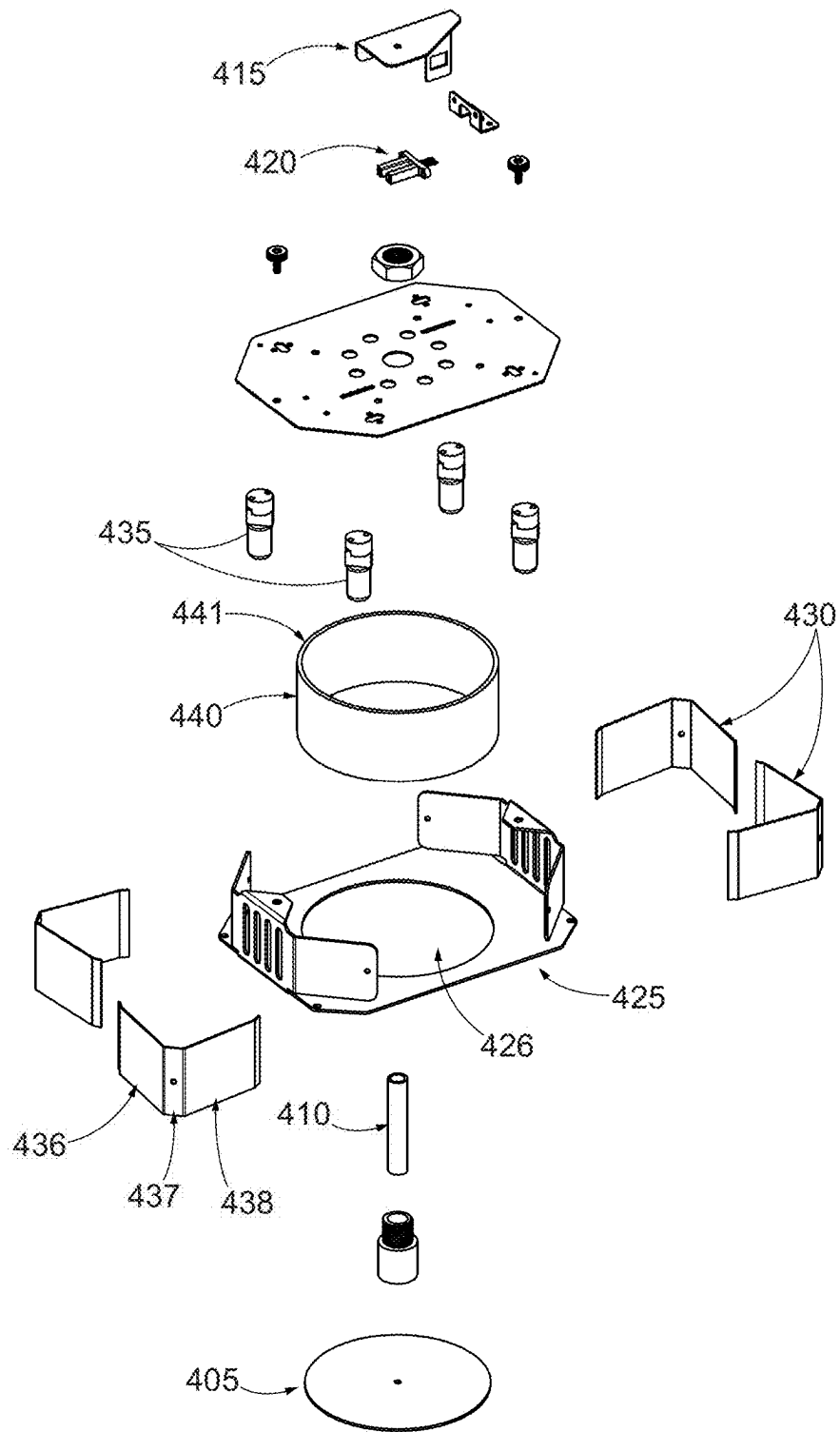


FIG. 6C

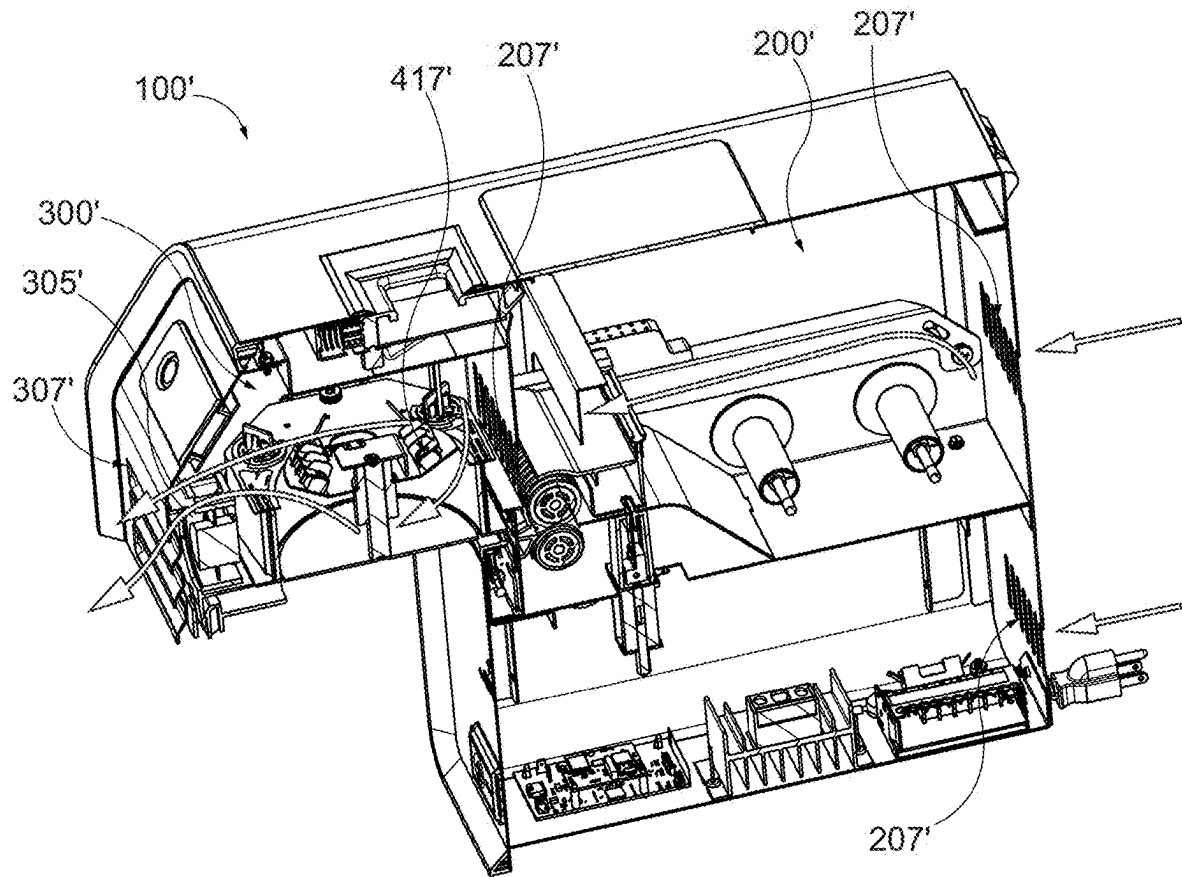


FIG. 7

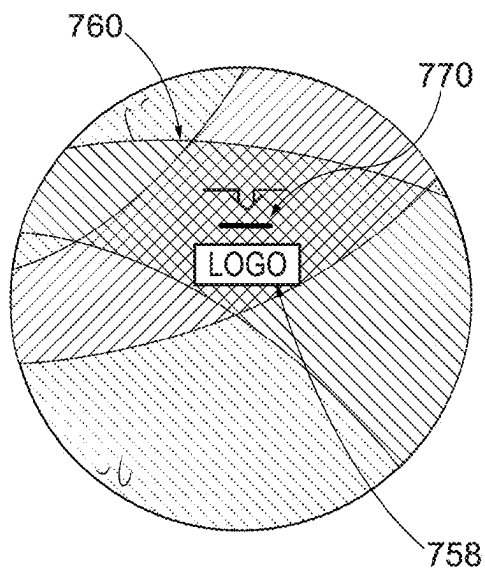


FIG. 8A

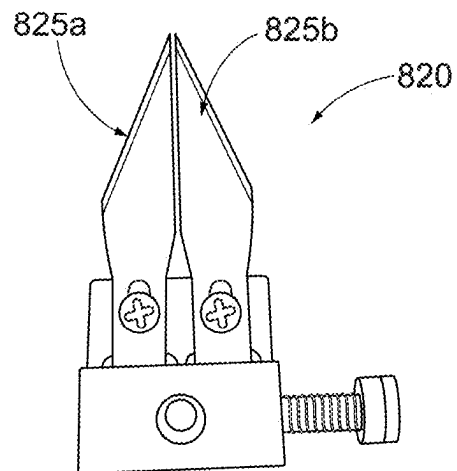


FIG. 8B

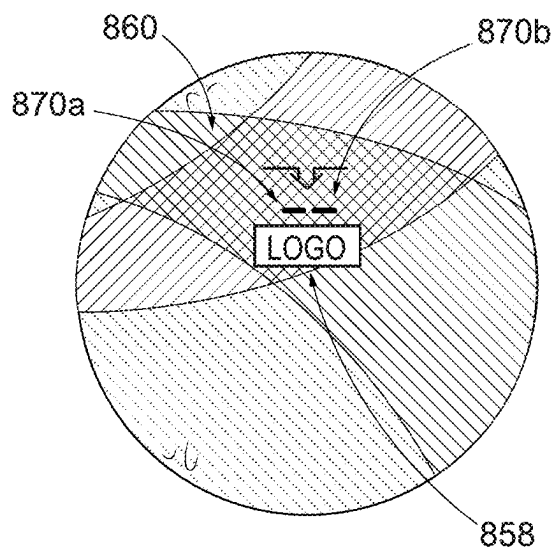


FIG. 8C

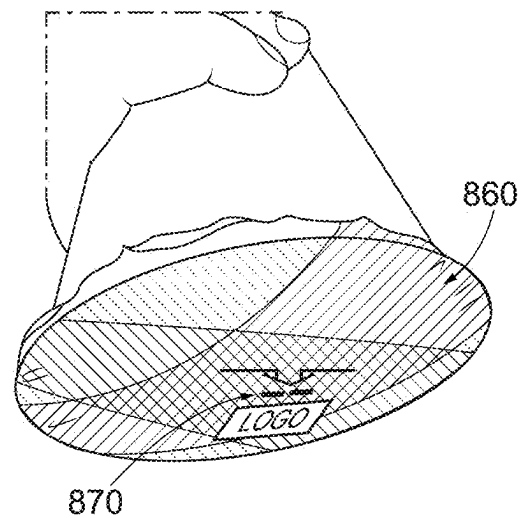


FIG. 8D

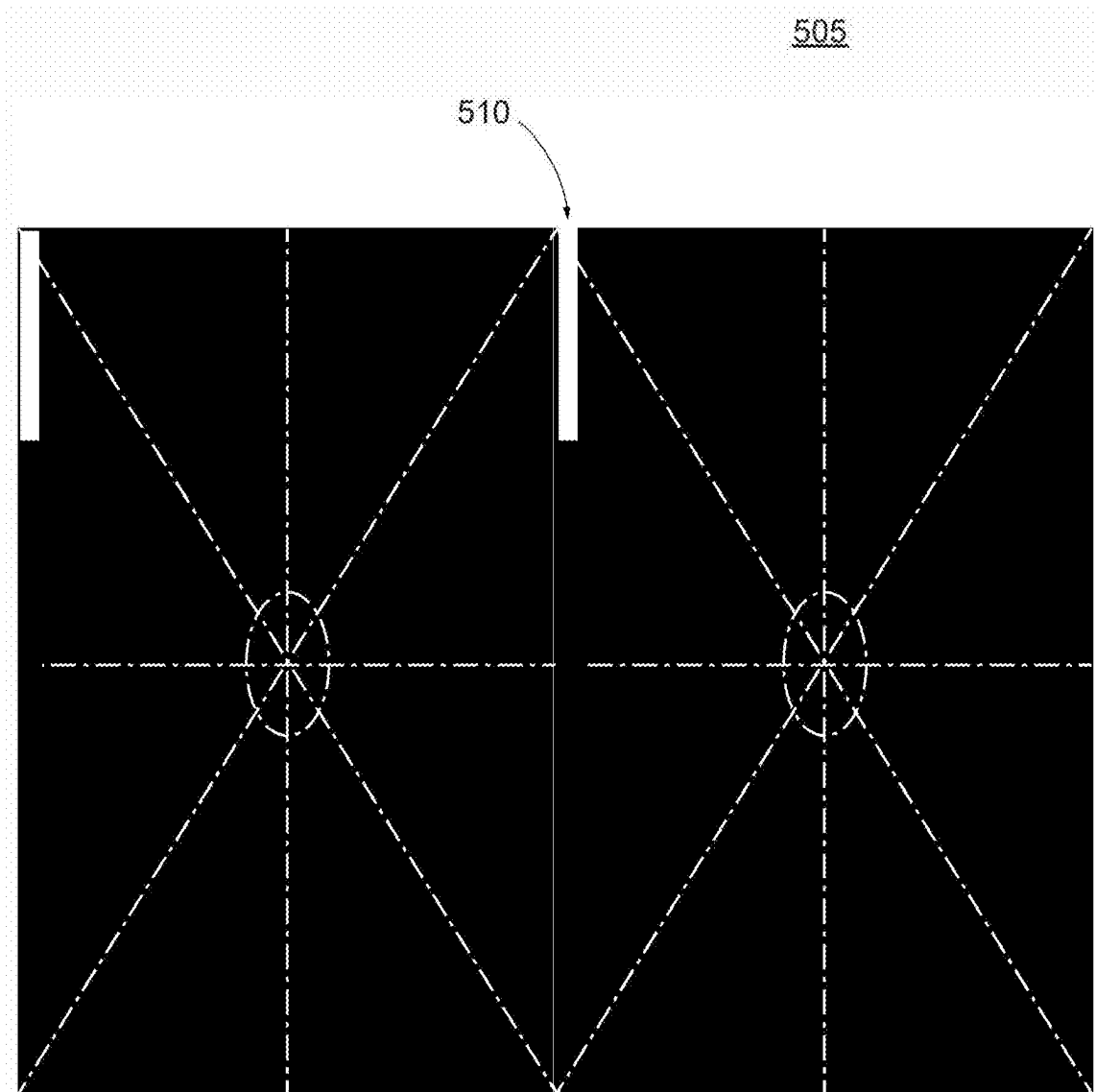


FIG. 9A

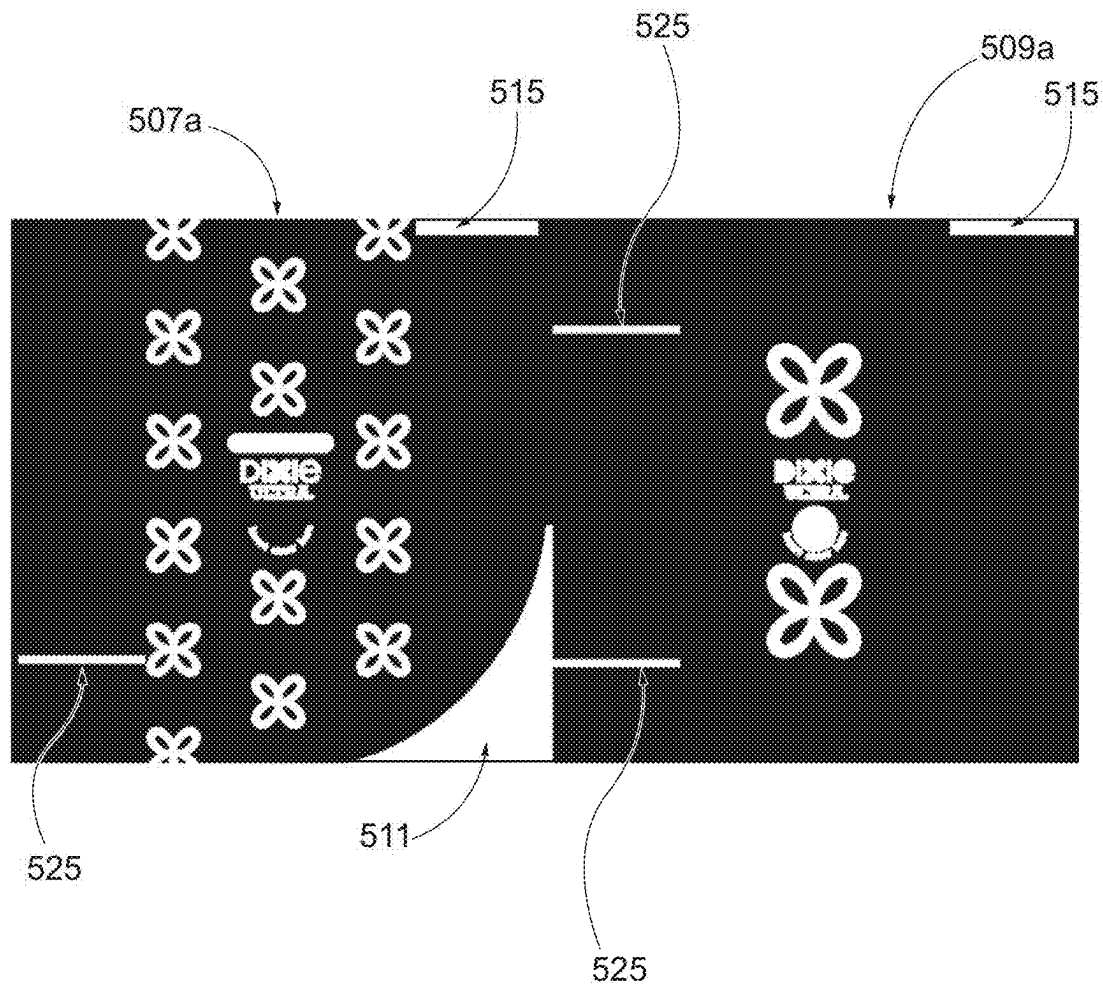


FIG. 9B

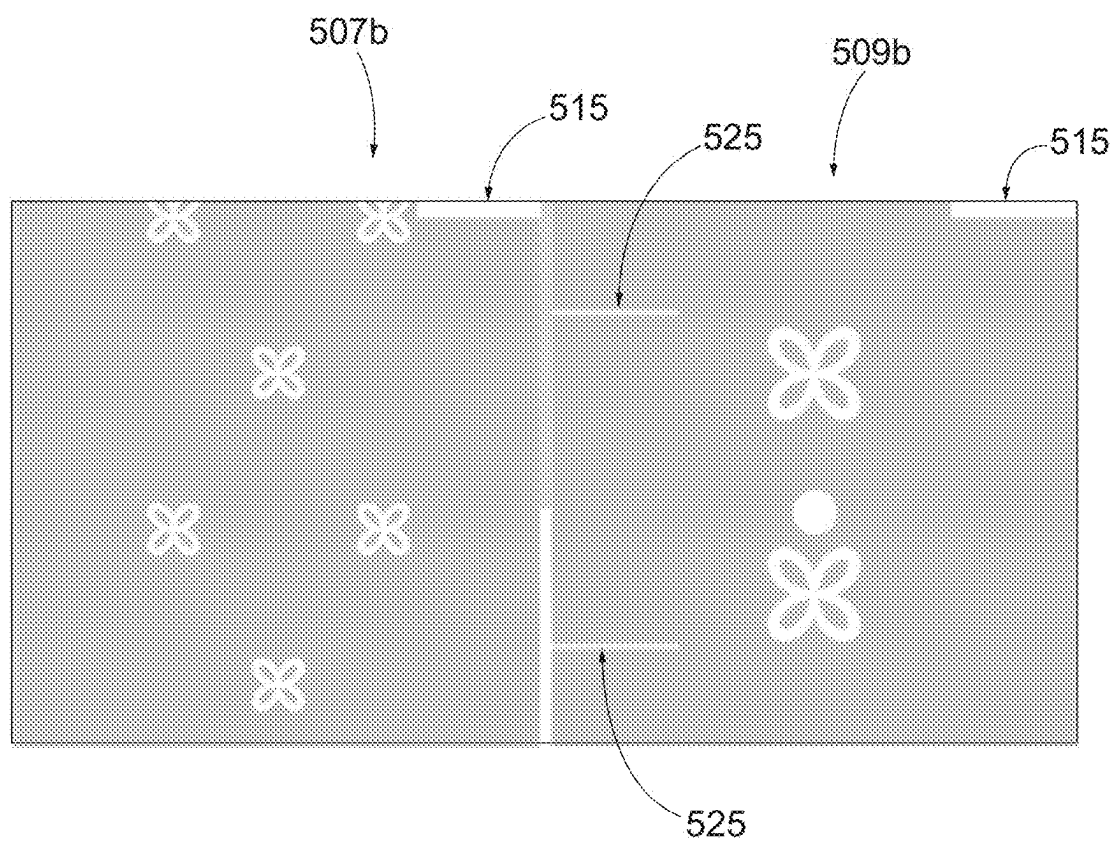


FIG. 9C

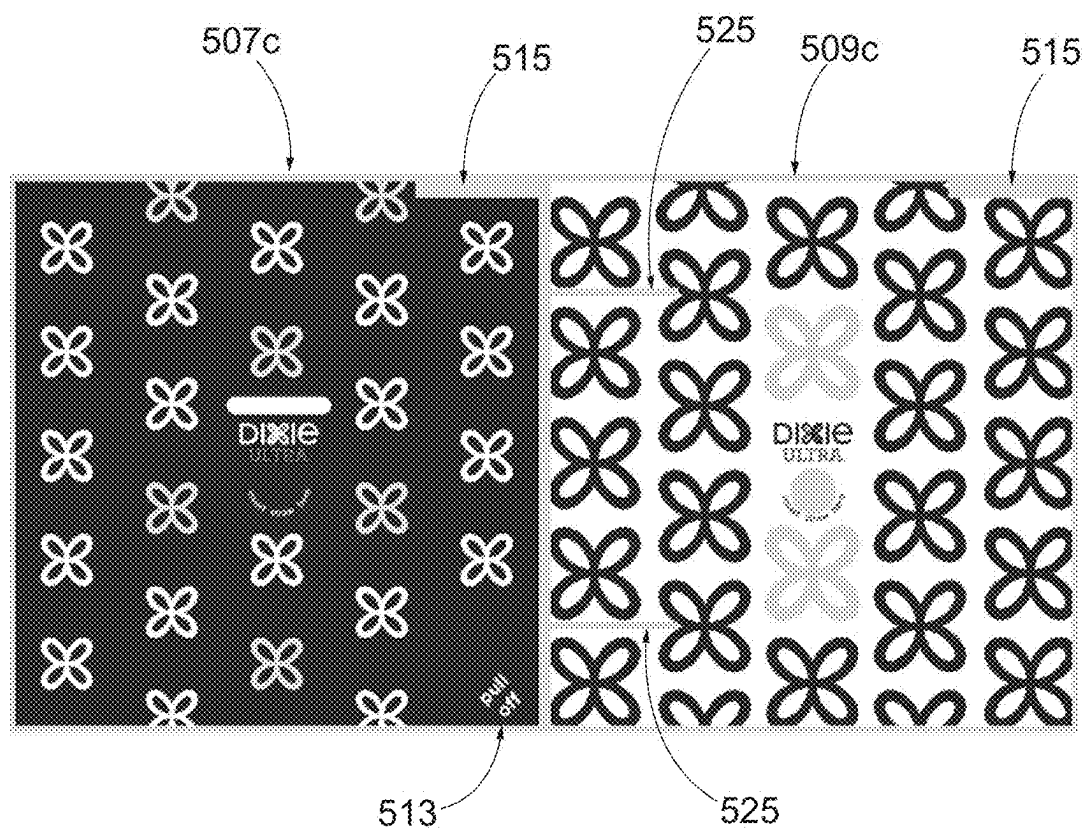
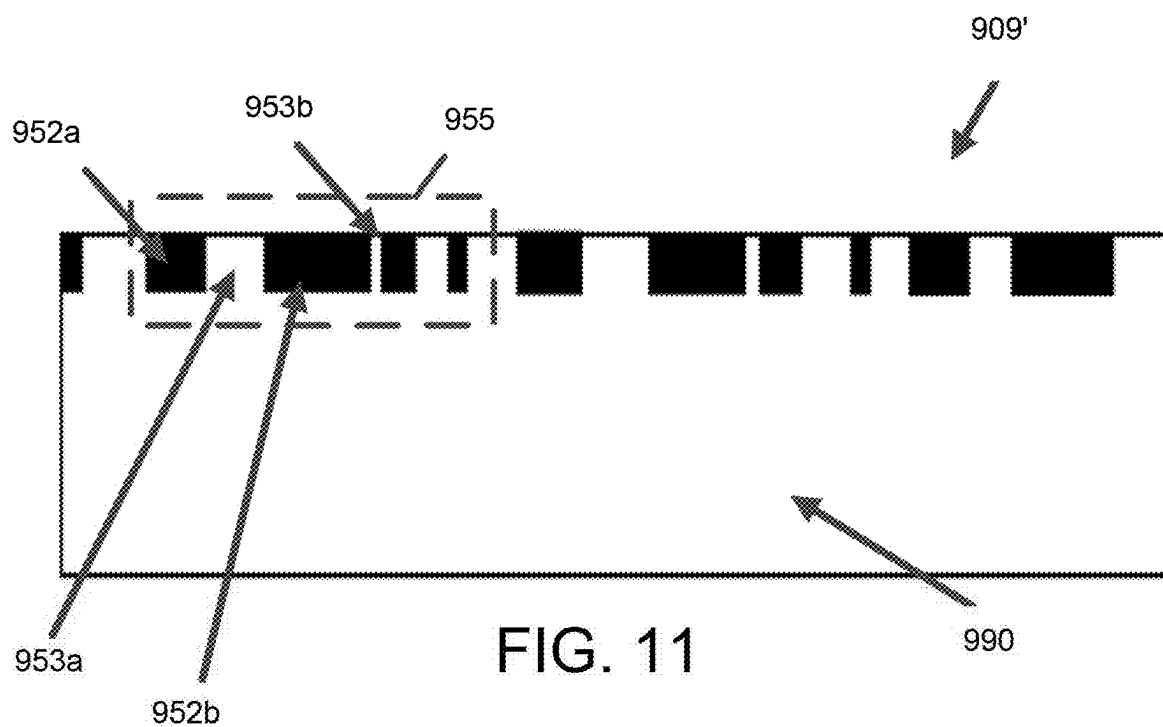
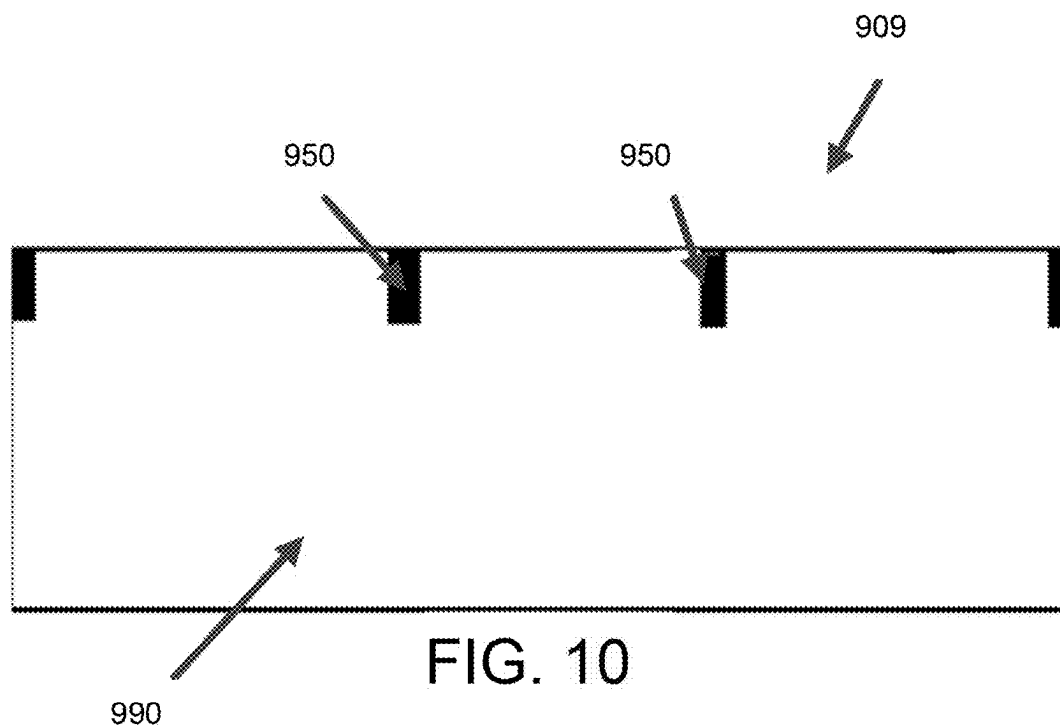


FIG. 9D



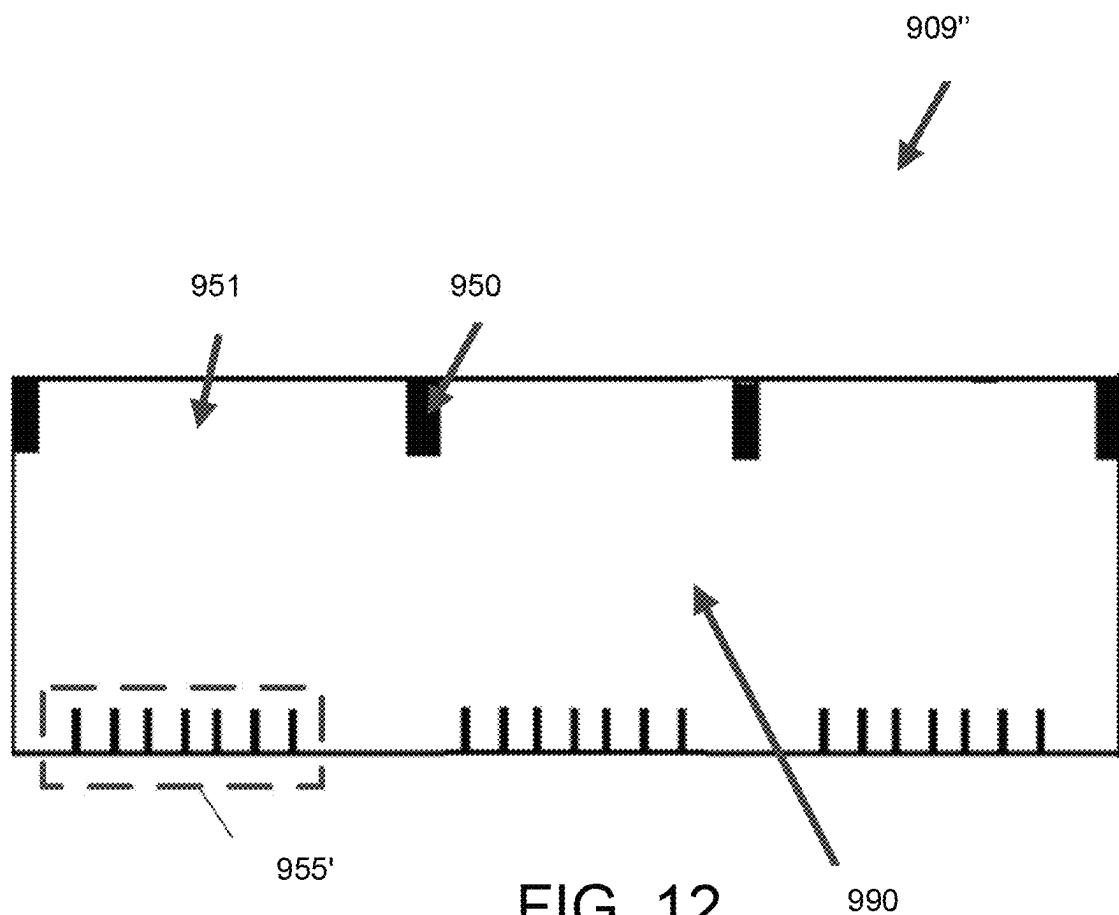


FIG. 12

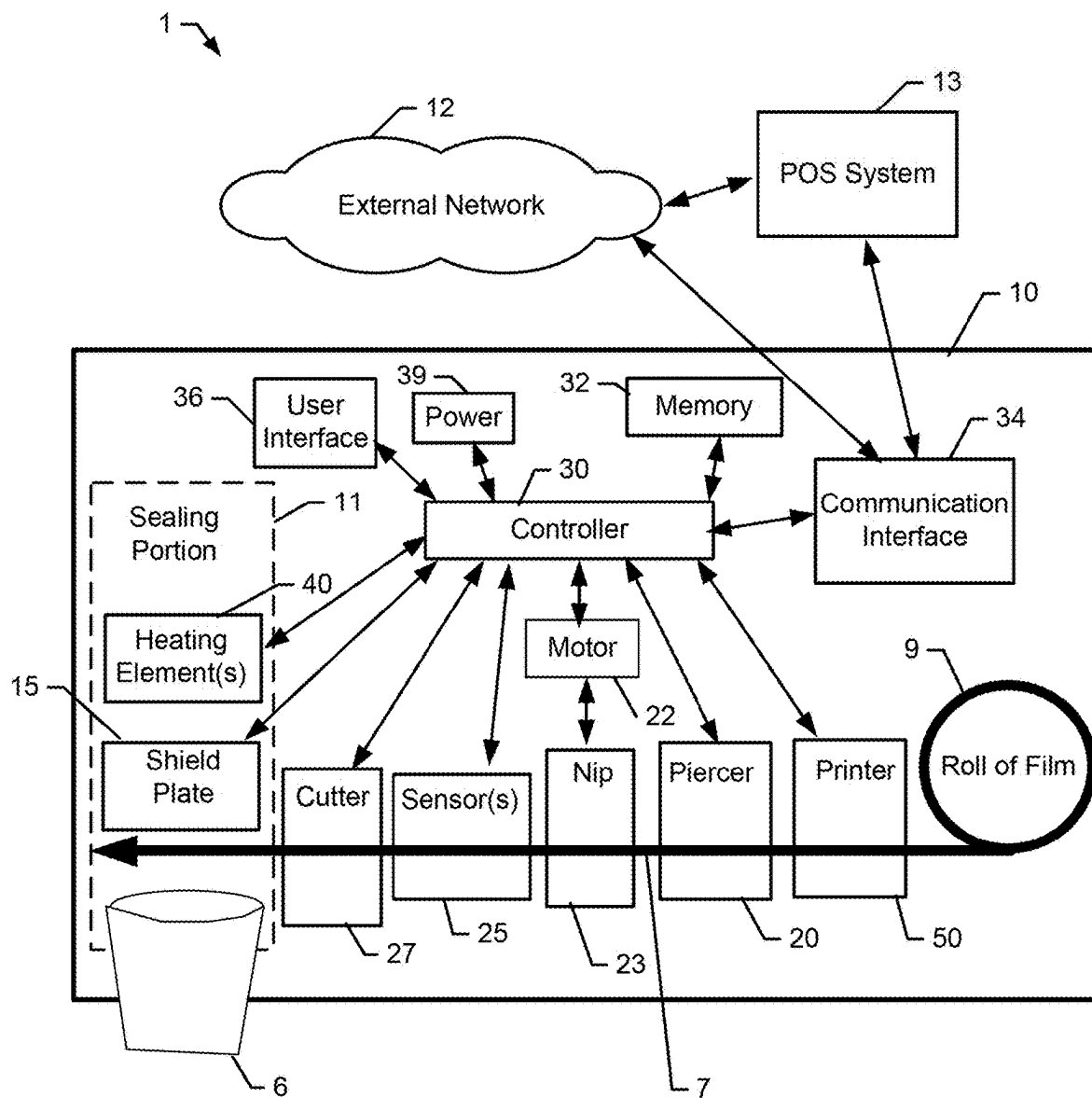


FIG. 13

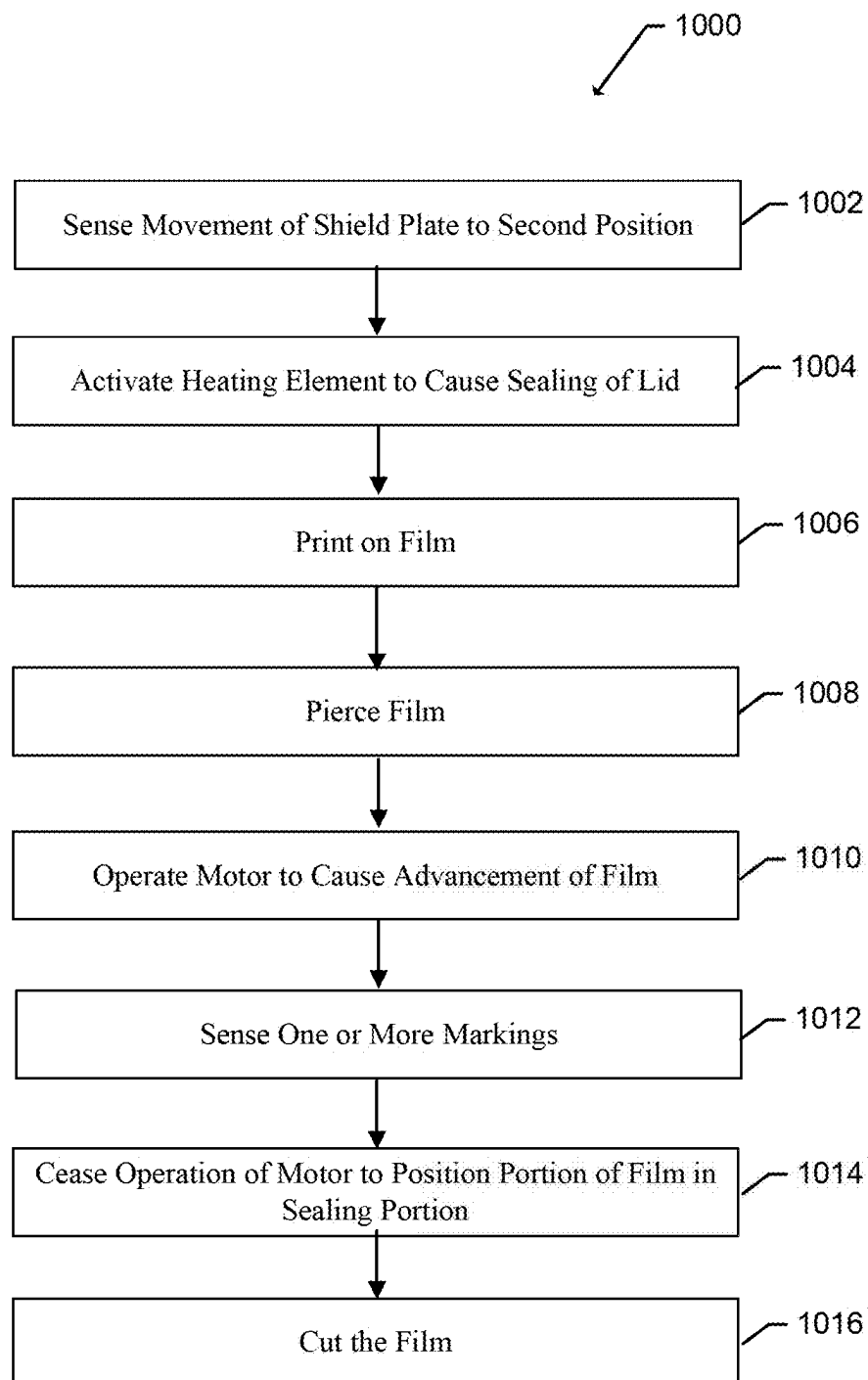


FIG. 14

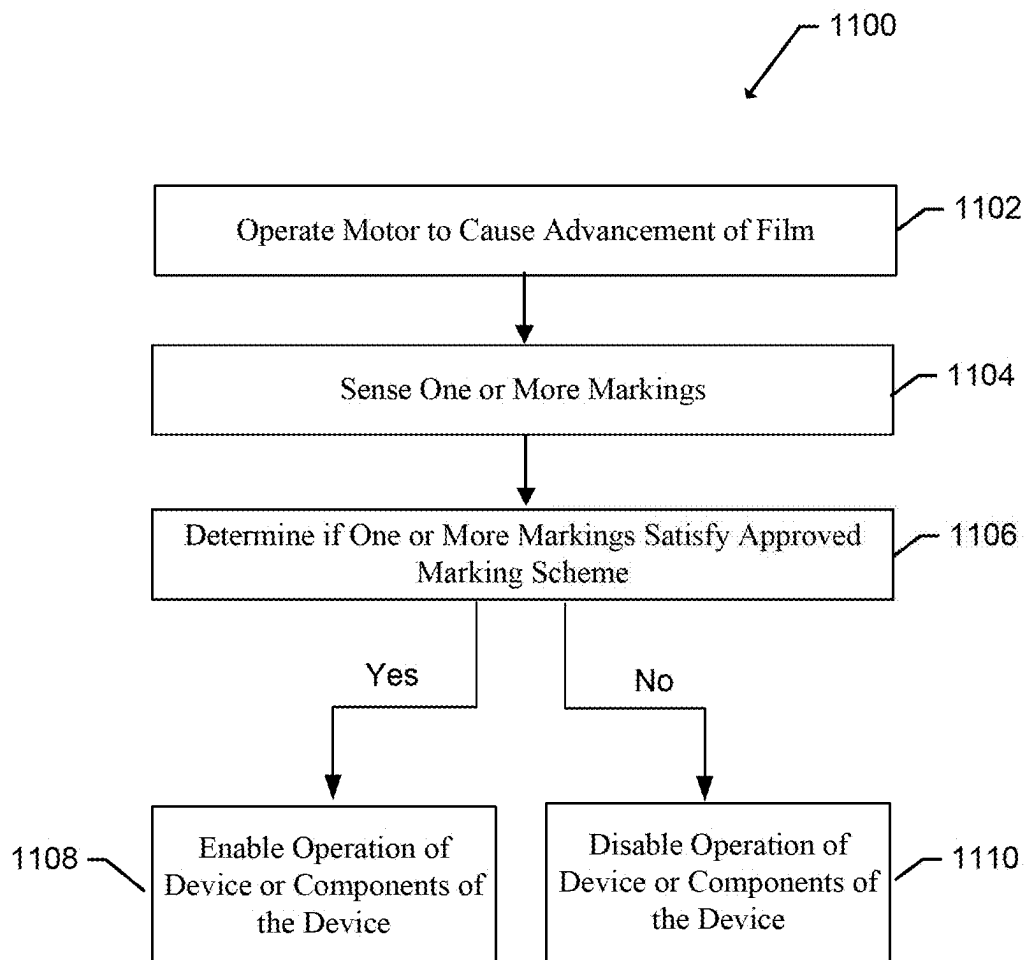


FIG. 15

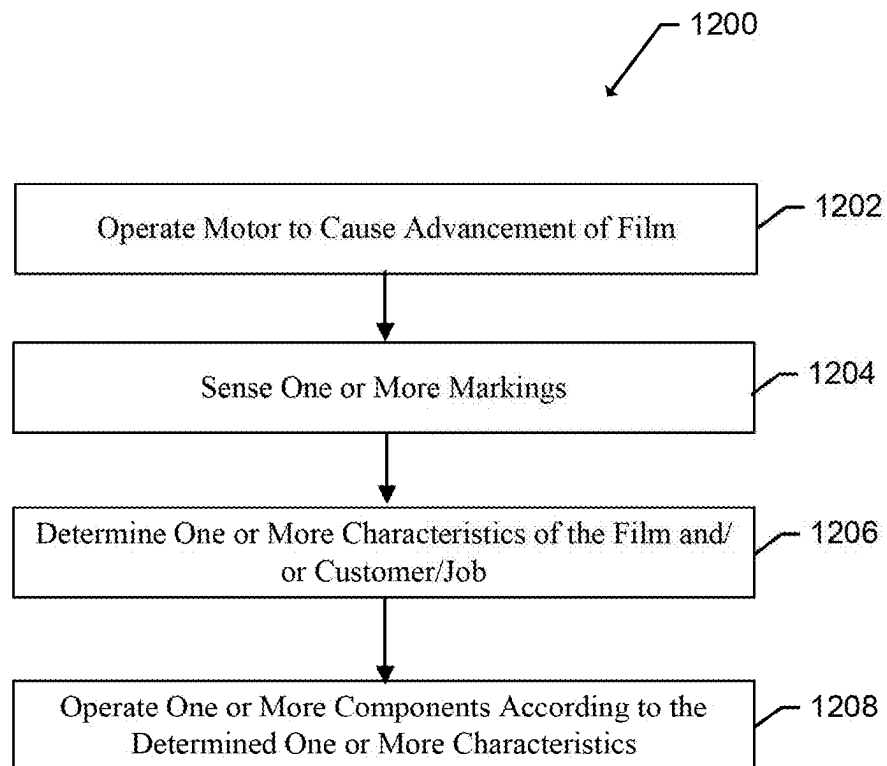


FIG. 16

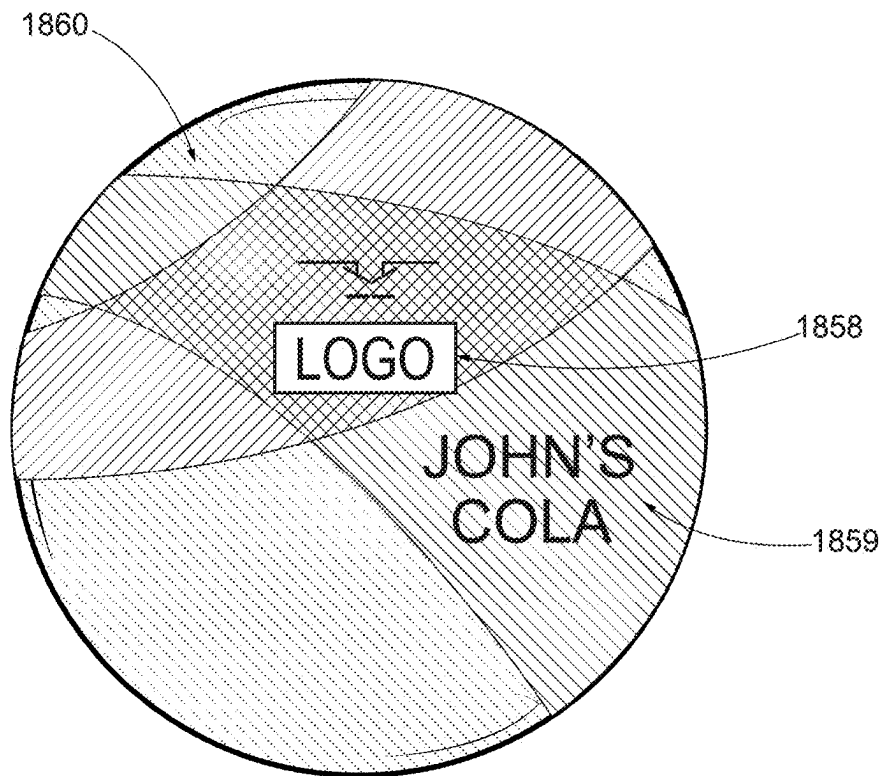


FIG. 17

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FILM SECURING APPARATUS AND METHOD**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 17/976,884 filed Oct. 31, 2022, which claims priority to U.S. application Ser. No. 16/701,453, filed Dec. 3, 2019, now U.S. Pat. No. 11,548,667, which is a continuation of U.S. application Ser. No. 16/700,164, filed Dec. 2, 2019, now U.S. Pat. No. 11,472,579, which claims priority to U.S. Provisional Application No. 62/848,735, entitled “Film Securing Apparatus and Method”, filed May 16, 2019, and to U.S. Provisional Application No. 62/775,227, entitled “Film Securing Apparatus and Method”, filed Dec. 4, 2018, the contents of each being incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

Example embodiments of the present invention generally relate to an apparatus and method of securing a film to a container to form a seal for the container.

BACKGROUND

A variety of systems to automatically secure lids and the like to containers are known. For example, a number of automatic sealers (e.g., lid sealing devices) are known in the beverage industry. However, many automatic sealers are poorly suited for user interaction without extensive training and review of laborious manuals. Further, such sealers are designed for industrial use and not well adaptable for commercial retail space, such as for individual use.

Another shortcoming associated with existing lid sealing devices relates to the limited ability of such systems to accommodate containers of alternate shapes, sizes, and materials. As such, known sealing devices are commonly tailored to operate with containers having only a single size and shape or a very limited deviation associated with the size and shape of the container. Furthermore, many sealers require containers of a particular material or the use of an adhesive to ensure proper securement of a lid to the container.

There is a need for a film securing (e.g., lid sealing) device that can be quickly and conveniently configured for use with containers having various sizes and shapes. Additionally, a need exists to improve the efficiency with which such securing devices can be manufactured, operated, and serviced as well as satisfy ever varying user demands associated with the production and presentation of the resultant sealed containers. There, thus, remains a continued need for an efficient and economic film securing apparatus and method. The presently disclosed subject matter satisfies these and other needs.

BRIEF SUMMARY

To achieve these and other advantages and in accordance with the purpose of the disclosed subject matter, as embodied herein and described, the disclosed subject matter includes systems, apparatuses, and methods related to example automatic sealing devices described herein.

Some example embodiments of the present invention provide a sealing apparatus and system that seals film from a supply of film (e.g., a roll of film) over the top of a

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container. Notably, the container may vary in size and shape, but may still be utilized with example sealing apparatuses. Further, the sealing apparatus may be automated and may simply require a user to position a top portion of a container (e.g., cup) into a sealing portion through an aperture. In response, the sealing apparatus may sense the presence of the cup and automatically seal a portion of film over the container—thereby providing an automatic seal. Some such example sealer devices provide a beneficial individual container sealer that can be quickly and easily employed by a user.

In some example embodiments, a portion of film may be cut and positioned within a loading zone in the sealing portion. A user may push the top of the container upwardly into the portion of the film. In response, heating element(s) may activate and cause the film to seal, such as through heat shrinkage to or around the top of the container to thereby form a sealed lid. In some cases, a shield plate may be positioned on the other side of the portion of the film, opposite the top of the container. A user may push the top of the container against the shield plate. A wall (e.g., a cylindrically shaped wall) may guide the film around the top of the container such that the application of heat from the heating element may form the sealed lid. In some cases, the shield plate may be movable within the sealing portion, with such movement triggering operation of the heating element(s). In this regard, a user may push the top of the container upwardly within the sealing portion to cause the shield plate to also move up—thereby initiating operation of the heating element(s) and sealing of the container.

Some example sealing apparatuses include various additional features that can be used in conjunction with sealing a lid to the container. For example, a motor may be configured to operate a drive roller to advance film from the roll of film to the loading zone.

In some embodiments, a printer may apply one or more messages or images to the film while being processed such that the messages/images appear on the lid of the container after the printed film is secured to the container. Such messages or images may be customized. In some embodiments, the sealing apparatus may be connected to a point-of-sale system and configured to provide a sealing lid for an ordered drink. In some such embodiments, information regarding the ordered drink may be printed on film that is used to form the sealing lid, such as an indication of what the drink is (e.g., a Cola) or who the drink belongs to (e.g., John’s Cola).

In some embodiments, a piercer may apply one or more slits (or perforations) to the film such that the resulting lid includes a ventilation capability and/or enables insertion of a straw. Notably, in some embodiments, the piercer may be a single piece with one or more protrusions (e.g., tips, blades, etc.). For example, the piercer may have two more spaced apart protrusions that provide two or more spaced apart slits in the lid for improved ventilation and weakness (e.g., for easy straw insertion) with still providing desired leakage protection. In some embodiments, the piercer may be actuated more than once in different locations on the film to form two or more spaced apart slits.

In some embodiments, one or more sensors may be utilized to determine a desired distance to advance the film into the loading zone. Further, one or more cutters may be used to cut the film to provide the portion of the film inside the loading zone for use with sealing the lid on the container.

In some embodiments, one or more sensors may be used to read one or more markings and verify that an approved film is being utilized with the sealing apparatus. In this

regard, there may be a desire to avoid usage of unauthorized rolls of film with the sealing apparatus, such as to avoid providing an unsatisfactory film or avoid potential maintenance issues. In some embodiments, the sealing apparatus may sense one or more markings on the film and confirm whether the film is approved for use with the scaling apparatus. If approved, various functions of the sealing apparatus may be enabled for use. If unapproved, various functions of the sealing apparatus may be disabled. Additionally, reports of the approved or unapproved film usage may be provided to a remote server for data generation and use (e.g., re-ordering, maintenance, etc.).

In some embodiments, the one or more markings may be read and used to provide information (e.g., characteristics) relevant to the installed film. For example, various characteristics of the film (e.g., thickness, pre-printed information, etc.) may be determined. Additionally or alternatively, various desired operational parameters of the scaling apparatus during use with the film (e.g., how long to activate the heating element(s), what to print on the film, whether or not to pierce the film, etc.) may be determined.

In some embodiments, depending on the desired configuration and the desired information to convey, the one or more markings may form marking schemes that utilize various marking characteristics (e.g., at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings). Such a marking scheme may be repeated along the film to enable confirmation of an approved film and/or the additional information/data relevant to the film. Various types of markings are contemplated, including for example, lines, rectangular shaped markings, barcodes, quick response (QR) codes, etc.

Some additional embodiments include apparatuses, systems, and methods including various example embodiments, such as described herein.

In an example embodiment, an apparatus to secure a film to a container is provided. The apparatus comprises a body portion to house film and a securing head assembly. The securing head assembly includes a housing defining an aperture to receive at least a top portion of a container. The securing head assembly further includes a wall disposed within the housing. The wall has a width dimension sized to receive at least a top portion of a container. The securing head assembly further includes a shield plate movable within the aperture and wall between a first position and a second position. The securing head assembly further includes at least one heating element disposed within the housing and positioned external to the wall, wherein the at least one heating element is activated to emanate energy when the shield plate is in the second position. The securing head assembly further includes a sensor assembly to sense movement of the shield plate and to activate the at least one heating element when the shield plate is in the second position. The securing head assembly further includes a loading zone to receive a predetermined dimensioned film from the body portion. The loading zone is positioned adjacent the shield plate in the first position. A top portion of a container is movable within the aperture and wall to move the shield plate to the second position and to secure a predetermined dimensioned film to a top of a container by energy from the at least one heating element.

In some embodiments, the width dimension of the wall is at least as large as a width dimension of the aperture.

In some embodiments, the wall is at least partially transparent or translucent to permit energy emanated from the at least one heating element through the wall.

In some embodiments, the apparatus further comprises at least one reflective device disposed within the housing and exterior to the wall. The at least one reflective device reflects at least a portion of energy emanated from the at least one heating element toward the wall. The at least one reflective device comprises at least one mirror having a first panel, a second panel, and a third panel. The first panel is disposed at a first angle relative to the second panel, and the third panel is disposed at a second angle relative to the second panel.

In some embodiments, the at least one heating element includes a plurality of heating elements. The plurality of heating elements are activated simultaneously when the shield plate is in the second position.

In some embodiments, the at least one heating element comprises a tungsten-halogen light bulb.

In some embodiments, the apparatus further comprises a sensor assembly having a signal emitter, a signal sensor, and a sensor flag therebetween. The sensor flag is coupled to the shield plate and the signal sensor detects a position of the sensor flag to activate the at least one heating element.

In some embodiments, the apparatus further comprises a securing portion, wherein the securing portion includes the securing head assembly, and a fan configured to cool the securing head assembly.

In some embodiments, the body portion comprises a drive nip having a drive roller and a pinch roller to progress film into the loading zone, wherein the pinch roller is adjustable in relation to the drive roller to insert film between the drive roller and the pinch roller. In some embodiments, the body portion further comprises a film support roller to support a length of film, wherein the drive roller rotates at a faster speed than the film support roller. In some embodiments, the body portion comprises a film sensor assembly having a film signal emitter and a film signal sensor to detect markers spaced at predetermined longitudinal distances along a length of film. In some embodiments, the housing further includes a guide support assembly having a ramp and a guide truss, the ramp being capable of receiving a film from the body portion and the guide truss capable of guiding a film to the loading zone. In some embodiments, the ramp has a surface extending from a first end to a second end, the surface oriented at an angle of inclination ranging up to 75 degrees, wherein the second end is coupled to a guide surface configured to receive a film from the surface of the ramp.

In some embodiments, the apparatus further comprises a guide assembly including an entry structure, wherein the entry structure comprises a funnel to receive a leading end of a film therethrough and to channel a leading end of a film into the loading zone.

In some embodiments, the body portion further comprises a piercer having an actuatable tip to pierce a film.

In some embodiments, the apparatus further comprises a printer to print on a film. In some embodiments, the apparatus further comprises a computer to receive at least one command, wherein the computer sends a signal to the printer to print a message associated with the at least one command when the at least one command is received.

In some embodiments, the wall is coupled to the shield plate and movable with the shield plate between the first and second positions.

In some embodiments, the wall is coupled to the housing, wherein the shield plate is movable with respect to the wall between the first position and the second position.

In some embodiments, the body portion comprises a film sensor assembly configured to detect one or more marking

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patterns along the film. The apparatus further includes a controller configured to: determine, based on sensor data from the film sensor assembly, if a detected marking pattern matches an approved marking pattern; and cause at least one of: in an instance in which the detected marking pattern matches an approved marking pattern, enabling operation of the apparatus or components of the apparatus; or in an instance in which the detected marking pattern does not match an approved marking pattern, disabling of operation of the apparatus or components of the apparatus.

In another example embodiments, a method of securing a film to a container is provided. The method comprises providing an apparatus having a body portion to house film and a securing head assembly. The securing head assembly includes a housing defining an aperture to receive at least a top portion of a container, a wall disposed within the housing, the wall having a width dimension sized to receive at least a top portion of a container, a shield plate movable within the aperture and wall between a first position and a second position, at least one heating element disposed within the housing and positioned external to the wall, wherein the at least one heating element is activated to emanate energy when the shield plate is in the second position, a sensor assembly to sense movement of the shield plate and to activate the at least one heating element when the shield plate is in the second position, and a loading zone to receive a predetermined dimensioned film from the body portion, the loading zone positioned adjacent the shield plate in the first position. The method further includes moving the top portion of the container to move the shield plate to the second position and to secure a predetermined dimensioned film to a top of the container by energy from the at least heating element.

In some embodiments, the at least one heating element includes a plurality of heating elements, and the method further comprises activating the plurality of heating elements simultaneously when the shield plate is in the second position.

In some embodiments, the method further comprises printing, by a printer operatively coupled to the body portion, on a length of film; and piercing, by a piercer having an actuatable tip, the film.

In some embodiments, the method further comprises progressing, by a nip having a drive roller and a pinch roller, the film into the loading zone.

In some embodiments, the method further comprises cutting, by a film cutter operatively coupled to the apparatus between the nip and the securing head assembly, a length of film from a film roll to form the predetermined dimensioned film.

In another example embodiment, an apparatus to secure a film to a container is provided. The apparatus comprises a securing head assembly including a housing defining an aperture to receive at least a top portion of a container. The housing includes a shield plate and at least one heating element disposed within the housing. The at least one heating element is configured to emanate energy when a container is in a predetermined position. The securing head assembly further includes a protective structure to surround the at least one heating element. The protective structure permits energy from the at least one heating element to disseminate therethrough. The securing head assembly further includes a sensor assembly disposed within the housing configured to activate the at least one heating element when the sensor assembly is tripped and the container is in the predetermined position. The securing head assembly further includes a loading zone to receive a predetermined dimensioned

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film. A top portion of a container is movable within the aperture to sandwich a film positioned in the loading zone between the shield plate and the top portion of a container. The at least one heating element emanates energy when the sensor assembly is tripped to secure a predetermined dimensioned film to a top of a container.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house film and a sealing portion configured to receive at least a top portion of the container. The sealing portion comprises an aperture sized to receive the top portion of the container therethrough. The sealing portion comprises a shield plate fixedly positioned within the sealing portion and a sensor configured to sense the presence of a container within the sealing portion. The sealing portion further includes at least one heating element positioned proximate the sealing portion and configured to activate to emanate energy in an instance in which the container is sensed within the sealing portion. The sealing portion further includes a loading zone configured to receive a portion of the film and cause positioning of the portion of the film so that the portion of the film is adjacent to the shield plate.

In some embodiments, the apparatus further comprises a fan configured to cool the sealing portion.

In some embodiments, the body portion comprises a film sensor assembly configured to detect one or more markings spaced along the film and a cutter configured to cut the film. The apparatus further includes a controller configured to: determine a desired length for the film; determine, based on sensor data from the film sensor assembly, an instance in which the desired length has been reached; and cause, in response to determining that the desired length has been reached, the cutter to cut the film to form a portion of the film with the desired length.

In some embodiments, the body portion comprises a film sensor assembly configured to detect one or more marking patterns along the film. The apparatus further includes a controller configured to: determine, based on sensor data from the film sensor assembly, if a detected marking pattern matches an approved marking pattern; and cause at least one of: in an instance in which the detected marking pattern matches an approved marking pattern, enabling operation of the apparatus or components of the apparatus; or in an instance in which the detected marking pattern does not match an approved marking pattern, disabling of operation of the apparatus or components of the apparatus.

In another example embodiment an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The sealing portion comprises an aperture sized to receive the top portion of the container therethrough. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus further includes a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus further includes a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus further includes at least one heating element configured to activate to emanate energy. The apparatus further includes a guide support assembly having a ramp and at least one guide truss. The ramp is configured to receive a portion of the film from the body portion. The ramp and the at least one guide truss are configured to guide the portion of the film to a loading zone within the sealing

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portion. The apparatus further includes a controller configured to: cause the motor to operate to cause the drive roller to advance the portion of the film into the sealing portion; and cause the at least one heating element to activate to emanate energy to cause the portion of the film within the scaling portion to seal the top portion of the container to form a lid for the container.

In some embodiments, the body portion further comprises a film support roller to support the supply of film.

In some embodiments, the ramp has a surface extending from a first end to a second end, the surface oriented at an angle of inclination ranging up to 75 degrees. The second end is coupled to a guide surface configured to receive the portion of the film from the surface of the ramp.

In some embodiments, the guide support assembly comprises an entry structure, wherein the entry structure comprises a funnel structured to receive a leading end of the film therethrough and to channel the leading end of the portion of the film into the loading zone.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The sealing portion comprises an aperture sized to receive the top portion of the container therethrough. The body portion defines a film path leading from the supply of film to the scaling portion. The apparatus comprises a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus further comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus further comprises at least one heating element configured to activate to emanate energy; a printer configured to print on the film; and a controller. The controller is configured to: cause the printer to print one or more messages or images on the film; cause the motor to operate to cause the drive roller to advance a portion of the film into the sealing portion; and cause the at least one heating element to activate to emanate energy to cause the portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container with the one or more messages or images printed thereon.

In some embodiments, the controller is configured to determine the one or more messages or images to print on the film.

In some embodiments, the apparatus further comprises a communication interface configured to communicate with a remote device. The controller is configured to receive data from the remote device and operate one or more components of the apparatus according to the received data.

In some embodiments, the controller is configured to determine the one or more messages or images to print on the film based on the received data. In some embodiments, the remote device is a point-of-sale system, and the determined one or more messages or images relate to an order received by the point-of-sale system. In some embodiments, the determined one or more messages or images comprise a description of contents of the container. In some embodiments, the determined one or more messages or images comprise a personalized or customized message or image related to a customer of the order.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The sealing portion

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comprises an aperture sized to receive the top portion of the container therethrough. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus comprises a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus comprises a piercer configured to pierce the film, wherein the piercer is configured to form at least two spaced apart slits in the film. The apparatus comprises at least one heating element configured to activate to emanate energy and a controller. The controller is configured to: cause the piercer to pierce the film to form at least two spaced apart slits in a portion of the film; cause the motor to operate to cause the drive roller to advance the portion of the film into the sealing portion; and cause the at least one heating element to activate to emanate energy to cause the portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container with the at least two spaced apart slits.

In some embodiments, the piercer comprises at least two spaced apart blades that are configured to form the at least two spaced apart slits in the film when the piercer is activated.

In some embodiments, the piercer is configured to form the at least two spaced apart slits in the portion of the film by activating two or more times at different locations on the portion of the film.

In some embodiments, the piercer is movable with respect to the film.

In some embodiments, the piercer is configured to activate as the film advances so as to form the at least two spaced apart slits in the film in a machine direction of advancement of the film.

In some embodiments, the at least two spaced apart slits formed on the lid are designed to enable venting and discourage leaking when the container is tilted such as due to the surface tension between contents of the container and the portion of the lid between the spaced apart slits.

In some embodiments, the at least two spaced apart slits formed on the lid form points of weakness in the lid that are designed to enable insertion of a straw therethrough.

In some embodiments, adjacent slits of the at least two spaced apart slits are separated by a portion of the lid therebetween.

In some embodiments, the at least two spaced apart slits define an overall length along the lid that, in comparison to a continuous length slit of similar overall length, provides equivalent ventilation for the container, weakness within the lid to enable insertion of a straw, and increased leakage prevention due to increased surface tension of liquid on an inside portion of the lid.

In some embodiments, the controller is configured to operate the motor to cause a different rotation speed of the drive roller in comparison to a film support roller for the supply of film so as to form tension in the film to aid with piercing thereof.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus comprises a film sensor positioned along the film path and configured to sense one or more markings on the film. The apparatus comprises a nip defined

by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus comprises at least one heating element configured to activate to emanate energy to cause a portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container. The apparatus comprises a controller configured to: determine, based on sensor data from the film sensor, if a detected one or more markings on the film satisfies an approved marking scheme; and affect, based on whether the detected one or more markings on the film satisfy the approved marking scheme, operation of one or more components of the apparatus.

In some embodiments, the controller is configured to cause at least one of: in an instance in which the detected one or more markings satisfies the approved marking scheme, enabling operation of the apparatus or at least one of the motor or the at least one heating element of the apparatus; or in an instance in which the detected one or more markings does not satisfy the approved marking scheme, disabling operation of the apparatus or at least one of the motor or the at least one heating element of the apparatus.

In some embodiments, the controller is configured to cause at least one of: in an instance in which the detected one or more markings satisfies the approved marking scheme, enabling operation of the one or more components of the apparatus; or in an instance in which the detected one or more markings does not satisfy the approved marking scheme, disabling operation of the one or more components of the apparatus.

In some embodiments, the controller is configured to affect operation of the one or more components of the apparatus by at least one of decreasing a speed of operation of the motor, decreasing a speed or temperature of operation of the at least one heating element, disabling a printer of the apparatus, causing the printer to print one or more messages or images in an off-center position on the film, causing the printer to print one or more messages indicating that an unapproved film is being utilized, increasing a delay time between sealing operations performed by the apparatus, or disabling a piercer of the apparatus.

In some embodiments, the controller is further configured to send a signal to a remote server indicating that the detected one or more markings does not satisfy the approved marking scheme.

In some embodiments, the controller is configured to determine a film marking scheme for the film based on the detected one or more markings. In some embodiments, the determined film marking scheme corresponds to a marking pattern comprised of a plurality of markings and determined spacing between each adjacent marking within the plurality of markings. In some embodiments, the determined film marking scheme is comprised of at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings. In some embodiments, the determined film marking scheme is repeated along the length of the supply of film such that the supply of film comprises a plurality of repeated film marking schemes.

In some embodiments, the apparatus further comprises a second film sensor configured to sense one or more markings along the film. The film sensor is positioned proximate a first edge of the film and the second film sensor is positioned proximate a second, opposite edge of the film. The controller is configured to control operation of the

motor to cease advancement of the film into the sealing portion based on sensor data from the second film sensor.

In some embodiments, the controller is further configured to control operation of the motor to cease advancement of the film into the sealing portion based on sensor data from the film sensor.

In some embodiments, the controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; determine a desired operation of one or more components of the apparatus based on the detected one or more characteristics; and cause operation of the one or more components of the apparatus based on the determined desired operation.

In some embodiments, the controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; and cause the at least one heating element to operate according to at least one of a specific amount of time or a specific heat based on the determined one or more characteristics.

In some embodiments, the controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; and cause the motor to operate according to at least one of a specific amount of time or according to a specific number of detected markings based on the determined one or more characteristics.

In some embodiments, the apparatus further comprises a piercer having at least one actuatable tip to pierce the film. The controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; and cause the piercer to operate based on the determined one or more characteristics.

In some embodiments, the apparatus further comprises a printer configured to print one or more messages or images on the film, wherein the controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; and cause the printer to operate based on the determined one or more characteristics.

In some embodiments, the controller is configured to determine the one or more messages or images to print on the film based on the determined one or more characteristics.

In some embodiments, the controller is further configured to: determine one or more characteristics of the one or more markings, wherein the one or more characteristics comprise at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings; and determine one or more characteristics of the film, wherein the one or more characteristics of the film comprise at least one of the thickness of the film, the associated customer, a time of operation of the heating element, a subset of printing

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options to present to a user for selection, or the amount of film remaining on the supply of film.

In some embodiments, the sealing portion comprises an aperture sized to receive the top portion of the container therethrough.

In some embodiments, the one or more markings are comprised of at least one of a Quick Response Code, a barcode, or a logo.

In some embodiments, the one or more markings are clear with respect to a remainder of the film.

In another example embodiment, a method for controlling operation of an apparatus configured to secure a film as a lid to a container is provided. The method comprises providing the apparatus, wherein the apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus comprises a film sensor positioned along the film path and configured to sense one or more markings on the film. The apparatus further comprises a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus further comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus further comprises at least one heating element configured to activate to emanate energy to cause a portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container. The apparatus further comprises a controller. The method further includes determining, based on sensor data from the film sensor, if a detected one or more markings on the film satisfies an approved marking scheme; and affecting, based on whether the detected one or more markings on the film satisfy the approved marking scheme, operation of one or more components of the apparatus.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus comprises a film sensor positioned along the film path and configured to sense one or more markings on the film. The apparatus comprises a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus comprises at least one heating element configured to activate to emanate energy to cause a portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container. The apparatus comprises a controller configured to: determine, based on sensor data from the film sensor, if a detected one or more markings on the film satisfies an approved marking scheme; and cause, in an instance in which the detected one or more markings satisfies the approved marking scheme, operation of one or more components of the apparatus.

In some embodiments, the controller is configured to cause, in an instance in which the detected one or more markings satisfies the approved marking scheme, operation of the motor to cause the drive roller to advance a portion of the film to the sealing portion.

In another example embodiment, an apparatus configured to secure a film as a lid to a container is provided. The

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apparatus comprises a body portion configured to house a supply of film and a sealing portion configured to receive at least a top portion of the container. The body portion defines a film path leading from the supply of film to the sealing portion. The apparatus comprises a film sensor positioned along the film path and configured to sense one or more markings on the film. The apparatus comprises a nip defined by a drive roller and a pinch roller, wherein the nip is positioned along the film path and configured to receive the film therethrough. The apparatus comprises a motor configured to operate the drive roller to cause advancement of the film along the film path. The apparatus comprises at least one heating element configured to activate to emanate energy to cause a portion of the film within the sealing portion to seal the top portion of the container to form a lid for the container. The apparatus comprises a communication interface and a controller configured to: cause, via the communication interface, a signal with sensor data from the film sensor to be sent to a remote server; receive, from the remote server, an indication as to whether a detected one or more markings on the film satisfies an approved marking scheme; and affect, based on whether the detected one or more markings on the film satisfy the approved marking scheme, operation of one or more components of the apparatus.

In another example embodiment, a supply of film for an automatic sealer for a container is provided. The supply of film includes a repeated marking scheme configured to be read by a film sensor of the automatic sealer to at least one of: in an instance in which the repeated marking scheme satisfies an approved marking scheme, enable operation of the automatic sealer or components thereof, or in an instance in which the repeated marking scheme does not satisfy the approved marking scheme, disable operation of the automatic sealer or components thereof.

In some embodiments, the repeated marking scheme comprises characteristics formed of at least one of the color of one or more markings of the repeated marking scheme, a width of the one or more markings of the repeated marking scheme, a length of the one or more markings of the repeated marking scheme, or a spacing between adjacent markings of the repeated marking scheme.

In some embodiments, the repeated marking scheme is designed to be read by the film sensor to control operation of one or more components of the automatic sealer based on one or more characteristics of the repeated marking scheme. The one or more characteristics comprise at least one of the color of one or more markings of the repeated marking scheme, a width of the one or more markings of the repeated marking scheme, a length of the one or more markings of the repeated marking scheme, or a spacing between adjacent markings of the repeated marking scheme.

In some embodiments, the repeated marking scheme is formed along an entire length of the supply of film.

In some embodiments, the supply of film is a roll of film.

In some embodiments, the marking scheme comprises a detectable non-visible marking, wherein the detectable non-visible marking is covered using an ink or other coating that is visible to a user and configured to enable detection of the marking therethrough.

In some embodiments, the supply of film is formed into a plurality of successive portions. The supply of film is formed of at least a first layer of ink and a second layer of ink. The first layer of ink includes a radiation-absorbing layer of ink and the second layer of ink includes a non-radiation-absorbing layer of ink. The first layer of ink is applied to a first portion of each of the plurality of successive

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portions such that a remaining corner portion of each of the plurality of successive portions does not include the first layer of ink so as to form a pull tab upon formation of a seal therefrom. The second layer of ink is applied to both the first portion and the remaining corner portion of each of the plurality of successive portions.

It is to be understood that both the foregoing general description and the following detailed description and drawings are examples and are provided for purpose of illustration and not intended to limit the scope of the disclosed subject matter in any manner. The accompanying drawings, which are incorporated in and constitute part of this specification, are included to illustrate and provide a further understanding of the devices of the disclosed subject matter. Together with the description, the drawings serve to explain the principles of the disclosed subject matter.

An apparatus may include a film support roller supporting a film roll and a housing that may include a sealing portion having an aperture sized to receive a top portion of a container. The housing may define a film path leading from the film roll to the sealing portion. A guide support assembly may have a ramp and at least one guide truss positioned along the film path. The ramp and the guide truss may guide the film across the aperture within the sealing portion, covering the aperture. The motor may advance the film along the film path. The heating element may emanate energy. The controller may cause the motor to advance the film onto the ramp and cause the heating element to emanate energy to cause the film to seal the top portion of the container.

BRIEF DESCRIPTION OF THE DRAWINGS

Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

FIG. 1A is a top perspective view of an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 1B is a top perspective view of another example sealing apparatus, in accordance with some embodiments described herein.

FIG. 1C is a bottom perspective view of the example sealing apparatus shown in FIG. 1B, in accordance with some embodiments described herein.

FIG. 2A is a cross-sectional side view of the example sealing apparatus shown in FIG. 1A, in accordance with some embodiments described herein.

FIG. 2B is a cross-sectional side view of the example sealing apparatus shown in FIG. 1B, in accordance with some embodiments described herein.

FIG. 3A is a partial side cross-sectional view of an example sealing apparatus with a shield plate in a first position, in accordance with some embodiments described herein.

FIG. 3B is a partial side cross-sectional view of the example sealing apparatus shown in FIG. 3A with a shield plate moving between a first position towards a second position, in accordance with some embodiments described herein.

FIG. 3C is a partial side cross-sectional view of the example sealing apparatus shown in FIG. 3A with a shield plate returned to the first position and with a film secured to a container, in accordance with some embodiments described herein.

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FIG. 4A is a side view of the example sealing apparatus shown in FIG. 1A with a body lid in an opened position, in accordance with some embodiments described herein.

FIG. 4B is a side view of the sealing apparatus of FIG. 4A, with the body lid open and the pinch roller adjusted away from the drive roller to enable loading of film from a roll of film, in accordance with some embodiments described herein.

FIG. 5A is an enlarged cross-sectional side view of a portion of an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 5B is an enlarged cross-sectional perspective view of another portion of an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 5C is an enlarged cross-sectional perspective view of yet another portion of an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 6A is a top perspective view of a portion of an example securing head assembly for an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 6B is an enlarged cross-sectional side view of the securing head assembly of FIG. 6A, in accordance with some embodiments described herein.

FIG. 6C is an exploded view of the securing head assembly of FIG. 6A, in accordance with some embodiments described herein.

FIG. 7 shows a cross-sectional view of the sealing apparatus shown in FIG. 2A, wherein air flow through the apparatus is illustrated, in accordance with some embodiments described herein.

FIG. 8A shows an example sealed lid of a container including a single slit.

FIG. 8B shows an example piercer with two spaced apart actuatable tips, in accordance with some embodiments described herein.

FIG. 8C shows an example sealed lid of a container utilizing the example piercer shown in FIG. 8B, in accordance with some embodiments described herein.

FIG. 8D illustrates example leakage protection provided by the spaced apart slits in the sealed lid shown in FIG. 8C, in accordance with some embodiments described herein.

FIG. 9A is a top view of an example film that can be used in accordance with some embodiments described herein.

FIG. 9B is a top view of a portion of a film supply, where a radiation-absorbing layer of ink has been applied to the film supply, in accordance with some embodiments described herein;

FIG. 9C is a top view of the portion of the film supply shown in FIG. 9B, where a first non-radiation-absorbing layer of ink has been applied to the film supply, in accordance with some embodiments described herein;

FIG. 9D is a top view of the portion of the film supply shown in FIG. 9C, where a second non-radiation-absorbing layer of ink has been applied to the film supply, in accordance with some embodiments described herein;

FIGS. 10-12 illustrates various example film marking schemes, in accordance with some embodiments described herein.

FIG. 13 shows a block diagram of an example system utilizing an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 14 illustrates a flowchart of an example method for operating an example sealing apparatus, in accordance with some embodiments described herein.

FIG. 15 illustrates a flowchart of an example method for confirming use of an approved film with an example sealing

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apparatus utilizing a marking scheme on the film, in accordance with some embodiments described herein.

FIG. 16 illustrates a flowchart of an example method for operating an example sealing apparatus utilizing a marking scheme on the film, in accordance with some embodiments described herein.

FIG. 17 shows an example sealing lid with a customized message printed on the lid, in accordance with some embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout.

In accordance with the disclosed subject matter, an apparatus to secure a film to a container to form a seal (e.g., a full seal, a partial seal, etc.) of the container to at least partially cover and/or inhibit flow from the container is provided. Solely for purpose of illustration, embodiments of an example sealing apparatus to secure a film to a container in accordance with the disclosed subject matter is shown in FIGS. 1A-1C. The examples herein are not intended to limit the scope of the disclosed subject matter in any manner. Particularly, and as illustrated, the apparatus 100, 100' can have a body portion 200, 200' and securing portion 300, 300'.

FIG. 2A is a cross-sectional side view of an example apparatus 100, in accordance with the disclosed subject matter. As depicted, the securing portion 300 can be positioned at a front of the apparatus 100 and adjacent to the body portion 200. The body portion 200 can be positioned at the back of the apparatus. As further described herein, the body portion 200 can house a length of film therein, such as a roll of film, that can feed into the securing portion 300. The securing portion 300 can further include a securing head assembly 400 with a sealing portion 301 that secures a film to a top of a container. FIG. 2B shows another example apparatus 100' with a body portion 200' and a securing portion 300'. Notably, FIG. 2B also illustrates an example film path 211' for the film to travel through the apparatus from the roll of film 500' to the loading zone within the sealing portion 301'.

For purposes of illustration, and not limitation, FIGS. 3A-3C show the operation of the apparatus with respect to a representative container, such as a disposable beverage cup. FIG. 3A depicts a length of film in the sealing portion in a position ready to be secured to a container 600. FIG. 3B depicts the container cooperating with the apparatus 100 to secure the film to the container during operation thereof. FIG. 3C depicts the film secured to the container prior to the apparatus feeding a next film into the loading zone. A method of operating the apparatus and reference to FIGS. 3A-3C is discussed throughout the specification.

Definitions

The terms used in this specification generally have their ordinary meanings in the art, within the context of this subject matter and in the specific context where each term is

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used. Certain terms are defined below to provide additional guidance in describing the disclosed subject matter.

As used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a component” can include a plurality of components.

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system.

Example Body of Apparatus

The body portion of the illustrated apparatus may contain the initial film loading, a piercer, film rollers, film sensors, a film cutter, and a guide support assembly, among other components, as described herein.

The apparatus 100, 100' can include a body portion 200, as shown in FIGS. 2A and 2B. FIGS. 4A-4B are side perspective views of various portions of the apparatus of FIG. 2A. FIG. 4C is side perspective view of another portion of the apparatus of FIG. 2B.

As shown in FIGS. 1A and 1B, the body portion 200, 200' can include a body cover 205, 205', and any other suitable structure to contain the machinery therein. The body cover 205, 205' can be coupled to the body portion 200, 200', for example pivotably coupled by a hinge, screws, positioning, or other coupling devices, and additionally or alternatively, by friction and/or gravity alone. The body cover 205, 205' can improve the usability, safety, aesthetics, and other properties of the apparatus 100. For example, the body cover 205, 205' can improve usability by reducing the amount of debris that enters the body portion 200, 200'. The body cover 205, 205' can contribute to the safe operation of the apparatus, such as by reducing the likelihood of bodily contact with internal machinery. The body cover 205, 205' can be sized and/or shaped to accommodate a roll of film to be used for securement to containers.

The body portion 200, 200' can comprise various features, including for example the body cover 205, 205', a first and second film support roller 210, 201', 215, 215', a piercer 220, 220' with a piercing tip 225, a printer 250, a nip including a drive roller 235, 235' and a pinch roller 230, 230', a motor 213' (such as for operating the drive roller 235, 235'), and one or more film sensors 240. Although depicted and described as being in the body portion 200, 200', any of the features can be located in any other suitable location. For example, at least one of the piercer 220, 220', the printer 250, the nip, and the film sensor 240 can be located in the securing portion 300, 300'.

As shown in FIGS. 2A, 3B, and 4B, the apparatus 100, 100' can comprise a first film support roller 210, 210' and second film support roller 215, 215' to cooperatively support a roll of film 500, 500'. That is, the roll of film 500, 500' can rest between the first film support roller 210, 210' and the second film support roller 215, 215' and be capable of rotation with respect thereto (although other forms of holding the roll of film are contemplated herein, such as with a roll holder). As shown in FIG. 4B, the first and second film support rollers 210, 215 can support the roll from underneath, whereas the movable arm 255 can ensure that the roll of film 500 is properly positioned transversely in the apparatus. The arm 255 and the body cover 205 can move independent to each other, as shown in FIGS. 4A and 4B, such as to enable replacement of an empty roll of film with

a replacement roll of film and loading of the leading edge of the roll of film into the nip (e.g., due to the arm **255** including the pinch roller **230** such that the pinch roller **230** is removed from the film path **211'**). In some embodiments, one or both of the film support rollers **210**, **210'**, **215**, **215'** is mechanically rotated. As such, the rotation of one or both of the film support rollers **210**, **210'**, **215**, **215'** imparts rotation to the supported roll of film.

In other embodiments, one or both of the film support rollers **210**, **210'**, **215**, **215'** is capable of passively rotating with non-negligible resistance. In those embodiments, rotation of the roll of film imparts a rotational force on the film support rollers **210**, **210'**, **215**, **215'**. The non-negligible resistance imparts rotational resistance to the roll of film, thereby imparting tension in the film between the roll of film and the drive roller **235**, **235'**. In still other embodiments, one or both of the film support rollers **210**, **210'**, **215**, **215'** can be prevented from rotating. In such embodiments, lack of rotation causes slippage to occur between one or both of the film support rollers **210**, **210'**, **215**, **215'** and the roll of film **500**, **500'**, imparting rotational resistance to the roll of film, thereby imparting tension in the film between the roll of film and the drive roller **235**, **215'**. In this regard, in some embodiments, the controller is configured to operate the motor to cause a different rotation speed of the drive roller in comparison to a film support roller for the supply of film so as to form tension in the film (which may aid with piercing and/or printing of the film). Although the one or more film support rollers are depicted upstream from the securing portion **300**, the one or more film support rollers can be positioned at a location in any direction along the path of the film **211'**.

In some embodiments, the apparatus **100**, **100'** may include a film supply window that may enable a user to determine/approximate an amount of film supply remaining on an installed roll, such as without opening the body cover **205**, **205'**. For example, the apparatus **100'** shown in FIG. 2B includes a film supply window **208'** that is positioned on the top of the body cover **205**, **205'**.

Example Piercer

In accordance with another aspect of the present invention, the apparatus **100**, **100'** may include a piercer **220**, **220'** having one or more protrusions **225** (e.g., tips, blades, etc.). In some embodiments, the piercer and the protrusions are monolithic such that the piercer is actuatable. The protrusions **225** can impart an impression upon (or through) a film to weaken the film at such a location for ventilation and/or for insertion of a drinking straw, for purposes of example. The piercer **220**, **220'** can actuate the protrusions **225** in any suitable way, for example by electromagnetism via a solenoid, by hydraulics, by a rotating arm actuator, by a linear actuator, or the like. In other embodiments, the entire piercer **220**, **220'**, including protrusions **225**, is actuatable with respect to the body portion. In other embodiments, the piercer **220**, **220'** does not contain protrusions **225**. Piercers without protrusions can include, for example, air jets, lasers, blast heaters, or any other suitable piercer.

Protrusions **225** can actuate with respect to the piercer **220**, **220'** and/or with respect to the body portion **200**, **200'**. The protrusions **225** can have a variety of shapes, for example, protrusions **225** can be comprised of one or more blades. Additionally or alternatively protrusions **225** can have a pyramid shape, such as a triangular pyramid, a square pyramid, a star pyramid, or other shapes as desired. Additionally or alternatively, protrusions **225** can be shaped to

have a series of needle-shaped protrusions. With such configurations, protrusions **225** can impart an opening or perforation pattern on a film. Additionally still, the protrusions **225** can be a circumferential blade having a closed shape. Accordingly, protrusions **225** can remove pieces of film **505** such as by way of kiss cutting. In some embodiments, the protrusions **225** are removable from the piercer **220**, **220'** to be exchanged for a piercing tip of a different construction.

The piercer **220**, **220'** can be positioned at any suitable location in the apparatus including being positioned in the securing portion **300**. For example, the piercer **220** can be positioned above the securing head assembly **400**. The piercer can be configured to pierce film **505** shortly before securement, during securement, or shortly after securement. In some embodiments, the piercer **220** includes a piercing rod coupled to the piercing tip **225**, wherein the piercing rod is coaxial with and movable with respect to the guide rod of the securing head assembly **400**.

In some embodiments, a controller (e.g., controller **30** described with respect to FIG. 13), may be configured to cause actuation of the piercer **220**, **220'** to cause the perforations or slits to be created in the film, such as it travels along the film path **211'**. In some embodiments, the controller may be configured to control the relative position of the perforations or slits on the lid, such as based on a desired operational parameter for the anticipated product or the utilized film.

In some embodiments, the piercer **220**, **220'** may be configured with two or more spaced apart protrusions (e.g., tips, blades, etc.). An example such piercer **820** is shown in FIG. 8B. The depicted piercer **820** includes a first protrusion **825a** and a second protrusion **825b** that are spaced apart. In some embodiments, the piercer may form a single device with two or more spaced apart protrusions. In some embodiments, the two or more spaced apart protrusions may be formed of two spaced apart protrusions extending from a single device. In other embodiments, there may be two separate devices that each form a protrusion—to thereby form the spaced apart protrusions. In some embodiments, a single protrusion may be used to form two spaced apart slits/perforations on the film. In such example embodiments the piercer with a single protrusion may move to a different location on the film in between punctures of the film to form the two spaced apart slits/perforations. Alternatively, the film may move to cause the piercer to puncture the film in different locations (e.g., in the machine direction).

In this regard, when applied to the film, such an example piercer **802** creates two spaced apart slits/perforations on the film. For example, FIG. 8C illustrates an example sealed lid **860** with two spaced apart slits **870a** and **870b** with a portion of the lid **860** remaining intact therebetween. Notably, the separation of the slit **870a** and **870b** provides for an elongated point of weakness—which provides for desirable ventilation and a greater weakness point for ease of insertion of a straw, while still providing preferable leakage protection, such as illustrated in FIG. 8D. For example, the extra lid material between the first slit **870a** and the second slit **870b** may keep the edges of the slit film sufficiently closed so as to allow sufficient surface tension to be created by liquid at surface—thereby minimizing/avoiding leakage through the slits **870**. In this regard, the two spaced apart slits are designed to enable venting and discourage leaking when the container is tilted such as due to the surface tension between the liquid and the portion of the lid between the spaced apart slits. In contrast, a single elongated slit, such as shown as slit **770** on the sealed lid **760** in FIG. 8A, of the same length as multiple slits may allow undesirable leakage

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when the container is tipped. Said differently, in comparison to a continuous length slit of similar overall length, the double spaced apart slits provide equivalent ventilation for the container, weakness within the lid to enable insertion of a straw, and increased leakage prevention due to increased surface tension of liquid on an inside portion of the lid. In some embodiments, the double spaced apart slits may provide slight increased resistance to insertion of a straw or other object to avoid/inhibit undesired and/or inadvertent tearing. In this regard, usage of the term equivalent is designed to account for some variation, but generally being similar in comparison.

Example Film Rollers

In some embodiments of the present invention, the example sealer device may include one or more film advancement mechanisms (e.g., nip **23** and motor **22** shown in and described with respect to FIG. **13**), such as for advancing the film from the roll of film along the film path. For example, with reference to FIGS. **2A** and **2B**, the illustrated apparatus **100**, **100'** comprises a nip having a drive roller **235**, **235'** and a pinch roller **230**, **230'**. In some embodiments, the nip functions to progress film **505** into the loading zone **325** (e.g., along the film path **211'**). In some embodiments, the pinch roller **230**, **230'** is adjustable in relation to the drive roller **235**, **235'** (such as being attached to the arm **255**) to enable insertion of film **505** between the drive roller **235**, **235'** and the pinch roller **230**, **230'**. For example, and as shown in FIG. **4B** for purpose of illustration and not limitation, the pinch roller **230**, **230'** can be coupled to a moveable lever arm **255**. Such a configuration can simplify a process of loading a length of film **505** into the nip. In other embodiments, the pinch roller **230** is adjustable relative to the drive roller **235** to accommodate different thicknesses of film **505** therebetween. Further, in some embodiments, other ways to load the film into the nip are contemplated, such as by feeding the film into the nip as the drive roller operates to prime the nip.

In some embodiments, the driver roller **235**, **235'** may be rotated via a motor **213'**, which may be controlled by a controller of the sealing device (e.g., controller **30** shown and described in FIG. **13**). In some embodiments, drive roller **235**, **235'** can be mechanically rotated. For example, drive roller **235**, **235'** can be mechanically rotated by a crank mechanism, a rotational driver, a drive shaft, a drive belt, a drive chain, or any other means of imparting rotational force. In some embodiments, pinch roller **230**, **230'** can freely rotate in response to a torque. For example, torque could be applied by friction between the pinch roller **230**, **230'** and the rotating drive roller **235**, **235'**, or by friction between the pinch roller **230**, **230'** and the film **505**.

In some embodiments, the drive roller **235**, **235'** can be mechanically coupled to at least one of the film support rollers **210**, **210'**, **215**, **215'**. For example, drive roller **235**, **235'** and the at least one film support roller can be mechanically coupled by a drive shaft, a drive belt, a drive chain, or other suitable couplings. The body portion **200**, **200'** can further comprise a film support roller **210**, **210'** to support a length of film, wherein the drive roller **235**, **235'** rotates at a faster speed than the film support roller **210**, **210'** such that the film can be positioned taught in along the film path. For example, the drive roller **235**, **235'** and film support roller **210**, **210'** can be geared and/or dimensioned such that their coupling imparts a different rotational speed to each roller.

Example Film Sensor

In some embodiments, the apparatus **100**, **100'** may include one or more film sensors **240**. In some embodiments,

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and as shown in FIG. **5A**, the film sensor **240** can comprise a film signal emitter and a film signal sensor to detect a film sensor flag **510** passing therebetween (although other types of sensors are contemplated). The film sensor flag **510** positioning can directly correspond to the position of the film as moved by the drive nip, and further discussed herein with respect to FIGS. **9A-9D**. The film signal emitter can continuously emit a light signal capable of being detected by the film signal sensor. Film **505** can be positioned between the film signal emitter **241** and the film signal sensor **242**, such that opaque portions of the film **505** block the light signal where opaque. However, when the film sensor **240** detects a change of the film, such as when detecting a transparent portion/window of the film or a film sensor flag **510**, the film sensor **240** can send a signal input indicating that the continuous emission of the signal emitter has been disrupted. Accordingly, as the film **505** moves between the film signal emitter **241** and film signal sensor **242**, the film sensor flag **510** or transparent window can allow transmission of the light signal to the film signal sensor. The film signal sensor, upon detecting the light signal or a change in the intensity of the light signal, can register that the film **505** has progressed upstream and generate a film sensor input. In other embodiments, a reverse set up can be provided. That is, the film can allow transmission of the light signal to the film signal sensor **242** until it is blocked or partially blocked by the film once the film has progressed. The film signal sensor, upon detecting the reduction of the light signal, can register that the film has progressed and generate a film sensor input to deactivate the drive nip.

In other embodiments, the film sensor **240** can be any sensor suitable for detecting the progression. For example, the film sensor **240** can include at least one of a light sensor as described, a mechanical sensor, a motion sensor, among others. A mechanical sensor can include a rotational sensor, that is rotated by progression of the film, and registers that the film has progressed upon rotation. For example, a mechanical sensor can register a degree of progression by the amount of rotation. In some embodiments, a rotational sensor can be coupled to the pinch roller **230** to detect rotation thereof.

In some embodiments, the drive roller **235**, **235'** can be can be mechanically rotated based on input from the film sensor **240**. That is, in response to film sensor inputs by the film sensor **240**, the drive roller **235**, **235'** can have its rotation initiated, continued, or ceased. For example, after a securement cycle, rotation of drive roller **235**, **235'** can be initiated, film sensor **240** can detect progression of film **505**, and film sensor **240** can provide a film sensor input. In response to the film sensor input, rotation of drive roller **235** can be ceased.

Solely for purpose of illustration, reference is now made to FIG. **9A**, which is a top perspective view of a film that can be used in conjunction with an apparatus of the disclosed subject matter. In some embodiments, and as shown for purpose of illustration and not limitation, the film **505** can include at least one film window **510** or other marking. As described above, with respect to the film sensor, the at least one film window **510** can allow at least partial transmission of the film sensor signal to indicate progression or the location of the film **505** relative to the dispenser.

In some embodiments, the film can include a plurality of film windows **510** (e.g., markings) spaced along a length of film at regular intervals. For example, in some embodiments, the plurality of film windows **510** can be spaced a distance corresponding to a predetermined length of film to be secured to a container, such as for purposes of example

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spaced at a distance of every 15 cm. In some embodiments using film windows spaced as above, the film sensor **240** can generate a film sensor input upon detecting a single film window to allow the film sensor to cooperate with a film cutter in creating the predetermined length of film. In other embodiments, the film window can alternatively be a suitable marking distinguishable from the remaining film area. In some embodiments, the windows (or markings) may be clear, printed, reflective, or printed with an ink that is not visible or readily visible to a human eye, such as phosphorescent ink. In this regard, in some embodiments, the one or more markings may be formed using ink or other coatings.

In other embodiments, the plurality of windows **510** are spaced at a distance corresponding to a small quantized value. In some embodiments using film windows **510** spaced at a small quantized distance, the film sensor **240** can generate a film sensor input upon sensing a predetermined plurality of film windows **510**. For example, the film windows **510** can be spaced at a distance of 25 mm, and the film sensor **240** can generate a film sensor input upon sensing the passage of six windows. As such, the apparatus can be programmed to generate different lengths of film for depositing within the loading zone to complement various sized containers thus minimizing film waste. In some embodiments, the film sensor **240** can be adjusted such that the number of film windows **510** needed to pass before a film sensor input is generated. For example, a command received prior to the initialization of film progression can determine the number of film windows **510** desired to pass. In some embodiments, the command can be received by the input device **315** and can be further processed such as by a computer operatively coupled with the apparatus, as further discussed herein. As detailed herein, the input device **315** may be used to provide many different types of input that can be recognized (such as by the controller **30**) and used to affect operation of the scaling apparatus (e.g., what temperature to operate the heating element(s) at, what to print on the film, etc.).

FIGS. 9B-9D illustrate an example portion of film **505** that includes a plurality of portions (e.g., FIG. 9B illustrates portion **507a** followed (e.g., along a machine direction) by portion **509a**). In order to form the aesthetic look of the seal, such as seal **760** shown and described with respect to FIG. 8A, one or more layers of ink may be applied to the film **505**. Notably, some of the layers of ink may be radiation-absorbing so as to be particularly susceptible to heat, so as to shrink (e.g., around the top of the container) to form the seal (such as described herein). In this regard, FIG. 9B illustrates a first layer of ink that includes radiation-absorbing ink. FIG. 9C illustrates an applied second layer of ink over the first layer of ink on both portions **507b**, **509b**. Notably, however, in some embodiments, the second layer of ink may be non-radiation-absorbing (or less radiation-absorbing) such that the ink is not particularly susceptible to heat and, thus, does not shrink during application of heat by the heating elements (such as described further herein). FIG. 9D illustrates an applied third layer of ink over the first and second layers of ink on both portions **507c**, **509c**. In this regard, the one or more images or patterns can be built-up based on colors applied via the layers of ink.

Further, however, in some embodiments, one or more markings (e.g., windows) may be provided such as described herein. For example, windows **515** and markings **525** may be unapplied portions of the radiation-absorbing layer of ink for portions **507a**, **509a** such that they are detectable (e.g., as described herein). Depending on the desired look of the film, such windows **515** or markings **525**

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may be covered with one or more layers of ink (e.g., non-radiation-absorbing layers of ink) and still be detectable. For example, with reference to FIG. 9C, the marking **525** on portion **507b** was covered with the corresponding layer of ink.

In some embodiments, the film sensor(s) **240** may be configured to sense one or more markings, marking schemes, and/or characteristics corresponding to the markings or marking schemes.

In this regard, in some embodiments, the one or more markings may correspond to a marking pattern comprised of a plurality of markings and/or determined spacing between each adjacent marking within the plurality of markings. In some embodiments, the film sensor may be configured to detect various characteristics of the markings or marking schemes, such as the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, or a spacing between adjacent markings. Notably, in some embodiments, the marking scheme may be repeated along the length of the roll of film such that the roll of film comprises a plurality of repeated film marking schemes. As described further herein, such information could be used to determine various characteristics of the film and/or control the device accordingly.

FIG. 10 shows an example film portion **909** with a marking scheme. The film sensor(s) may be configured to read the markings **950** and/or various characteristics regarding the read markings, such as their width, their length, or the distance between adjacent marks. FIG. 11 illustrates another example film portion **909'** with a more complex marking scheme that could be used to convey even more information. In this regard, the various widths and distances between the markings may be sensed and used to convey that information to the device. In the illustrated embodiment, a repeatable series of individual markings could be considered a single repeated marking scheme/pattern (indicated by the dashed line box **955**). In particular, the illustrated marking scheme **955** includes a series of black markings **952a**, **952b** (or color markings) of varying widths that are each separated by white space **953a**, **953b** (or white markings, blank markings, etc.) of varying widths. FIG. 12 illustrates another example film portion **909''** with marking schemes on both edges of the film (e.g., markings **950** and **951** on one edge and marking scheme **955'** on the other edge). In such an example embodiment, more than one film sensor could be used to gather the desired information. Further, different edges of the film could convey different information. For example, the markings **950** may be used to provide data regarding the length of film being conveyed (such as for determining when to turn off the motor) and the marking scheme **955'** may provide information used to determine various characteristics of the film and/or control the device accordingly. Likewise, a type of cut marking could be maintained on one side (such that it could be used with a standard cut-to-mark system), while the other side could be used to provide other marking schemes and convey additional information.

Notably, in the illustrated embodiments the markings are positioned proximate the edges of the film, which may enable various features to be accomplished in the center of the film **990**, such as may translate to the portion of the film that will form the lid of the container. In this regard, such as described further herein, a printer may print one or more messages or images on the film in the center section **909** and/or the piercer could form the slits/perforations in the center section **990**.

Though the depicted markings are shown as black markings and white/blank markings, other types of markings are contemplated, such as barcodes, color markings, quick response (QR) codes, among others. In some embodiments, logos or other images on the film may be utilized as markings. In some embodiments, the film sensor may also be configured to measure or sense other types of markings, such as holes, bumps, or other features of the film. In this regard, the film sensor(s) may be configured to read any such markings or subsets/combinations of any such markings.

Example Film Cutter

In some embodiments, the apparatus **100**, **100'** can further include a film cutter to cut a predetermined length of film from a film roll. For example, FIG. 5C illustrates an example cutter **214'** that includes driving mechanism **216'**. The predetermined length of film can be sized to cover the top opening of a container to allow the film to be secured to the container for a suitable seal thereon. In some embodiments, the film cutter is positioned along the film path **211'** between the loading zone **325** and the nip comprised of the drive roller **235**, **235'** and the pinch roller **230**, **230'**. In some embodiments, the film cutter comprises at least one of a guillotine cutter and a shear cutter. Additionally or alternatively, the film cutter can comprise any cutter suitable for cutting the film **505** or at least perforating the film to create segments of film. For example, the film cutter can include a rotary cutter, an air jet cutter, a laser cutter, a blast heater cutter, or any other suitable cutter.

In some embodiments, the film **505** can travel along a longitudinal path. For example, the film **505** can travel along a longitudinal path within the body portion **200** of the apparatus **100**. In some embodiments, the film support roller **210**, the drive nip, and the loading zone **325** are arranged along the longitudinal path. In other embodiments the path is non-linear.

In some embodiments, a controller (e.g., the controller **30**) may be configured to control actuation of the film cutter to cause the portion of the film to be cut from the roll of film. As described herein, such control may be based on a measured distance of travel of the film along a film path and may be based on data conveyed from one or more markings.

In some embodiments, with reference to FIG. 2B, the apparatus **100**, **100'** may include an edge cutter, such as a manual sliding cutter **251'**. The edge cutter may be configured to cut a clean edge of the film. In this regard, in some cases, the film may form a jagged or wrinkled edge that makes it difficult to load or operate the sealing apparatus **100**, **100'**. For example, after a film jam incident, the film may be bunched up, wrinkled, or have an uneven edge. The edge cutter may be used to cut the film and re-load it through the sealing apparatus **100**, **100'**. In some embodiments, the edge cutter may be manually actuated such that an operator must directly operate it accordingly. In some embodiments, the edge cutter comprises a sliding cutter that includes a movable knife that can be moved within a track across a width of the film to thereby cut the film and form a clean edge.

Example Guide Assembly

As shown in FIGS. 5A-5B, the apparatus can further include a guide assembly to guide the film between the body portion **200**, **200'** to the securing portion **300**, **300'**. The guide assembly can comprise an entry structure **270** and a guide support assembly **445**.

The entry structure **270** can be downstream the drive nip and channel the film towards the loading zone **325** in the sealing portion **301**, **301'**. The entry structure **270** can define a funnel **272** with a wider entry opening and narrower exit to steer a leading end of the film through the entry structure **270** and towards the loading zone **326**. In an example embodiment, the entry opening is approximately $\frac{1}{8}$ th inch.

Once the film enters through the entry structure **270** the film can progress to a guide support assembly **445** of the apparatus. The guide support assembly **445** can include a ramp **446** and at least one guide truss **447** (e.g., ribs) to guide the film from the body portion to the securing portion. The ramp is configured to receive a predetermined length of film (e.g., film cut from the roll) from the body portion **200**, **200'** to guide a leading end of the film to the loading zone **325**. The ramp **446** has a surface extending from a first end to a second end thereof, as shown in FIGS. 5A and B. The ramp surface is oriented at an angle of inclination that can be less than or equal to 85 degrees, or more preferably in a range from about 10-65 degrees. In some embodiments, the second end of the ramp is coupled to a guide surface **448** configured to receive a film **505** from the surface of the ramp, as shown in FIGS. 5A and B.

The guide assembly can serve to facilitate proper progression of the film **505** into the loading zone **325**. For example, the guide support ramp can promote movement of the leading end of a film in the direction upward of the inclination. In this manner, the film **505** can have a reduced likelihood of progressing in a direction other than toward the loading zone **325** and reduce the likelihood the film from falling into a gap (e.g., the aperture for receiving the container). In particular, the progressing film can intersect the ramp and progress therealong (e.g., along the guide trusses **447**, which may be spread along the width direction of the film path), thus progressing toward the loading zone. In this regard, in some embodiments, the combination of the ramp **446** and the one or more guide trusses **447** may form a guide for directing the leading edge of the film such that the leading edge of the film clears the gap/aperture and rests on the other side of the gap/aperture.

In some embodiments, the shield plate may be positioned to cover a second aperture leading into the sealing portion, such as shown in FIG. 5B. In such an embodiment, the shield plate may form a top boundary guide for directing the leading edge of the film through the loading zone **325**.

Furthermore, the entry structure together with the guide support assembly creates the guide assembly structure to enable the film to be positioned in the loading zone at a close vertical proximity to the shield plate in a ready position, thus limiting the gap distance between the film of the loading zone and the shield plate, as further discussed herein. Reducing the gap distance reduces the distance a container may be required to move further within the apparatus before activation of the heating elements. Accordingly, the film of the loading zone would only need to travel a distance between a top of a truss and the bottom surface of the shield plate to initially engage the shield plate.

In some embodiments, the guide support assembly **445** can comprise a plurality of guide trusses, each capable of receiving a film from the body portion. In some embodiments, the guide trusses can further include an end portion thereof disposed at an angle of inclination, similar to the ramp. For example, in some embodiments each guide truss can extend in a direction parallel to the direction of film progression into the loading zone **325** and have at least a portion that is disposed at an angle of inclination with respect thereto.

In accordance with some embodiments, the apparatus **100** can be used in conjunction with any suitable type of film. In operation, once heating elements are activated and energy is channeled towards the free ends of the film, the free ends of the film change shape and heat shrink about the top portion of the container, such as about a lip **602** of the container **600**, as shown in FIG. **3C** and further discussed below. Suitable films include those that will shrink in the presence of heat or radiant energy. For example, the film can be a plastic wrapping film which has the capability of shrinking when it is heated, and in some cases to near the melting point of the film. These films are commonly manufactured from plastic resins such as polyvinyl chloride (PVC); polypropylene (PP); linear-low density polyethylene (LLDPE); low density polyethylene (LDPE); high density polyethylene (HDPE); copolymers of ethylene and vinyl acetate (EVA); copolymers of ethylene and vinyl alcohols (EVOH); ionomers (e.g., SURLYN™, by E.I. du Pont de Nemours and Company of Wilmington, Del.); copolymers of vinylidene chloride (e.g., PVDC, SARAN™ (“SARAN” is a trademark of The Dow Chemical Company of Midland, Mich.)); copolymers of ethylene acrylic acid (EAA); polyamides (PA); polyester, polystyrene, nylon and copolymers of ethylene and octene. Additionally or alternatively, the film can be a bi-axially oriented thin shrink film having a thickness of between 40 to 120 gauge (1.02 mm to 3.05 mm). In another embodiment, the film can be a bi-axially oriented thin shrink film having a thickness of between 60 to 100 gauge (1.52 mm to 2.54 mm).

In accordance with some embodiments, the film can further include at least one energy absorbing substance (e.g., a radiation-absorbing layer of ink) on at least part of the film. In some embodiments, the one or more energy absorbing substances can be pre-applied to the film, such as by printing, brushing, spray coating, electrostatic coating, electrodeposition coating, flow coating, roller coating, dip coating, among others. Additionally or alternatively, the substances can be incorporated into the film during formation or manufacture thereof. In other embodiments, the one or more substances can be printed onto the film during operation of the disclosed apparatus. Such substances can allow or enable shrinking of the film at the desired locations to create a suitable seal.

The film can be dimensioned to operate within the confines of the apparatus. In one embodiment, the film can have a width dimension of between approximately 8 and approximately 30 cm. In some embodiments, the film can be cut to a predetermined length dimension of between approximately 8 to approximately 30 cm. In some embodiments, the width and length dimensions of the predetermined dimensioned film can be the same. In some embodiments, the predetermined dimensioned film can be square, although other shapes are contemplated herein.

In some embodiments, the film can have a degree of elasticity. As such, when the film is secured to a top of the container and thereafter removed from the container, the removed film maintains its shape with respect to the heat shrunk edges, and the film can be reapplied to the top of the container for securement.

In some embodiments, the film may be pre-printed prior to installation into the sealing device. For example, such as described herein, the film may include one or more markings pre-printed thereon. In this regard, as is consistent with various embodiments described herein, the one or more markings may be configured to be read by the film sensor(s)

or other components to provide information/data for use in operation of the sealing device. For example, with reference to FIGS. **9-12** and the corresponding description herein, the roll of film may include a repeated marking scheme configured to be read by a film sensor of the automatic sealer. Depending on the desired functionality, the marking scheme may be utilized to affect operation of the automatic sealer (e.g., sealing device) and/or one or more components thereof. For example, the marking scheme may be checked against an approved marking scheme list/database and corresponding affecting (e.g., disabling, enabling, changing, etc.) operation of the automatic sealer and/or components thereof may occur.

In some embodiments, the repeated marking scheme on the film may comprise characteristics formed of at least one of the color of one or more markings of the repeated marking scheme, a width of the one or more markings of the repeated marking scheme, a length of the one or more markings of the repeated marking scheme, or a spacing between adjacent markings of the repeated marking scheme. In some embodiments, the repeated marking scheme is designed to be read by the film sensor to control operation of one or more components of the automatic sealer based on one or more characteristics of the repeated marking scheme.

In some embodiments, the repeated marking scheme is formed along its entire length. In some embodiments, various types of markings may be used, such as windows, ink, paint, a Quick Response Code, a barcode, or a logo. In some embodiments, the markings may be clear such that they are detectable by the film sensor. However, such as described herein with respect to FIGS. **9B-9D**, it may be desirable to cover the clear markings with an ink that visually covers the clear marking to a user looking through the ink, but also enables detection of the clear marking through the ink. In this regard, the ink (e.g., specific layers of ink) may have specific properties that enable detection by the film sensor therethrough without allowing a user to see therethrough (e.g., the ink may be a detection transparent ink). As noted, this provides a marking that is detectable by a film sensor through the ink, but that is non-visible by a user through the ink (e.g., the marking may be a detectable non-visible marking). Said differently, in some embodiments, the marking scheme comprises a detectable non-visible marking, wherein the detectable non-visible marking is covered using an ink that is visible to a user from the top of the film and still configured to enable detection of the marking by a film sensor from the top of the film. Further, utilizing such a detection transparent ink may provide a more consistent shrinkage of the film during formation of the lid on the container, such as opposed to when no ink was present in the marking (e.g., window).

In some embodiments, the portions of the film that are designed to form individual seals may be formed with a “pull tab” or “peel tab”. For example, with reference to FIGS. **9B-9D**, a corner portion **511** of the portion **507a** may be unapplied with the radiation-absorbing layer of ink (e.g., with respect to a remainder of the portion **507a**). In this regard, upon application of heat, the corner portion **511** may not shrink like the remainder of the portion **507a** and, thus, it may remain relatively flat and available for a user to grasp and use to peel back the seal. As noted herein, additional layers of ink (e.g., non-radiation-absorbing layers of ink) may be applied over the corner portion **511** and still the corner portion **511** may not shrink. In this regard, indications or instructions to the user, such as “pull off” **513**, could be printed over the corner portion **511** to direct the user.

Notably, while the above example describes a corner portion **511**, other portions of the film are contemplated to form a pull or peel tab.

Example Printer

In accordance with some embodiments, the apparatus **100**, **100'** can further include additional components. For example, the apparatus can further comprise a printer **250** to print any suitable information (e.g., messages or images) on the film **505**, such as the type of drink or product (or symbol or emoji thereof) disposed within the container as noted below (e.g., an example image (e.g., LOGO **758**) is shown printed on the sealed lid **760** in FIG. **8A** and another example image (e.g., LOGO **858**) is shown printed on the sealed lid **860** in FIG. **8C**). The printer **250** can be any suitable type of printer to make markings on film **505**. For example, the printer can make markings by laser printing, ink jet printing, laser etching, or any other type of printing suitable for film **505**.

In some embodiments, the printer **250** can be used to print the same message on film **505** during every securement cycle. Additionally or alternatively, printer **250** can be used to print a message on film **505** depending on an input. For example, the apparatus **100**, **100'** can further comprise a computer (e.g., controller **30**) to receive at least one command. The computer can be operatively coupled with the apparatus. As such, the apparatus can contain the computer therein or the apparatus can cooperate with a wireless or remote computer/server. In any arrangement, the computer can send a signal to the printer **250** to print a predetermined message on film **505** associated with the at least one command when the at least one command is received. For example, the message can indicate at least one of the type of contents secured within the container, a trademark, a safety message, or any other suitable message. Alternatively or in addition thereto, the printer can be configured to receive commands through a network and wirelessly.

The printer **250** can be positioned at any suitable location within the apparatus and is not limited to placement within the body portion **200**, **200'**. In some embodiments, the printer **250** can be positioned in the securing portion **300**, **300'**. For example, the printer **250** can be positioned above the securing head assembly **400**. The printer can be configured to print on film **505** shortly before securement, during securement, or shortly after securement. In some embodiments, the securing head assembly **400** and the shield plate each include a printing window through which printer **250** can print a message on film **505**.

In some embodiments, such as described herein, the controller may be configured to print one or more messages or icons on the film based on input from an external device/network. For example, the controller may be in communication with a remote device/server and receive instructions or other data that may cause the controller to cause printing on the film, such as using the printer **250**. As an example, the sealing apparatus **100**, **100'** may be in communication with a point-of-sale (POS) system (e.g., the POS system **13** shown in FIG. **13**). In such an embodiment, the POS system may receive an order that may include one or more drink orders. In response, the controller **30** of the sealing apparatus **10** may be configured to receive data corresponding to the order and control its operations accordingly, such as by printing a label on the film to indicate the drink order (e.g., "Cola" or "Diet Cola"). In some embodiments, the printed message or image may be customized, such as indicating the customer (e.g., "John's Cola"). In

such a manner, the corresponding sealed lid may include the appropriately printed data that can be used with fulfilling the order. In some embodiments, the determination as to what to print may occur remotely from the sealing apparatus **10** and communicated thereto, such as at the POS system **10** or some other remote system. An example customized lid **1860** with a logo **1858** and a printed customized message ("JOHN'S COLA") **1859** is shown in FIG. **17**. Example systems for use with a point-of-sale system, including example print determinations and data management, may be found in U.S. patent application Ser. No. 16/212,047, entitled "Personalized Food Service Material Printing Systems", which published as U.S. Publication No. 2019/0180392, and which is owned by the Assignee of the present invention and incorporated by reference herein.

In some embodiments, the sealing device and/or the components thereof may form part of a beverage forming apparatus or system. For example, the beverage forming apparatus/system may create (e.g., mix, satisfy, dispense, form, etc.) a beverage order, such as pour a cup of soda, water, or juice, from one or more supplies. In addition to forming the beverage, the beverage forming apparatus/system may employ various components/systems described herein (e.g., a sealing apparatus) in conjunction with creating the beverage such that the beverage is produced to a customer with a sealed lid. In some embodiments, such as described herein, the sealed lid may include one or more identifying messages or images appropriate for the created beverage.

In some embodiments, the controller may be configured, such as in response to certain operation conditions, to have the printer print a specific code on the film to provide one or more indications to an operator, such as the position of a splice, an amount of film remaining in the supply, or other indication. In the case of the splice, this may allow the sealing device to alert the operator that a certain film and/or components within the sealing device may be non-functional (e.g., due to presence of splice). In the case of an amount of film remaining, the operator may be alerted that the scaling apparatus may need a replacement supply of film (e.g., due to an indication of a low amount of film supply remaining).

Example Securing Head Assembly

The securing head assembly of the apparatus can contain a housing, at least one heating element, and sensor assembly among other components as described herein. The securing head assembly can further include a plate, such as a shield plate or bearing plate, as further discussed herein. Although described herein as an assembly, in some embodiments, one or more components may not be part of an assembly and may be generally a part of the overall device.

FIGS. **6A-6C** respectively depict a top perspective view of the securing head assembly in partial phantom, an enlarged cross-sectional side view of the securing head assembly, and an exploded view of a securing head assembly. As shown, the securing head assembly includes a housing **425** defining an aperture **426** sized to receive at least a top portion of a container therethrough so that the top portion of the container may fit within the sealing portion **301**, **301'** (shown in FIGS. **2A** and **2B**). As shown, a wall **440** is disposed within the housing which has a width dimension (e.g., diameter) sized to receive at least a top portion of a container. In the illustrated embodiment, a shield plate **405** is movable within the sealing portion **301**, **301'** and with respect to the wall **440** between a first position and a second position, although other configurations are contemplated

herewith as further discussed. The first position of the shield plate **405** is shown in FIGS. 3A and 3C. FIG. 3B and FIG. 6B depict the shield plate **405** moving from the first position and towards the second position. In some embodiments, when the shield plate **405** reaches a predetermined distance, such as a top portion **441** of the wall **440**, the shield plate is in the second position, as further discussed herein. As described herein, however, in other embodiments, the shield plate may be fixedly positioned within the sealing portion.

Example Housing

The housing **425** of the securing head assembly defines a sealing portion **301**, **301'** that can house the shield plate **405** therein when in the first position. Furthermore, the aperture **426** can be sized to receive at least a top portion of a container when the shield plate is translated. The aperture **426** and the sealing portion **301**, **301'** can be dimensioned and shaped to accommodate at least a top portion of a variety of containers. For example, the aperture **426** defined by the housing **425** can be a circular aperture with a diameter larger than the average diameter of a container, such as a disposable drinking cup. In one embodiment, the diameter can range up to approximately 30 cm, and in one embodiment can be approximately 15 cm. The aperture **426** can have any suitable shape such as polygonal, square, rectangular, elliptical, rectilinear, ovoid, circular, or irregular. The aperture **426** of the securing head assembly can coincide with any additional apertures of the apparatus **100** to facilitate movement of the container within the sealing portion of the securing portion **300**, **300'**.

As referenced above, the securing head assembly **400** can further include a wall **440** that can be disposed within the housing **425**. The wall can create a barrier between the heating element(s) and the film disposed on the container. As such, the wall can protect the heating element from having the film melt onto the heating element. In certain embodiments, the wall can also protect the container and user from broken heating elements, e.g., broken bulbs. As shown in the example of FIG. 5B and FIG. 6C, the wall **440** is embodied as a glass tubular structure that tracks along a periphery defining the aperture **426**. Such structure allows the energy to emanate through the wall, but still protect the heating elements, as discussed below. The tubular structure complements the shape of the container inserted therein. The wall **440** can have any suitable shape, for example defining a polygonal, square, rectangular, elliptical, circular, ovoid, circular, or irregular tube. The wall **440** can have a width dimension sized to receive at least a top portion of a container. For example, in some embodiments, the width dimension (e.g., diameter) of the wall **440** is at least the width dimension of the aperture **426** defined by the housing **425**. The wall can furthermore deflect corners of a film disposed on a container downward toward a rim of the container to better position the film for securement to the container, as further discussed herein.

In an alternative example, the at least one wall can be disposed about the at least one heating element. Accordingly, in one embodiment, the wall can be embodied as a glass tubular protective structure that houses the heating element therein. Any inadvertent rupturing of heating elements, such as broken bulbs, can be contained within the glass tubular structure and be contained therein. In such an example, a portion of the glass tubular wall can have a reflective surface or coating, and the remaining portion of the glass tubular wall can be configured to reflect energy from the heating element in a desired direction. In some

embodiments, the wall (e.g., a protective structure) may also be configured to filter out undesirable wavelengths of electromagnetic radiation produced by the heating elements (e.g., the UV component of energy emitted by the tungsten-halogen light bulbs)

The wall **440** can be made of any suitable material. For example, the wall **440** can be made of glass, plastic, or metal or fiber screening. In some embodiments, the wall **440** may be at least partially transparent or translucent to permit energy emanated from the at least one heating element **435** through the wall **440**. In other embodiments, the wall can include opaque sections and transparent sections to focus the energy through the transparent sections. In some embodiments of the disclosed subject matter, the wall **440** can be coupled to the shield plate **405** (when provided) and also be movable with the shield plate **405** between the first and second positions. That is, as the shield plate **405** moves position, the wall **440** can move therewith. However, in the embodiment of FIGS. 6A-6C, the wall **440** remains static and the shield plate moves with respect to the wall between positions. As such, the wall **440** may be coupled to the housing **425** as shown in FIG. 6B, wherein the shield plate **405** is movable with respect to the wall **440** between the first position and the second position. That is, movement of the shield plate **405** does not move the wall **440**. As shown in FIG. 6B for purpose of illustration and not limitation, the shield plate **405** is moveable within the aperture and the wall **440** between a first position and a second position along an axis defined by the wall **440**. In some embodiments, the wall **440** can be coupled to the housing **425** along a circumference of the aperture.

Example Heating Elements

As shown in the example of FIGS. 6A and 6B, at least one heating element **435** is disposed within the housing and positioned external to the wall **440**. In the embodiment of FIG. 6A, the at least one heating element **435** comprises four heating elements as further discussed herein, but any suitable number of heating elements is contemplated herein. The at least one heating element **435** is activated to emanate energy towards the wall **440** when the shield plate is in the second position. Furthermore, the at least one heating element can deactivate while still in the second position after a predetermined length of time to ensure safety of the apparatus. The emanated energy can interact with a film disposed between the shield plate and a container to couple the film to the container. The securing head assembly may further include a sensor assembly **420** to sense movement of the shield plate **405** and to activate the at least one heating element **435** when the shield plate **405** is in the second position. A loading zone **325** is configured to receive a predetermined dimensioned film from the body portion (e.g., portion **505a**), as shown in FIG. 3A and as further discussed herein. The loading zone **325** is positioned adjacent the shield plate **405** in the first position. As shown by FIGS. 3A-3C, a top portion **602** of a container **600** is movable within the aperture **426** and wall **440** to move the shield plate **405** to the second position and to secure a predetermined dimensioned film portion **505a** to a top of a container **600**. This is accomplished by energy from the at least one heating element **435** causing the film portion **505a** to shrink around the top portion **602** (e.g., a lid) of the container **600**—thereby forming a seal, such as shown in FIG. 3C. As noted herein, the film portion **505a** may include one or more radiation-absorbing layers of ink that shrink upon application of heat.

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The at least one heating element can comprise any suitable device that can be configured to emanate a suitable amount of energy to cause the film to secure to a container to form a seal thereon. In one example, the at least one heating element comprises a light bulb containing a tungsten filament (e.g., a tungsten-halogen light bulb). Other examples include, but are not limited to, a resistor, a cathode ray tube, a light emitting diode, a carbon filament bulb, a ceramic heater, amongst others. In some embodiments, the heating element(s) may be chosen/tailored to the specific component of the film which absorbs the radiant heat (e.g., carbon black pigment or other near infrared absorbing pigment/dye which could be a component of the applied coatings/inks). For example, an appropriate heating element (e.g., a bulb, an LED, a heater, etc.) may be chosen that radiates a type of heat that maximizes efficient absorption from the type of film and/or ink/coating thereon for sealing purposes. In some embodiments, two or more different types of heating elements may be installed in the sealing device and the most efficient/effective type of heating element may be chosen to operate based on the film that is currently being utilized (such as described with respect to various embodiments herein).

The at least one heating element **435** can be activated, such as via a controller (e.g., controller **30**) to emanate energy. In this regard, the term “emanate” may encompass any type of heat transfer to the film (e.g., conduction, convection, radiation, transfer, etc.). In this regard, though the described example light bulbs radiate energy, some embodiments of the present invention contemplate other types of heating elements that emanate energy via other methods, such as convention, conduction, heat transfer, radiation, etc.

For example, the at least one heating element **435** can comprise a light bulb that illuminates when the shield plate **405** moves between a first position and a second position. The emanated energy from the at least one heating element **435** can contact and heat the film, causing it to shrink. For example, energy from the at least one heating element can pass through wall **440** to contact the film. As discussed further below, the shrinkage of the film causes the film to secure or tightly secure to or around a lip, rim, or edge of a container. As used herein, shrinkage of the film creates a seal around the container opening where the film has substantially sealed the opening. In some embodiments, the film is not adhered or heat glued to the container, but rather is heat-shrunked to itself. In some embodiments, the film can cooperate with the container to melt and fuse together. For example, the container can include a coating that interacts with the film upon the activation of energy thereto.

In embodiments that include a plurality of heating elements, the plurality of heating elements can be activated simultaneously when the shield plate **405** is in the second position to effectuate a uniform melting of the film about the container. As used herein, the term “simultaneously” or “simultaneous” means at the same time or nearly at the same time such that any difference in activation is not readily apparent or detectable. The at least one heating element **435** can be positioned at any suitable location within the apparatus **100**. For example, in one embodiment, the at least one heating element **435** can be disposed between the wall **440** and the surrounding housing of the securing head assembly **400**. In other embodiments having a plurality of heating elements, the heating elements **435** can be equally radially spaced around a central axis defined by the wall **440**.

In some embodiments, each of the plurality of heating elements may be activated individually. Similarly, in some

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embodiments, groups of the plurality of heating elements may be activated together (e.g., in the case of four heating elements, a first group of two heating elements can be activated at the same time and then the second group of heating elements can be activated at a same second time). In some embodiments, the heating elements may be activated individually or in groups to achieve a desired effect (e.g., reach different temperatures, heat different sections of the film, etc.), such as for efficiently securing the film to the container. In some embodiments, the size or shape of the container may be detected and/or determined and the heating elements may be activated individually or in groups according to the detected and/or determined size or shape. In some embodiments, the ink applied to the film may be detected and/or determined and the heating elements may be activated individually or in groups according to the detected and/or determined ink.

The at least one heating element can be activated for a predetermined time once the heating element receives a signal for activation, such as from a controller. Alternatively, the at least one heating element can be activated while the shield plate remains in the second position.

In some embodiments, the heating element(s) may be configured to be replaceable to facilitate quick and easy replacement thereof. For example, the connection of the heating element(s) to the sealing apparatus may comprise a quarter turn connection or other type of connection.

In accordance with some embodiments, the apparatus **100**, **100'** can comprise at least one reflective device **430** disposed within the housing **425** and exterior to the wall **440**. The reflective device **430** reflects at least a portion of energy from the heating element **435** toward the wall **440** when the shield plate is in the second position. This energy propagates through the wall and is imparted upon the film ready to be secured to the container, as noted above. In some embodiments, the at least one reflecting element **430** comprises a mirror. In other embodiments, the at least one reflecting element **430** can comprise any suitable mechanism to allow energy to be redirected in a desired direction, such as but not limited to a metal piece, a plastic piece, a painted piece, a reflective coating, amongst others.

As shown in the illustrated embodiments of FIG. **6A** and **6B**, the apparatus comprises a plurality of reflecting elements **430** that can be arranged to form a continuous perimeter about the at least one heating element **435**. As shown, there are four reflecting elements in FIGS. **6A-6C**. In some embodiments, and as shown for example in FIG. **6C**, each reflecting element **430** comprises a first panel **436**, a second panel **437**, and a third panel **438**, wherein the first panel **436** is disposed at a first angle relative to the second panel **437**, and the third panel **438** is disposed at a second angle relative to the second panel **437**. In some embodiments, the first angle can be the same as the second angle. In alternate embodiments, the first and second angles can differ. Either or both of the first and second angles can be any suitable angle, for example in a range of about 20-70 degrees. In other embodiments, the reflecting element can include a monolithic curved plate embodied as the first, second, and third panel.

Example Shield Plate

As shown in FIG. **6B**, the apparatus can include a movable shield plate **405** (although in some embodiments the shield plate may be fixed so as to not be movable within the sealing portion). As depicted in the example of FIG. **6B**, the shield plate **405** is movable and can be coupled to a guide

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rod **410**. The guide rod **410** can facilitate movement of the shield plate **405** from the first position to the second position. The guide rod **410** can be of any suitable shape and can guide the shield plate **405** along any suitable movement trajectory between the first position and the second position within the sealing portion **301**, **301'**. For example, and as shown in FIGS. **6B** and **6C** for purpose of illustration and not limitation, guide rod **410** can have a longitudinal shape and can guide shield plate **405** along a vertical axis. In some embodiments, the vertical axis can be parallel to an axis defined by a center of the wall **440**. In other embodiments, guide rod **410** can have a different shape and can guide the shield plate **405** along a different trajectory between the first position and the second position, such as a curved trajectory. The shield plate in embodiment of FIG. **6B** is illustrated at a distance **X** above a portion of the housing for purposes of discussion. In some embodiments, it is contemplated herewith that the shield plate only need to move a minimal distance suitable to signal the sensors of the housing in order to activate the heating element. Thus, the shield plate can have a short stroke distance to activate the apparatus. As noted above, the guide assembly enables the minimal distance required for activation as the guide assembly loads the film in a ready position into the loading zone.

In an alternative embodiment, the shield plate is static and disposed at a distance above the aperture **426** to allow a top portion of the container to press a film there against. Such distance required for operation may be minimized due to the guide assembly, as referenced above. In such embodiments, the apparatus can include a sensor assembly adjacent the aperture **426** that detects when an object, such as the film and top of the container, breaks a plane of the aperture (or at least breaks a plane at a distance above the aperture) to signal the heating element to activate. In this embodiment, the static shield plate can be disposed beyond the aperture to provide a surface for the film to be pressed against by a top of the container while the heating element is activated. In yet another embodiment, the sensor assembly can be disposed in any suitable location to signal when the film is pressed against the shield plate by a top of the container and in a ready configuration, as further discussed herein. In some embodiments, a top wall of the securing head assembly may perform the same function as the fixed shield plate such that no shield plate is needed.

The shield plate **405** can be made of any suitable material that absorbs minimal energy from the heating element or is at least partially retardant thereto. In some embodiments, the shield plate is nonconductive and does not absorb energy. For example, the shield plate can be made of certain plastics, metals, wood, or other suitable materials. In one embodiment, the shield plate comprises aluminum. The shield plate **405** is preferably at least partially opaque with respect to energy emanated by the at least one heating element **435**. Thus, the shield plate **405** can protect a portion of the film from energy emanated by the at least one heating element. The shield plate, in some embodiments, can be biased towards the first position by gravity and by the weight of the shield plate itself. In other embodiments, the apparatus can include a biasing mechanism such as a spring to further bias the shield plate toward the first position to ensure the shield plate returns to the first position after use.

Example Sensor Assembly

In accordance with some embodiments, a sensor assembly can be operatively coupled to the shield plate **405** when the shield plate is embodied as a movable part. In other

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examples, such as when the shield plate is embodied as a static part, the sensor assembly can be positioned within the apparatus beyond the aperture, as discussed above.

In the embodiment shown in FIG. **6A**, the sensor assembly includes a signal emitter **421**, a signal sensor **422**, and a sensor flag **415** therebetween. The sensor flag positioning can directly correspond to the position of the shield plate as moved by the container. The signal emitter can continuously emit a signal capable of being detected by the signal sensor. The sensor flag **415** can be positioned between the signal emitter and the signal sensor, such that sensor flag **415** blocks the signal where opaque. However, the sensor flag **415** can further define a window therein at a predetermined location in order to allow the signal to trigger the signal sensor. Accordingly, as the signal flag **415** moves between the signal emitter and signal sensor, the window can allow transmission of the signal to the signal sensor. The signal sensor, upon detecting the signal, can register that the shield plate **405** has moved between the first position and the second position and generate a sensor input. In other embodiments, a reverse set up can be provided. That is, the sensor flag **415** can allow transmission of the signal to the signal sensor until it is blocked by the flag once the shield plate **405** has moved between the first position and the second position. The signal sensor, upon detecting the reduction of the signal, can register that the shield plate **405** has moved between the first position and the second position and generate a sensor input to activate at least one heating element **435**. Similarly, the signal sensor can deactivate the heating element(s) once the signal flag **415** moves back between the signal emitter and signal sensor, or as the signal sensor returns to its first state.

In other embodiments, the sensor can be any sensor suitable for detecting the movement of the shield plate **405** between the first position and the second position, or detect when the film and top of the container breaks a threshold plane when the shield plate is embodied as a static part. For example, the sensor **420** can include at least one of a signal sensor as described, a mechanical sensor, a motion sensor, amongst others. A mechanical sensor can include a rotational sensor, that is rotated by movement of the shield plate, a guide rod, a sensor flag, or another part coupled to the shield plate, and registers movement of the shield plate upon rotation, or registers other movements as contemplated herein. For example, a mechanical sensor can register a degree of movement by the amount of rotation.

The sensor input generated by sensor **420** can activate the at least one heating element **435** when the shield plate **405** has moved between the first position and the second position, or detect when the film and top of the cup breaks a threshold plane when the shield plate is embodied as a static part. Thus, at least one heating element **435** can be activated to emanate energy when the shield plate **405** is in the second position, initiating a securing cycle.

Example Additional Components

The securing portion **300** can further include any additionally desired components. Accordingly, the securing portion can include a base portion **320** as shown in FIG. **1A**. In some embodiments, the base portion **320** can be reflective to assist a user in aligning a container in the proper position. As such, the reflective base portion **320** can assist a user to insert a container into an approximate center of the aperture or can otherwise assist in positioning the container within the aperture in the proper location and ready for securement.

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As shown in FIGS. 2A and 2B, the securing portion **300**, **300'** can further contain a fan **305**, **305'** or a cooling mechanism to at least partially regulate the temperature of the securing portion **300**, **300'**. In such embodiments, the securing portion **300**, **300'** can further include a temperature sensor, a time keeper, or the like to either sense or calculate the temperature of the securing portion **300**, **300'**. The temperature regulation allows the apparatus **100**, **100'** to have improved repeat usability and to enhance safety. For example, the temperature regulation reduces the ability of the securing portion **300**, **300'** to overheat, which may cause damage to various components of the sealing apparatus and/or the container or may cause the film to improperly melt in an undesired manner. Additionally or alternatively, the partial temperature regulation allows for improved safety, for example by decreasing the risk that a person that comes into close proximity or contact with the securing portion **300**, **300'** may be exposed to undesirable heat. The fan **305**, **305'** can function to channel air toward the interior of the securing head assembly, or alternatively, away from the interior of the securing portion **300**, **300'**. Furthermore, the securing portion **300**, **300'** can contain more than one fan **305**, **305'** or other cooling devices known in the art.

In some embodiments, various vents or louvers may be employed within the apparatus **100**, **100'** to regulate heat within body portion **200**, **200'** and/or securing portion **300**, **300'**. For example, with reference to FIGS. 2A and 7, the securing head assembly **400'** includes two louvers **417'** that are configured to vent heat from within the sealing portion **301'**. Further, the apparatus **100'** may include various vents **307'**, **207'** that are configured to enable air flow through the various components of the apparatus **100'**. For example, FIG. 7 illustrates an example air flow (shown by arrows) through the apparatus **100'**, such as may be due to the fan **305'** pulling air from within the apparatus **100'** and pushing the air out to the environment through vent **307'**.

In some embodiments, the fan **305**, **305'** may be configured to operate or not operate at certain points during a life cycle of a sealing operation. For example, the controller may be configured to cease operation of the fan **305**, **305'** during operation of the heating elements **435** so as to enable efficient heat transfer to the film portion **505a** for shrinking thereof (and forming a seal around a top of a container). In some embodiments, after deactivation of the heating elements **435**, the controller may be configured to cause the fan **305**, **305'** to operate to cool down the inside of the apparatus **100**, **100'**. Further, in some embodiments, the controller may be configured to cease operation of the fan **305**, **305'** after a certain amount of time, such as to preserve power for the apparatus **100**, **100'** and/or prolong the overall life of the fan **305**, **305'**.

Example User Interface

The apparatus **100**, **100'** can be configured to receive inputs and commands. Such inputs and commands can be effectuated by way of a user interface operatively coupled with the apparatus. Alternatively or additionally thereto, the apparatus can be configured to receive inputs and commands remotely or wirelessly from a user or other electronically coupled devices.

In some embodiments, the securing portion **300**, **300'** includes a user interface to receive inputs and commands from a user. The user interface can include an indicator **310**, **310'** to display information to a user and at least one input device **315** to receive information. Furthermore, the indicator and input device can be combined as one device. In some

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embodiments, the indicator **310**, **310'** can include but is not limited to at least one of an electronic display (such as LED, OLED, LCD and the like), a light, a rotating indicator, sound device, an actuating indicator, a touchscreen, a smart device, or the like. The indicator **310**, **310'** can indicate the status or mode of the apparatus **100**, **100'**. For example, the indicator **310**, **310'** can indicate that the apparatus **100**, **100'** is in a ready state, a securing state, a preparing state, or another state. Additionally or alternatively, the indicator **310**, **310'** can indicate how many cycles the apparatus has performed since reset, the status of adjustable settings, repair information, a warning such as to replace a roll of film therein, and/or other information about the apparatus, as desired. Upon operation of the apparatus **100**, **100'** for example, the indicator **310**, **310'** can indicate completion of a securement cycle.

In some embodiments, the at least one input device **315** can include at least one of a push button, a lever, a dial, a virtual input on a graphical user interface, or the like. The at least one input device **315** can adjust at least one adjustable setting of the apparatus **100**. For example and not limitation, the at least one input device **315** can adjust the operation of the securing head assembly **300**. Additionally or alternatively, the at least one input device **315** can include a command capable of being understood by a computer (e.g., controller **30**). The indicator **310**, **310'** and the input device **315** can, in some embodiments, work in parallel and are not dependent on each other. In some embodiments, the input device **315** can display the value of the information received, for example, the input device **315** can comprise a dial that displays the value associated with each radial position of the dial. Additionally or alternatively, the apparatus can operatively cooperate with an external device, such as a smartphone, tablet, or external computer. As such, the apparatus can send and/or receive commands by such external devices. Thus, any indications contemplated by the indicator **310** or inputs by the input device **315** can be sent to and from the external device.

In some embodiments, and as further discussed herein, the apparatus further includes a computer (e.g., controller **30**) therein or be configured to cooperate with an external computer or device. The computer can be configured to at least one of control the apparatus, control the indicator **310**, **310'**, receive information from the at least one input device **315**, store adjustable settings, and communicate with an external device. Additionally or alternatively, the computer can be configured to process information, for example warning information, status information, and mode information, among others. In some embodiments, the computer can coordinate the operation of the apparatus **100**, for example by receiving sensor inputs and activating certain functionalities in accordance with the adjustable settings. The computer can be internal or external to the apparatus. The apparatus can furthermore include any devices necessary to ensure the apparatus carries out its operation, such as a circuit board **260**, **260'**.

In some embodiments, the user interface may be configured to display and receive user input, such as one or more user selections. For example, in some embodiments, a user may be able to select and/or provide instructions for the apparatus **100**, **100'**. As an example, the user interface may display printing options for a user to select from for printing on the film (such as with the printer). In some embodiments, the user may enter a desired message for printing on the film.

In such embodiments, the resulting sealed lid will then include the user selected message and/or image.

Example Methods of Use

Some example methods of operating the apparatus are discussed throughout the description and in relation to the figures. According to some embodiments, the method comprises providing an apparatus having a body portion to house film, and a securing head assembly. The securing head assembly includes a housing defining an aperture to receive at least a top portion of a container, a wall disposed within the housing, a shield plate movable within the aperture and wall between a first position and a second position, at least one heating element disposed within the housing and positioned external to the wall, a sensor assembly to sense movement of the shield plate and to activate the at least one heating element when the shield plate is in the second position, and a loading zone to receive a predetermined dimensioned film from the body portion. The wall has a width dimension sized to receive at least a top portion of a container. The at least one heating element is activated to emanate energy when the shield plate is in the second position. The loading zone is positioned adjacent the shield plate in the first position. The method further includes moving the top portion of the container relative to the aperture to move the shield plate to the second position and to secure a predetermined dimensioned film to a top of the container by energy from the at least one heating element. The method may further include moving the shield plate from the second position toward the first position, such as to deactivate the at least one heating element and/or after a period of time has elapsed.

In accordance with the disclosed subject matter, containers that can be used in conjunction with the disclosed subject matter can be of any suitable shape and size. For example, in some embodiments, containers that can be used in conjunction with the disclosed subject matter have a rim, lip or flange at the top portion of the container. In some embodiments, the film can shrink around the lip or flange and thereby be secured to the top portion of the container. In other embodiments, containers that can be used in conjunction with the disclosed subject matter have at least one of tabs, threads, ribs, panels, or other features to facilitate securement of the film to the containers.

As disclosed herein, the devices presented herein can be used for securing film to containers. The containers can contain a product held therein. Such products can include any type of product suitable for the containers, such as fluids or flowable products, beverages, food products and the like. The products can also include non-flowable products like solids and semi-solids of any of the above categories and more.

In accordance with the disclosed subject matter, in some embodiments containers that can be used in conjunction with the disclosed subject matter can be made of plastic, paper, metal, biodegradable materials, recycled materials, and/or reusable materials, among others. The material construction of the containers can complement the kind of film used to affect a secure film attachment. In some embodiments, the disclosed subject matter can secure a film to the top of containers ranging from about 200 mL to about 800 mL in capacity.

The disclosed subject matter can be combined with other features as well. For example, in some embodiments, the disclosed subject matter can be combined with a filling assembly in an integral unit. For example, the apparatus of

any of the embodiments of the disclosed subject matter can further comprise a fill nozzle, chute, funnel, or tube, among others, capable of filling product into the container before securing the film. Additionally or alternatively, in other embodiments, product can be filled into the container after securing of the film, for example through a portion of film pierced by the piercer **220**. In this manner, known filling assemblies can be incorporated with the disclosed subject matter.

In some embodiments, the disclosed subject matter can be combined with a cup moving assembly. For example, the apparatus of any of the embodiments of the disclosed subject matter can further comprise a mechanically movable gripper to hold and move containers. The mechanically moveable gripper can be of any suitable format. Alternatively, the apparatus can include a container seat to move containers disposed therein, as known in the art. In other embodiments, the apparatus can comprise a plurality of moveable levers for moving a container, amongst others. In some embodiments, the mechanically moveable gripper can move a container to interact with the apparatus of the disclosed subject matter, for example by rotational, linear, or other actuation of the gripper. In this manner, known container moving assemblies can be incorporated with the disclosed subject matter.

Example Marking Scheme System

Some embodiments of the present invention may provide a system for ensuring that an approved film is utilized with the sealing device. In this regard, there may be a desire to avoid usage of unauthorized rolls of film with the scaling apparatus, such as to avoid providing an unsatisfactory film or avoid potential maintenance issues.

For example, one or more film sensors (e.g., film sensor **240**) may be used to read one or more markings on the film, once the film is loaded into the apparatus **100**, **100'**. The sensor data may be sent to a controller (e.g., controller **30**), which may then determine, based on sensor data from the film sensor, if a detected one or more markings on the film satisfies an approved marking scheme. For example, the controller may be configured to access a database (such as in memory) of approved marking schemes and determine if the detected markings and/or marking scheme match (or sufficiently match) one of the approved marking schemes. In some embodiments, the controller may determine that certain marking characteristics of the detected markings matches (or sufficiently matches) one of a set of approved marking characteristics. Then, if approved, the apparatus **100**, **100'** and/or various functions/components (e.g., such as described herein) of the apparatus **100**, **100'** may be enabled for use. If unapproved, the apparatus **100**, **100'** and/or various functions/components (e.g., such as described herein) of the apparatus **100**, **100'** may be disabled. Additionally, reports of the approved or unapproved film usage may be provided to a remote server for data generation and use (e.g., re-ordering, maintenance, etc.). In some embodiments, the controller may cause the sensor data to be sent to a remote server to determine whether or not the detected one or more markings satisfy an approved marking scheme at the remote server.

In some embodiments, the controller may be configured to determine a film marking scheme for the film based on the detected one or more markings. For example, the determined film marking scheme may correspond to a marking pattern comprised of a plurality of markings and determined spacing between each adjacent marking within the plurality of

markings. In some embodiments, the controller may determine the film marking scheme based on at least one of the color of the one or more markings, a width of the one or more markings, a length of the one or more markings, a spacing between adjacent markings, etc.

In some embodiments, the determined film marking scheme may be repeated along the length of the roll of film such that the roll of film comprises a plurality of repeated film marking schemes. In such an embodiment, the film can be checked (e.g., continuously or periodically) and reconfirmed to avoid switching to an unapproved film after an initial loading takes place. Additionally or alternatively, in some embodiments, a body lid switch may be employed that provides an indication to the controller when the body lid has been opened. In response, the controller may check the marking scheme to confirm that the installed film includes an approved marking scheme.

In some embodiments, the determined film marking scheme may be checked against an approved marking scheme, such as via a controller. In response, one or more operations of the sealing device may be affected based on whether or not the detected marking scheme satisfies an approved marking scheme. For example, the controller may affect operation by enabling operation, disabling operation, or changing operation of the sealing device and/or one or more components of the sealing device. In some embodiments, the current or future operation (e.g., cycles) of the sealing apparatus may be affected.

In some embodiments, the controller may be configured to cause, in an instance in which the detected one or more markings satisfies the approved marking scheme, enabling operation of the sealing device and/or components thereof. For example, the controller may enable power to transfer to the sealing device (or its components); enable operation of the motor, enable operation of the heating element(s), enable operation of the printer, enable operation of the piercer, and/or enable operation of other various components.

Additionally or alternatively, in some embodiments, the controller may be configured to cause, in an instance in which the detected one or more markings does not satisfy the approved marking scheme, disabling operation of the sealing device and/or components thereof. For example, the controller may cut off power to the sealing device; disable the motor, disable the heating element(s), disable the printer, disable the piercer, or other various components.

Additionally or alternatively, in some embodiments, the controller may be configured to change or alter operation of one or more components of the sealing device. For example, if the detected one or more markings does not satisfy the approved marking scheme, the controller may cause one or more components to operate in a reduced or hindered capacity to discourage further use of the unauthorized film. For example, the controller may decrease a speed of operation of the motor, cause the film to misalign with the top of the container, decrease a speed or temperature of operation of the at least one heating element, cause the printer to print one or more messages or images in an off-center position on the film, cause the printer to print one or more messages indicating that an unapproved film is being utilized, increase a delay time between sealing operations performed by the sealing device, or cause the piercer to pierce the film in a undesired location. As more examples, the controller may cause an increased or over anticipated amount of film to be used to cause faster depletion of the unauthorized supply of film and/or misalignment of features to provide a less desirable end product.

In some embodiments, the one or more markings may be read and used to provide information (e.g., characteristics) relevant to the installed film. For example, various characteristics of the film (e.g., thickness, pre-printed information, etc.) may be determined and/or various desired operational parameters of the sealing apparatus during use with the film (e.g., how long to activate the heating element(s), what to print on the film, whether or not to pierce the film, etc.) may be determined. For example, an installed roll of film may have a thickness that would require a longer than normal time of operation for the heating element(s) to remain active for providing a sealed lid. Similarly, the planned lid may be printed with a lighter ink that may require a different than normal amount of time of operation for the heating element(s) to remain active for providing a sealed lid.

For example, in response to determining one or more characteristics of the one or more markings, the controller may further determine a desired operation of one or more components of the apparatus based on the detected one or more characteristics and cause operation of the one or more components of the apparatus based on the determined desired operation. As an example, the controller may cause the at least one heating element **435** to operate according to at least one of a specific amount of time or a specific heat based on the determined one or more characteristics (e.g., the specific film may require a certain heat for proper shrinkage and scaling). As another example, the controller may cause the motor **213'** to operate according to at least one of a specific amount of time or according to a specific number of detected markings based on the determined one or more characteristics (e.g., the film may correspond to a specific product and/or may be designed for use with a specifically-sized container—which may lead to a desire to provide a portion of film of a specific length to the sealing portion **301, 301'**). As yet another example, the controller may cause the piercer **220, 220'** to operate based on the determined one or more characteristics (e.g., the associated product may not require the piercer to operate or it may be desirable to utilize the piercer to provide a slit/perforation in a specific location or of a specified dimension). As yet another example, the controller may cause the printer **250** to operate based on the determined one or more characteristics, such as by printing one or more messages or images on the film based on the determined one or more characteristics (e.g., the associated product that is utilized with the film may be associated with a certain logo to be printed on the film). In addition to the above examples, other example operations/features that would be controllable are contemplated, such as providing a delay between dispenses of the film portion, controlling the fan air flow, controlling the motor operation speed, controlling which ones and how many heating elements to activate, controlling the user interface, among many others.

In some embodiments, the controller may determine one or more characteristics of the film based on the detected one or more marking characteristics. For example, the controller may determine at least one of the thickness of the film, the associated customer for the container, the associated product for use with the film, a time of operation of the heating element, a subset of printing options to present to a user for selection, or the amount of film remaining on the roll of film. In the case of being able to determine the amount of film remaining on the roll of film, the marking scheme may further include a differentiating characteristic from among at least some of the marking schemes to thereby indicate a

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relative position along the roll of film (e.g., there may be countdown type characteristic applied to some of the marking schemes).

In some embodiments, the controller may be configured to communicate with a remote server. In some such example embodiments, the controller may be configured to receive an update (e.g., a software update) and update various functionality accordingly. In some example embodiments, the controller may communicate any gathered information to the remote server, such as with respect to usage data or other types of data.

Example System Architecture

FIG. 13 illustrates an example system/environment in which some example embodiments of the present invention may be employed. The system 1 may include an example sealing apparatus 100, 100', such as described herein. The example scaling apparatus 10 may comprise hardware and/or software capable of performing functions described herein. In this regard, the apparatus 10 may include a roll of film 9 (although the film may be in stacks in some embodiments) that includes a leading edge that extends through the scaling apparatus along a film path 7. Additionally, the apparatus 10 may include a controller 30, a motor 22, a memory 32, a communication interface 34, a user interface 36, and a power source 39. Further, the apparatus 10 may include a printer 50, a piercer 20, a nip 23, one or more film sensors 25, a cutter 27, and a sealing portion 11 that are positioned along the film path 7. The scaling portion 11 may include one or more heating elements 40 and a shield plate 15. A user may position a container 6 within the sealing portion 11 to cause sealing thereof. In this regard, various described components and features of the example system 1 may correspond to components and features described herein, such as with respect to apparatus 100, 100' (among the other described embodiments).

The controller 30 may be any means configured to execute various programmed operations or instructions stored in a memory device such as a device or circuitry operating in accordance with software or otherwise embodied in hardware or a combination of hardware and software, thereby configuring the device or circuitry to perform the corresponding functions of the controller 30 as described herein. In this regard, the controller 30 may be configured to receive (such as via the communication interface 34 or user interface 34) and/or determine (such as based on sensor data) one or more instructions for operating one or more components of the apparatus 10. In this regard, the controller 30 may be connected to and control or cause operation of the user interface 36, memory 32, communication interface 34, motor 22, printer 50, piercer 20, film sensor(s) 25, cutter 27, shield plate 15, and/or heating element(s) 40. Though shown as a single controller, in some embodiments, various separate controllers (whether alone or in communication with each other) may perform functions described herein.

The one or more motors 22 may be used to drive (e.g., through rotation or otherwise) the nip 23 (e.g., the drive roller) to cause advancement of the film from the roll of film 9 along the film path 7. In some embodiments, the controller 30 may be configured to operate the motor 22 accordingly.

The printer 50 may be configured to print on the film. In some embodiments, the printer 50 includes ink that enables the printing, such as provided from one or more ink cartridges, tanks, reservoirs, etc. In some embodiments, the controller 30 may be configured to instruct the printer 50 to cause printing of various messages and/or images.

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The film sensor(s) 25 may be configured to detect one or more markings and/or marking characteristics of a marking scheme on the film, such as described herein. The film sensor(s) 25 may be configured to provide the sensor data to the controller 30, such as described herein.

The cutter 27 may be configured to operate to cut the film, such as in response to instruction from the controller 30.

The heating element(s) 40 may be configured to operate, such as based on instructions from the controller 30 or other sensor assembly, such as described herein. In particular, in some embodiments, the heating element(s) 40 may be configured to emanate energy to cause shrinkage of the film to or around the top portion of the container 6 to form a sealed lid thereon. In some embodiments, the formed seal may be configured to be absolute such that no liquid may leak around the seal. In other embodiments, the formed seal may be configured to be partial such that a portion of the top container is unsealed or lightly sealed. In such an example embodiment, a user may be able to drink from the unsealed portion and/or peel back the unsealed or lightly sealed portion (e.g., to facilitate drinking therefrom and/or pouring therethrough).

The shield plate 15 may be configured to provide a shielded surface for positioning of the container 6 within the sealing portion 11. The shield plate 15 may provide physical resistance and/or heat dispersion or other characteristic to aid in sealing of the container 6.

The memory 32 may be configured to store instructions, computer program code, approved marking schemes and/or characteristics, and other data/information associated with the apparatus 10 in a non-transitory computer readable medium for use, such as by the controller 30.

The communication interface 34 may be configured to enable connection to external systems (e.g., an external network 12 and/or one or more other system(s)/device(s), such as another apparatus 10). In some embodiments, the communication interface 34 may comprise one or more transmitters configured to transmit, for example, one or more signals according to example embodiments described herein. Likewise, the communication interface 34 may include at least one receiver configured to, for example, receive data according to example embodiments described herein. In some embodiments, the transmitter and receiver may be combined as a transceiver. In this regard, the apparatus 10 may be configured for wired and/or wireless communication. In some embodiments, the communication interface 34 may comprise wireless capabilities for WiFi, Bluetooth, or other wireless protocols. In some embodiments, the apparatus 10 may be connected to one or more point-of-sale system(s) 13 to aid in performance of food orders (such as by forming sealing lids for use with the order).

The user interface 36 may be configured to receive input from a user and/or provide output to a user. The user interface 36 may include, for example, a display, a keyboard, keypad, function keys, mouse, scrolling device, input/output ports, touch screen, or any other mechanism by which a user may interface with the system. Although the user interface 36 is shown as being directly connected to the controller 30 and within the apparatus 10, the user interface 36 could alternatively be remote from the controller 30 and/or apparatus 10. Likewise, in some embodiments, other components of the apparatus 10 could be remotely located.

The power source 39 may be any type of power source, such as a battery (or batteries) and/or an external power source (e.g., a plug 209, such as shown in FIG. 1A, may be used to receive power from an outlet). Power from the power

source 39 may be used to provide power to any of the components/devices utilized in the apparatus 10.

Example Flowchart(s)

Embodiments of the present invention provide methods, apparatuses and computer program products for operating example sealer devices according to various embodiments described herein. Various examples of the operations performed in accordance with embodiments of the present invention will now be provided with reference to FIGS. 14-16.

FIG. 14 illustrates a flowchart according to an example method for operating an example sealer device according to an example embodiment. Notably, while FIG. 14 provides a flow of various operations, the order of occurrence of the operations is not meant to be limited to that illustrated in FIG. 14 and may vary within embodiments of the present invention. The operations illustrated in and described with respect to FIG. 14 may, for example, be performed by, with the assistance of, and/or under the control of one or more of the components of example systems/device described herein, such as apparatus 10, 100, 100' among other things.

The method 1000 may include sensing movement of the shield plate to a second position within the sealing portion at operation 1002. At operation 1004, the method may include activating one or more heating element(s) to cause sealing of the film to a container. At operation 1006, the method may include printing on the film, such as including one or more messages or images. Then, the method may include piercing the film with a piercer at operation 1008. The method may further include operating the motor to cause advancement of the film along the film path at operation 1010. At operation 1012, the method may include sensing one or more markings on the film. Then, at operation 1014, the method may cause ceasing operation of the motor to position portion of the film in the sealing portion. Next, at operation 1016, the method may include cutting the film, thereby putting the sealing device back into a ready status.

As noted above, in some embodiments, different orders of operations of FIG. 14 may occur in various embodiments described herein. For example, the printing and/or piercing may occur before the heating elements are activated. Likewise, the printing and/or piercing may occur after the motor has operated to advance the film. In some embodiments, the operations may occur at the same time (e.g., the piercing may occur while the motor is operating to advance the film, the sensing may occur while the motor is operating to advance the film, etc.).

FIG. 15 illustrates a flowchart according to an example method for enabling or disabling operational ability of the sealer device based on whether the installed film is an approved roll of film according to an example embodiment. The operations illustrated in and described with respect to FIG. 15 may, for example, be performed by, with the assistance of, and/or under the control of one or more of the components of example systems/device described herein, such as apparatus 10, 100, 100' among other things.

The method 1100 may include operating the motor to cause advancement of the film along the film path, such as across the film sensor, at operation 1102. At operation 1104, the method may include sensing one or more markings on the film. At operation 1106, the method may include determining if the one or more markings satisfy an approved marking scheme. Then, if the one or more markings do satisfy an approved marking scheme, the method may include enabling operation of the sealing device and/or

various components of the sealing device at operation 1108. However, if the one or more markings do not satisfy an approved marking scheme, the method may include disabling operation of the sealing device and/or various components of the sealing device at operation 1110.

FIG. 16 illustrates a flowchart according to an example method for operating an example sealer device according to an example embodiment. The operations illustrated in and described with respect to FIG. 16 may, for example, be performed by, with the assistance of, and/or under the control of one or more of the components of example systems/device described herein, such as apparatus 10, 100, 100' among other things.

The method 1200 may include operating the motor to cause advancement of the film along the film path, such as across the film sensor, at operation 1202. At operation 1204, the method may include sensing one or more markings on the film. At operation 1206, the method may include determining one or more characteristics of the film and/or the planned operation of the sealer device based on the sensor data. Then, the method may include operating one or more components of the sealer device according to the determined one or more characteristics at operation 1210.

FIGS. 14-16 illustrates an example flowchart of a system, method, and computer program product according to various example embodiments described herein. It will be understood that each block of the flowcharts, and combinations of blocks in the flowcharts, may be implemented by various means, such as hardware and/or a computer program product comprising one or more computer-readable mediums having computer readable program instructions stored thereon. For example, one or more of the procedures described herein may be embodied by computer program instructions of a computer program product. In this regard, the computer program product(s) which embody the procedures described herein may be stored by, for example, the memory and executed by, for example, the controller 30. As will be appreciated, any such computer program product may be loaded onto a computer or other programmable apparatus to produce a machine, such that the computer program product including the instructions which execute on the computer or other programmable apparatus creates means for implementing the functions specified in the flowchart block(s). Further, the computer program product may comprise one or more non-transitory computer-readable mediums on which the computer program instructions may be stored such that the one or more computer-readable memories can direct a computer or other programmable device to cause a series of operations to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus implement the functions specified in the flowchart block(s).

CONCLUSION

Many modifications and other embodiments of the inventions set forth herein may come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the embodiments of the invention are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the invention. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combina-

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tions of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the invention. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated within the scope of the invention. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

The invention claimed is:

1. An apparatus comprising:

a film support roller to support a film roll that has a length of film;

a housing with a sealing portion including an aperture sized to receive a top portion of a container therethrough, wherein the housing defines a film path leading from the film roll to the sealing portion;

a guide support assembly having a ramp and at least one guide truss positioned along the film path within the housing, wherein the ramp is configured to receive a portion of the film, and wherein the ramp and the at least one guide truss are configured to guide the portion of the film to extend across the aperture within the sealing portion such that the portion of the film covers the aperture;

a motor configured to operate to cause advancement of the film along the film path;

at least one heating element configured to activate to emanate energy; and

a controller configured to:

cause the motor to operate to advance the portion of the film onto the ramp; and

cause the heating element to activate to emanate energy to cause the portion of the film within the sealing portion to seal the top portion of the container by forming a lid for the container.

2. The apparatus of claim 1, wherein the ramp has a surface extending from a first end to a second end, the

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surface oriented at an angle of inclination ranging up to 75 degrees, wherein the second end is coupled to a guide surface configured to receive the portion of the film from the surface of the ramp.

3. The apparatus of claim 1, wherein the guide support assembly comprises a funnel structured to receive a leading end of the film therethrough and to channel the leading end of the portion of the film onto the ramp.

4. The apparatus of claim 1 further comprising a printer configured to print on the film; and wherein the controller is configured to cause the printer to print one or more messages or images on the film prior to the lid being formed on the container, such that the lid includes the one or more messages or images printed thereon.

5. The apparatus of claim 4, wherein the controller is configured to determine the one or more messages or images to print on the film.

6. The apparatus of claim 5, wherein the controller receives information from a point-of-sale system that controls the one or more messages or images to be printed on the film.

7. The apparatus of claim 5, wherein the one or more messages or images to be printed on the film is a personalized message related to an order.

8. The apparatus of claim 5, wherein the one or more messages or images to be printed on the film is based on an order.

9. The apparatus of claim 8, wherein the one or more messages or images comprise a description of contents of the container.

10. The apparatus of claim 1, further comprising a communication interface configured to communicate with a remote device, wherein the controller is configured to receive data from the remote device and operate one or more components of the apparatus according to the received data.

11. The apparatus of claim 10, wherein the remote device is a point-of-sale system.

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